GE Healthcare

LightSpeed 5.X Pro16

Pre-Installation Manual

OPERATING DOCUMENTATION

2349181-100
Rev 12

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IMPORTANT PRECAUTIONS

LANGUAGE	
Предупреждение (BG)	• ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК. • АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД. • НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА. • НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.
警告 (ZH-CN)	• 本维修手册仅提供英文版本。 • 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。 • 未详细阅读和完全理解本维修手册之前，不得进行维修。 • 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。
VÝSTRAHA (CS)	• Tento provozní návod existuje pouze v anglickém jazyce. • V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. • Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. • V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	• Denne servicemanual findes kun på engelsk. • Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. • Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået. • Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
WAARSCHUWING (NL)	• Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. • Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. • Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is. • Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)	• This Service Manual is available in English only. • If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.
HOIATUS (ET)	• Käesolev teenindusjuhend on saadaval ainult inglise keeles. • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	• Tämä huolto-ohje on saatavilla vain englanniksi. • Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. • Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. • Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	• Ce manuel de service n'est disponible qu'en anglais. • Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire. • Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris • Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	• Diese Serviceanleitung existiert nur in Englischer Sprache. • Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen. • Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. • Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch stromschläge, Mechanische oder Sonstige gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	• Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο. • Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης. • Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εαν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις. • Εαν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS
(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése.
- Ne próbálja elkezdeni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN
(IS)

- Þessi þjónustuhandbók er eingöngu fáanleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA
(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告
(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고
(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다 .
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 .
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 .
- 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

BRDINJUMS
(LV)

- Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā.
- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS
(LT)

- Šis eksploatavimo vadovas yra prieinamas tik anglų kalba.
- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.
- Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote.
- Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.

ADVARSEL
(NO)

- Denne servicehåndboken finnes bare på engelsk.
- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE
(PL)

- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.

ATENÇÃO
(PT-BR)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado ecompreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode por em perigo a segurança do técnico, operador ou paciente devido a choques elétricos, mecânicos ou outros.

ATENÇÃO
(PT-PT)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado ecompreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros..

ATENȚIE (RO)	• Acest manual de service este disponibil numai în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării şi înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	• Данное руководство по обслуживанию предлагается только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	• Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilacke usluge. • Ne pokušavajte da opravite uredaj ako niste procitali i razumeli ovo servisno uputstvo. • Zanemarivanje ovog upozorenja može dovesti do povredivanja servisera, rukovaoca ili pacijenta usled strujnog udara ili mehanickih i drugih opasnosti
UPOZORNENIE (SK)	• Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednos″ou zákazníka. • Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže vyústi″ do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.
ATENCION (ES)	• Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	• Den här servicehandboken finns bara tillgänglig på engelska. • Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DIKKAT
(TR)

- Bu servis kilavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kilavuzu ingilizce dişinda bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kilavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyariya uyulmamasi, elektrik, mekanik veya diğer tehlikelerden dolayi teknisyen, operatör veya hastanin yaralanmasina yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “damage in shipment” written on all copies of the freight or express bill before delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on http://egems.med.ge.com/edq/home.jsp
- Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION

Risk of Explosion.

Danger of explosion if battery is incorrectly replaced.

Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

ATTENTION

Danger d'Explosion

Il y a danger d'explosion s'il y a replacement incorrect de la batterie.

Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rébut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the Healthcare PQR Process to report all omissions, errors, and defects in this publication.

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Revision History

Revision	Date	Reason for change
12	09/26/08	Updated **Important Precautions**, Language section to add new languages (PT-BR, PT-PT, SK). **PQR13168182**: Updated, Chapter 9, Section 5, Table 9-6 to add information about A1 Disconnects for Europe. **SPR FCTge36010**: • Chapter 4, Section 5.3: removed bulletpoint allowing for remote mount of the console. • Chapter 4, Section 3.1.4: Changed table headings in Table 4-7 to match those in LS 5.X Installation mnaual. • Chapter 7, updated Section 4.0 Storage Requirements; updated Section 5.0 "Extreme Temperature Transportation and Deliveries"; Section 7.0: update the URL for the umi-dolly shop.
11	10/12/07	**Important Precautions:** Updated to include new language–Bulgarian **Chapter 4:** • Section 1.0: Added bullet point concerning cable storage. • Section 2.1.1: Corrected *Regulatory Minimum Working Clearance by Major Subsystem* tables to match what is in the LS 5.X Installation Manual. Added tables with information for the UPS and A1 Disconnect. • Section 4.1: Added bullet point concerning clear access for service components. • Section 8.6 Floor levelness: added Note about specifications for the floor/gantry to prevent cable crushing. • Section 9.0: Added the network specifications. • Section 10, Corrected scatter plots for body and head filters. **Chapter 5:** Section 4.0: Added note concerning EMI testing near power sub-stations. **Chapter 9:** Section 6.1: Added information concerning LOTO and UPS. **General updates to wording and grammar.**
10	2/23/07	Updated measurements to metric to meet European compliance. General revision of manual. Updated Warnings section to include new languages. **PQR 13092823,13093839:** **Chapter 8**: Updated Section 2.1 - Power Source Configuration
9	10/03/06	Updated Chapters 1 - 4 for “Limited Access” Sites.

Revision	Date	Reason for change
8	2/23/06	**Chapter 1**: • Updated Section 1.0 - Site Readiness • Updated Section 2.0 - Responsibility of Purchaser • Deleted old Section 3.0 - Site Preparation Prior to Equipment Delivery • Added/Updated new Section 3.0 - Pre-Installation Checklist (moved from Chapter 2) • Added "Pre-Installation Block Diagram" (moved from Chapter 2) **Chapter 2**: • Moved Checklist and Block Diagram to Chapter 1. • Added Notice to review DVD prior to beginning Pre-Install work. • Added "Floor Levelness Specification" **Chapter 4**: Major updates, including (but not limited to) definition of *Regulatory* vs. *Service* Clearances and updates to "Structural Requirements". **Chapter 5**: • Updated Section 1.0 - Temperature and Humidity Specifications • Updated Section 2.0 - Cooling Requirement *This Page Intentionally Blank.* **Chapter 6**: • Added Table 6-2 - Gantry and Table Mounting Requirements • Updated Figure 6-5 - Power Distribution Unit (NGPDU) **Chapter 7**: • Added Notice regarding lifting of CT Gantry. • Updated Section 4.0 - Storage Requirements • Updated Section 5.0 - Extreme Temperature Transportation and Deliveries • Updated Section 6.0 - System Transportation **Chapter 9**: • Updated Section 4.0 - Contractor Supplied Components • Updated Section 5.0 - UPS Interconnect • Updated Section 6.0 - Typical Customer Supplied Wiring Added **Appendix** "FCT Installation Site Ready Form" Updated formatting to latest "branding" standards.
7	8/28/05	PSR 13050339: Added Note to Figure 9-3: Interconnection Runs
6	07/21/05	PSR 13033111: Changed Scatter Survey values (shown in Chapter 4) from mR/scan to µGy/scan.
5	10/22/04	EMC Ed2 Update. Chapter 6 - Floor Anchor updates. Chapter 9 - Added UPS Compatibility Matrix.
4	05/18/04	CTCge87943: Corrected Floor Anchoring Information. Chapter 5 - Updated Cooling Requirements Worksheet. Chapter 6 - Corrected Gantry Floor Loading Spec.
3	12/15/03	Chapter 5 - Updated cooling requirements Chapter 9 - Added A1 panel cat num.

Revision	Date	Reason for change
2	09/16/03	Chapter 4 – Updated notice in Sec. 3.2 regarding floor requirements and the need to contact GEMS Engineering Chapter 5 – Updated Table 5-1: Cooling Requirements (Worksheet) Chapter 6 – Updated Table 6-1: LightSpeed 5.0 System Floor Loading General Updates
1	04/29/03	General Updates
0	02/14/03	Initial Release of Document

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Table of Figures

Figure 2-1
Hole Locations 37
Figure 3-1
Base Scanner System.... 40
Figure 4-1
Regulatory Clearance Requirements for LightSpeed Pro16 System Configurations.... 42
Figure 4-2
Room Layout with cabinets 53
Figure 4-3
Room Layout without cabinets 53
Figure 4-4
Minimum Service Clearance 54
Figure 4-5
Typical Control Room Layout.... 57
Figure 4-6
Gantry Front Cover with Service Dolly Dimensions 59
Figure 4-7
Table and Gantry (Side View) 60
Figure 4-8
Gantry shown tilted +30º (top) and -30º (bottom).... 61
Figure 4-9
Power Distribution Unit (NGPDU-4) 62
Figure 4-10
Power Distribution Unit (NGPDU-4) 62
Figure 4-11
Operator's Console 63
Figure 4-12
Room Layout, with Injector Option 65
Figure 4-13
Console Rear Bulkhead 69
Figure 4-14
Typical Scatter Survey (Body Filter).... 74
Figure 4-15

Typical Scatter Survey (Head Filter) 75
Figure 5-1
HVAC Air Vent Placement - Scan Room 80
Figure 5-2
HVAC Air Vent Placement - Control Room 80
Figure 5-3
Sample Room Layout, showing approximate EMI requirements 82
Figure 6-1
Typical Floor Anchor, Gantry and Table 87
Figure 6-2
Gantry (CT2) 88
Figure 6-3
Gantry Anchor Locations.... 88
Figure 6-4
Patient Table (CT1).... 89
Figure 6-5
Power Distribution Unit (NGPDU) 90
Figure 6-6
Operator Console.... 91

Figure 7-1
Gantry Strap Location 93
Figure 7-2
Gantry with Shipping Dollies and Side Rails........ 96
Figure 7-3
Console ready to be unloaded from shipping skid. 97
Figure 8-1
System Ground Map 103
Figure 9-1
System Interconnect Diagram 107
Figure 9-2
System Interconnect Diagram 109
Figure 9-3
Interconnection Runs 111
Figure 9-4
Typical UPS 113
Figure 9-5
Typical UPS Interconnect 113
Figure 9-6
Primary Power Disconnect (A1) 114
Figure 9-7
Primary Power Disconnect (A1) – Fusible Disconnect and Magnetic Contactor 115
Figure 9-8
TS6 X-Ray Warning Light Connections 116
Figure 9-9
TS6 Room Door Interlock Connections - Without a Door Interlock 116
Figure 9-10
TS6 Room Door Interlock Connections - With a Door Interlock 116

Table of Tables

Table 1-1
Schedule Date Commitments 31
Table 1-2
General Site Planning 32
Table 1-3
References for Specific Site Requirements 32
Table 1-4
Equipment Compatibility 33
Table 1-5
Network Connections 33
Table 1-6
Miscellaneous Tasks 33
Table 3-1
System Options Catalog (Part) Numbers 39
Table 4-1
Console Subsystem 43
Table 4-2
NGPDU Subsystem 43
Table 4-3
Gantry Subsystem 44
Table 4-4
Table Subsystem 44
Table 4-5
UPS Subsystem 45
Table 4-6
A1 Disconnect Subsystem 46
Table 4-7
System Specifications (Pro16) 49
Table 4-8
Equipment to be stored in storage cabinet 56
Table 4-9
Dimensions of Components 58
Table 4-10
Minimum Dimensions and Clearances 64
Table 4-11
Shielding Requirements Scaling 73
Table 5-1
Cooling Requirements (Worksheet) 79
Table 6-1
LightSpeed Pro16 System Floor Loading 85
Table 6-2
Gantry and Table Mounting Requirements 86
Table 7-1
Size of Gantry & Dollies, with and without Side Rails 96
Table 8-1
Nominal Line Voltage 101
Table 8-2
Purchasable Options 101
Table 8-3
Minimum Feeder Wire Size 102

Table 8-4
Minimum Sub-Feeder Wire Size 102
Table 9-1
Component Designators 105
Table 9-2
GE Healthcare Supplied Cables (Standard Run) - UL Information 106
Table 9-3
GE Healthcare Supplied Cables (Optional, Long Run) - UL Information 108
Table 9-4
Runs 1, 2, 3, 4 and 5 Connections 110
Table 9-5
Contractor-Supplied Components 112
Table 9-6
LightSpeed 5.X Partial UPS Back-up Options 113

Table of Contents

Chapter 1
Introduction 25

Section 1.0
Site Readiness 25

Section 2.0
Responsibility of Purchaser 26

2.1 Customer Room Prep Items 26
2.2 Purchaser Site Preparation Work 27
2.3 Manufacturer's System Level Siting Requirements 27
2.3.1 Meeting Site Ready Requirements 28
2.3.2 Quick Installs 28
2.3.3 Two-Step and Upgrade Installs 28
2.3.4 Site Ready Inspection Visit 29

Section 3.0
Pre-Installation Checklist 31

Chapter 2
Pre-Installation Overview 35

Section 1.0
Dust/Dirt Contamination 35

Section 2.0
Chemical Contamination 35

Section 3.0
Walls, Ceiling, and Floor 35

Section 4.0
Broad-band 36

Section 5.0
Phone Line (for optional modem) 36

Section 6.0
Review 36

Chapter 3
System Catalog 39

Section 1.0
Option Catalog Numbers 39

Section 2.0
Base Scanner System 40

2.1 Application 40
2.2 Configuration 40

Chapter 4
Room Planning 41

Section 1.0
Required Systems Clearances 41

Section 2.0
Regulatory and Service Clearances 42
2.1 Regulatory Clearances 42
2.1.1 Regulatory Minimum Working Clearance by Major Subsystem 43
2.1.2 Terms and Definitions 46

Section 3.0
Additional Regulatory Clearance Information 48
3.1 Minimum Room Size (Limited Access) 48
3.1.1 Regulatory Caution 48
3.1.2 Egress Clearance 48
3.1.3 Operational Caution 48
3.1.4 System Specifications (Pro16) 49
3.1.4.1 Preferred Room Size 49
3.1.4.2 Typical Room Size 49
3.1.4.3 Minimum Room Size 49
3.1.5 Room Size for PET Upgrades 49
3.2 How to Measure 50
3.3 Minimum Room Size & Requirement Layouts 51
3.4 Preferred and Typical Room Size & Requirement Layouts 53

Section 4.0
Service Clearances 54
4.1 Service Clearances for Single Service Engineer 55
4.2 Power Distribution Unit (NGPDU) Service Clearance 55
4.3 Console Service Clearance 55

Section 5.0
Recommended Layouts 56
5.1 Storage Cabinet 56
5.2 Advantage Workstation (AW) 56
5.3 Control Room Considerations 57

Section 6.0
Component Dimensions 58
6.1 Table and Gantry 60
6.2 Power Distribution Unit 62
6.3 Operator Console 63

Section 7.0
Minimum Dimensions and Clearances 64
7.1 Injector Control 64

Section 8.0
Structural Requirements 66
8.1 Table and Gantry Mounting Requirements 66
8.2 Minimum Floor Thickness 66

8.3 Floor Anchors 67
8.4 Non-Concrete Floors 67
8.5 Floor Strength 67
8.6 Floor Levelness 67
8.7 Floor Vibration 67
8.7.1 Steady State Vibration 68
8.7.2 Transient Vibration 68
8.7.3 Equipment Location 68
8.8 Walls: Scan Window 69

Section 9.0
Network Connections 69
9.1 US Broad Band Process Overview 70
9.2 Customer Broad-Band Responsibilities 70

Section 10.0
Radiation Protection 73

Chapter 5
Environmental Conditions 77

Section 1.0
Temperature and Humidity Specifications 77
1.1 Temperature (Scan Room & Control Room) 77
1.2 Humidity (Scan Room & Control Room) 78
1.3 Other Guidelines 78

Section 2.0
Cooling Requirement 79

Section 3.0
Altitude 80

Section 4.0
Electro-Magnetic Interference (EMI) 81
4.1 Gantry 81
4.2 Color Monitor 81
4.3 Console / Computer Equipment 81
4.4 Magnetic Media 81
4.5 PDU 81
4.6 EMI Reduction 81
4.7 UPS 82
4.8 Equipment EMI “Envelopes” 82
4.9 EMC Edition 2 Compliance 83
4.9.1 General Scope 83
4.9.2 USE LIMITATION: External Components 83
4.9.3 INSTALLATION REQUIREMENTS & ENVIRONMENT CONTROL 83

Chapter 6
Floor Loading and Weights 85

Section 1.0
Floor Loading 85

Section 2.0
Mounting Data, Including Seismic 86

Chapter 7
Delivery Data 93

Section 1.0
Van Delivery 93

Section 2.0
Delivery/Shipping Requirements & Considerations 93

Section 3.0
Site Environmental Considerations 94

3.1 Dust/Dirt Contamination 94
3.2 Chemical Contamination 94

Section 4.0
Storage Requirements 95

4.1 Short-term Storage (Less than Six Months) 95

Section 5.0
Extreme Temperature Transportation and Deliveries 95

Section 6.0
Gantry Considerations 96

Section 7.0
Table Considerations 97

Section 8.0
Console Considerations 97

Chapter 8
Power Requirements 99

Section 1.0
Introduction 99

Section 2.0
System Input Power 99

2.1 Power Source Configuration 99
2.2 Rating 99
2.3 Regulation 100
2.4 Phase Imbalance 100
2.5 Sags, Surges & Transients 100
2.6 Grounding 100

Section 3.0
Recommended Power Distribution System 101

Section 4.0
Ground System 103

Chapter 9
Interconnection Data 105

Section 1.0
Introduction 105

Section 2.0
Component Designators 105

Section 3.0
Interconnect Runs, Wiring and Cables 106

3.1 GE Healthcare Supplied (Standard Length) 106
3.2 GE Healthcare Supplied (Optional, Long Run) 108
3.3 Contractor (Customer) Supplied 110

Section 4.0
Contractor Supplied Components 112

Section 5.0
UPS Interconnect 113

Section 6.0
Typical Customer Supplied Wiring 114

6.1 Primary Power Disconnect 114
6.2 Scan Room Warning Light & Door Interlock 116

End of Section.

Chapter 1 Introduction

This direction contains physical and electrical data necessary for planning and preparing a site.

Pre-installation work is defined as site preparation for installation of the GE CT scanner. It is the responsibility of the purchaser to arrange and pay for this work. Pre-installation work includes:

- Installation of electrical conduit, junction boxes, ducts, outlets, and line safety switches.
- Installation of interconnection wiring that is AWG stranded copper. The electrical contractor shall ring out and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Ground wires must conform to local codes.
- Any site renovation.
- Alterations and modifications to products not specifically included in the sales contract.

All work must conform to local building and safety codes. Unless specifically mentioned, GE Healthcare does not provide or install wires, conduits, junction boxes, and ducts as illustrated in this publication.

All General Electric Headquarters Architectural Planning must review all CT site plans, preliminary concepts and final working drawings prior to construction or approval.

Contact your local General Electric sales representative or Project Manager of Installation for complete information regarding your site-specific room layout.

Section 1.0 Site Readiness

Site ready is a requirement that must be achieved to install a CT product. For your convenience, a site ready visit inspection is performed at least three (3) days prior to the installation date. The site inspection must conclude with a minimum of a conditional pass status to be ready on the requested installation delivery date. Site ready inspections on the delivery date are not acceptable unless prior arrangements have been made.

Pre-Installation and Site Ready Tools:

- Pre-Installation Video
- Floor template
- Pre-Installation Check List
- Pre-Installation Block Diagram
- Site Room Layouts
- Power and Grounding Inspection
- Pre-Installation Support

Section 2.0 Responsibility of Purchaser

2.1 Customer Room Prep Items

The CT air intake is near the bottom of the gantry and draws air in through a filter in the gantry heater assembly. Fine dust, as listed below, clogs the filter and is deposited throughout the gantry, table, operator console and PDU electronics. This fine dust cannot be completely removed and can damage electronic components.

For these reasons, the scanner should be the last item installed in your CT suite area.

"Pre-installation" is work necessary to plan and prepare a site for installation of equipment.

Pre-installation work helps the user (customer) avoid:

- Application delay and scheduling
- Surprise siting discoveries
- Installation confusion
- Waste of manpower

The following MUST be completed before installation work can begin for a GE CT scanner:

- Completely finished:
  - Wall painted or have final wall covering
  - Ceiling tiles installed and no remaining ceiling work is required
  - Final floor covering installed with no remaining dust-causing floor work required
  - All room millwork installed as shown on the site print
  - All plumbing work in the CT suite is completed
  - No construction in or around the scan suite AREA that will produce:
    * Concrete dust
    * Drywall dust
    * Ceiling tile dust
    * Wood sawdust or shavings
    * Dust tracked into the CT suite area
- Active Broadband connection
  - A completed network connection is required for ALL CT installations.
  - A GE Healthcare network specialist may be required to complete the VCN connection. This may take a week or longer to schedule.
- Power available to the A1, with provision for Lockout/Tagout at the A1 disconnect.

  If a UPS is required, a GE A1 breaker* is needed to complete this installation. Refer to the electrical section for more details.

NOTICE **SERVICE NOTICE: An improperly prepared site (i.e., one that is in a state of construction) can result in increased installation time.**

A CT scanner installed in a dirty environment is more prone to contamination, which can result in decreased reliability and increased scanner downtime.

2.2 Purchaser Site Preparation Work

The following list describes many of the items to consider when planning for a system replacement or designing a room for new equipment:

- Determine room dimensions and verify that doorways are large enough for the scanner system.
- Install appropriate conduits and duct work for system cables. If additional components are required in the CT suite, their connection consideration must be determined and completed.
- Install junction boxes of correct size with covers at locations shown in installation plan.
- Install an A1 main disconnect.
- Install a power supply of correct voltage output and adequate KVA rating.
- Install local disconnects, including proper over-current protection.
- Install “steelwork” or other suitable support work for mounting equipment on walls or from the ceiling.
- Camera should be on-site at the time of installation.
- Complete all suite and room alterations and modifications.
- Verify that room shielding is adequate for the system being installed.
- Review structural requirements, including floor vibration, levelness, and thickness.
- Review HVAC requirements, including system regulation and patient comfort.
- Review operational clearances to see if your daily used items fit, such as beds and carts.
- Consider emergency medical equipment.
- Show storage cabinets and sink (if required) on the site print.
- The following contractors, and others, may be required to help confirm that the site meets all installation requirements:
  - Structural Engineer and /or Architect
  - HVAC contractor
  - Electrical contractor
  - Qualified radiological health physicist
- The project manager or sales will deliver a copy of the pre-installation quick start kit. The above items can be found in chapter 2 through 9 in this manual. This work should be completed at least three (3) days prior to delivery.

2.3 Manufacturer’s System Level Siting Requirements

The following minimum siting requirements must be met to install a new or replacement system:

- Network Communication in place and active
- Meets all scan room regulatory and service requirements
- Meets all minimum scan room structural requirements
- Meets minimum scan room HVAC requirements
- Meets minimum scan room electrical requirements
- Reviewed radiation protection section in the Pre-Installation manual
- All in-room items are shown on the final GE Healthcare site print, and the final print is on-site
- No construction in the scan room or neighboring suite areas

This work should be completed at least three days prior to delivery

2.3.1 Meeting Site Ready Requirements

The site ready visit will take place at least three days prior to the delivery date. The site ready visit is intended to verify that all of the siting requirements are met and the site is ready for installation.

The site ready visit results in a report to the project manager indicating one of the following:

Pass - All required items are present, completed, and the site is ready for installation.

Conditional Pass - Eighty percent (80%) of all of the tasks are completed and all parties agree that the remaining 20% will be completed by the installation delivery date.

If a Conditional Pass is granted on the inspection date, the project manager must present conclusive evidence that unfinished tasks are completed and that the site is ready for delivery one business day prior to delivery.

Fail - Less than 80% of the tasks are completed and all parties cannot agree that the remaining work will be completed by the requested installation delivery date. Failed sites will be rescheduled when all items are completed.

2.3.2 Quick Installs

Quick installations are described as sites with minimum room improvements required. These include, but are not limited to, the following items:

- Existing electrical disconnect device, wire size, and grounds meet all of the above requirements.
- Existing structural items, including floor thickness, meet all of the above requirements
- Existing HVAC capacity and regulation meet all of the above requirements
- Existing CT suite meets all of the above regulatory and minimum size requirements
- Existing facility can accommodate the delivery and meet all of the above delivery requirements

Quick Installs are subject to the following restrictions:

- Quick installs must have a new room print that accurately reflects the rooms to be upgraded.
- Existing floor anchors from a non-5.X system CANNOT be reused.
- New floor anchors must be a minimum of 10 cm (4 in.) from any existing floor penetrations.

Quick Installs typically involve a weekend de-install and room prep completion, with a next business day delivery and install.

2.3.3 Two-Step and Upgrade Installs

A Two-Step installation is the practice of temporarily installing one CT system in a site with the intention of upgrading the site to a different CT system at a later date.

- For a two-step installation to be considered, the room must meet the minimum room requirements for the upgraded/final project.
- As with any upgrade installation, “two-steps” are subject to ALL of the siting requirements imposed by the upgraded/final system. This includes the recommended room size as well as electrical, structural, and HVAC requirements.
- Two-Step Installations and other upgrades may be done as “Quick Installs.” In this case, all requirements described in Section 2.3.2 (above) also apply.
- It is the customer’s responsibility to check that all requirements are met.
- Sites that do not meet the minimum room requirements for the final product must either upgrade (or enlarge) their room, or consider the “Left-Side Limited Access” option.

2.3.4 Site Ready Inspection Visit

The GE Healthcare Project Manager of Installation (GE PMI) will review the site delivery process with you to determine how to best transfer the equipment from the transportation truck to your room. At the time of the site ready visit, the site must meet all requirements in Section 2.1 and Section 2.2, plus the following items.

- Delivery information
  - Determine delivery route into the scan room.
  - Determine if tilt dollies or riggers are required.
  - Determine if elevators, doorways, and hallways are adequate for delivery.
  - Determine if floor protection is required.
  - Determine if a tilt bed truck is required for ground delivery and ordered.
- Regulatory Requirements
  - Room size meets the minimum requirements.
  - Site print is present and accurately reflects the room size and layout.
  - No grounded walls are present in the regulatory clearance areas.
  - All regulatory clearances space is met.
  - Room meets all local codes.
- Manufacturer Requirements: As listed in Section 2.0, all requirements are met
- Purchaser's Site Preparation Work: As listed in Section 1.0, all actions are completed

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Section 3.0
Pre-Installation Checklist

Required Information for Site

Must be completed before the scheduled delivery date

Hospital Name as it appears on the system screens:

____

Network ID numbers / IP addresses Camera:______________________

PACS: ______________________ AW:______________________

Other - Specify type & ID:______________________________________

Other - Specify type & ID:______________________________________

Camera setup information:______________________________________

____

____

AW Direct Connect address:______________________________________

Do you want HIPAA enabled? No___ Yes ___

Do you want automatic downloads enabled? No___ Yes ___

:

Table 1-1 Schedule Date Commitments

GE	Cust	Dates
Y N	Y N	
	❐ ❐	Has the project schedule been verified with facilities department, contractor, and GE?
	❐ ❐	Will the committed site-ready date be met?
❐ ❐	❐ ❐	Does the completion date for any/all construction meet or preceed the delivery date?
❐ ❐	❐ ❐	Is the Power & Ground survey complete? Date: ______________ Hospital contact: ______________________________
	❐ ❐	Site-Ready visit is scheduled. Date: ______________
	❐ ❐	Delivery date is scheduled. Date: ______________
	❐ ❐	Installation date is scheduled. Date: ______________
❐ ❐	❐ ❐	Installation timing: A: Weekdays___ B: Weekend___ C: Quick Install___ If B or C, have all sub-contractors been notified? No___ Yes ___
❐ ❐	❐ ❐	Does the delivery and/or installation date need to be adjusted?
	❐ ❐	First-Use date is scheduled. Date: ______________
❐ ❐	❐ ❐	Applications/Training dates: On-Site Training Date: ______________ Healthcare Institute Training Date: ______________

Table 1-2 General Site Planning

GE	CUST	**General / Site Requirements**
Y N	**Y N**	*Must be completed 5 weeks before scheduled delivery date*
❐ ❐	❐ ❐	Have final drawings been approved and distributed to the contractors?
❐ ❐	❐ ❐	Are final drawings “signed off” to approve equipment layout / orientation?
❐ ❐	❐ ❐	Do the actual room dimensions match those on the final drawings?
	❐ ❐	Has the radiologist health physician reviewed and approved the room layout and shielding requirements?
	❐ ❐	Have any additional requirements or questions about the installation been discussed with GE? List: ____________________ ____________________ ____________________ ____________________
	❐ ❐	Is there a person assigned to review and verify that all installation requirements are met? Name: ____________________
	❐ ❐	Have the specific site requirements been discussed with the contractor? Refer to the GE final drawings specifications. (See Table 1-3 below.)
	❐ ❐	Has the responsibility of cabling, installing, and interfacing accessories not on the order been discussed?
	❐ ❐	Are all third-party vendors identified, notified and scheduled? (Examples: Netcom, Medrad, etc.)
	❐ ❐	Have all regulatory requirements been met per *Regulatory and Service Clearances*, on page 42?
	❐ ❐	Will existing network, broadband, and camera cable drops reach new locations and will they meet the requirements and function with LightSpeed 5.X Pro16? If not, what are the requirements? List: ____________________ ____________________ ____________________

Table 1-3 References for Specific Site Requirements

Sections for Specific Requirements	
Room Planning - Chapter 4 (page 41)	Floor Loads & Weights - Chapter 6 (page 85)
Radiation Protection - page 73	Delivery - Chapter 7 (page 93)
Environment - Chapter 5 (page 77)	Power - Chapter 8 (page 99)
All work by contractors must be completed before the scheduled delivery date.	

Table 1-4 Equipment Compatibility

GE	Cust	**Equipment**
Y N	**Y N**	*Must be completed 5 weeks before scheduled delivery date*
❒ ❒	❒ ❒	Has the order been reviewed for completeness and compatibility with existing equipment? Typical equipment: Remote monitors ____ AW relocation ____ Cardiac option ____ Injectors ____
❒ ❒		Are interfaces to existing and/or new accessories ordered and planned for accordingly?
❒ ❒	❒ ❒	Have the following peripheral locations been included in the site drawings? EKG monitor___ Injector control___ Laser camera ___ UPS ___ 2nd Monitor___
❒ ❒	❒ ❒	Will GE Healthcare provide additional services per contract negotiations?
❒ ❒		Are correct length cables on order?

Table 1-5 Network Connections

GE	Cust	**Network Installation and Setup**
Y N	**Y N**	*Must be completed 5 weeks before scheduled delivery date*
❒ ❒	❒ ❒	Have IP addresses and Host Names been obtained?No___ Yes ___
❒ ❒	❒ ❒	Will a network camera be used?No___ Yes ___
❒ ❒	❒ ❒	**Mandatory:** Is the network installed, are network jacks installed, and is the entire network tested?
❒ ❒	❒ ❒	**Mandatory:** Broadband VPN installed/setup?
❒ ❒	❒ ❒	**Mandatory:** Are network software options ordered ____ HIS RIS option ___ DICOM print ___ AW ____
	❒ ❒	**Optional:** Has modem option ordered? _____ (Requires a site escalation)
❒ ❒	❒ ❒	**Optional:** Is the LightSpeed 5.X Pro16 service telephone line identified and installed for InSite? *(Electrical, mechanical, etc.)*

Table 1-6 Miscellaneous Tasks

GE	Cust	**Other**
Y N	**Y N**	*Must be completed before the scheduled delivery date*
	❒ ❒	Arrangements made in the schedule to allow adequate time for remodeling, if required (such as wall, floor, or ceiling repair work, painting, other cosmetic finishes)
	❒ ❒	Have arrangements been made to clean the floor *after* equipment removal and *prior* to the installation of the new equipment?
❒ ❒	❒ ❒	Is de-installation of existing equipment required? No__ Yes ___ Removal date ___________

1 - Introduction

Table 1-6 Miscellaneous Tasks (Continued)

GE	Cust	Other
Y N	**Y N**	*Must be completed before the scheduled delivery date*
❒ ❒	❒ ❒	Is there a trade-in of existing equipment? No __ Yes __ GoldSeal _____________
❒ ❒	❒ ❒	Delivery route identified and verified with the proper hospital personnel? No__ Yes __ Elevators and doors checked for size and weight constraints? No__ Yes __
❒ ❒	❒ ❒	Have appropriate arrangements been made with traffic for delivery?No__ Yes __
❒ ❒	❒ ❒	Will acceptance/performance testing or bio-medical testing be required?No__ Yes __ Date: ___________
❒ ❒	❒ ❒	Are trash and/or recycling bins available for the removal of papers, boxes, etc. during the installation? No__ Yes __

Chapter 2
Pre-Installation Overview

Before installing a LightSpeed 5.X system, all pre-installation requirements must be complete.

- Chapter 1, Section 1.0 Site Ready Visit
- Chapter 4, Section 8.0 and Section 9.0 Structural Requirements
- Chapter 4, Section 9.1 Broadband standard
- Chapter 4, Section 10.0 Radiation Protection
- Chapter 5, Section 1.0 Environmental conditions
- Chapter 9, Section 3.0 Interconnections

Site-specific items must be verified before the installation can begin.

Section 1.0
Dust/Dirt Contamination

The LightSpeed 5.X system (consisting of: console, PDU, table, and gantry) is highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site.

NOTICE **Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.**

Section 2.0
Chemical Contamination

Never install wet film processors in the same room as the scanner, due to the possibility of chemical contamination of the LightSpeed Series components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consider the location of this equipment and chemical fumes relative to human contact, as it relates to locating this equipment and chemicals in the control room.

Section 3.0
Walls, Ceiling, and Floor

All walls, ceiling, and flooring must be completed before installation can begin.

LightSpeed 5.X scanners can only be installed on a 10 cm (4 in.) concrete floor.

Section 4.0
Broad-band

For information on broadband requirements, refer to Chapter 4, Section 9.0, on page 69.

Section 5.0
Phone Line (for optional modem)

Two phone lines must be installed at or near the operator console and be operational prior to installation.

- One analog line (for modem use)
- One voice line

Section 6.0
Review

LightSpeed 5.X systems use adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads. The Table has five (5) leveling pads.

Using the GE print to establish the room layout, make sure all the operating and service clearances shown on the print are observed. Using the template (P/N 2363851) shipped with the system, locate the anchor holes. Make sure they clear structural interferences in the floor.

Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

1.) Lay out the two (2) floor templates.
2.) Start with the gantry template—align it per the GE print.
3.) Place the table template over the top of the gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues.
4.) Check the level of the floor (See Figure 2-1) across the templates.

Note: Tiles (or other resilient flooring) around all holes will be cut during the installation process.

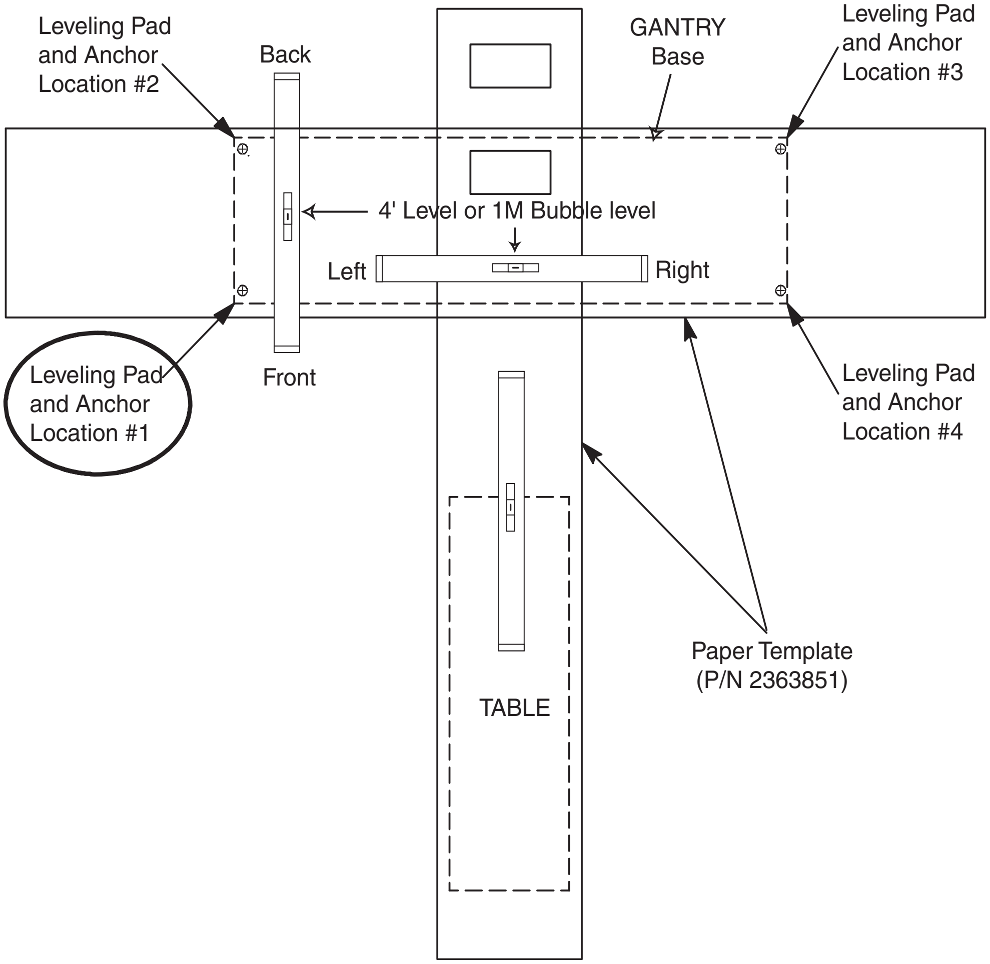

Figure 2-1 Hole Locations

Floor levelness Specification

3 mm (1/8 in.) over 3048 mm (10 ft.)

This should be measured on the template over the table/gantry area, as shown in Figure 2-1, above.

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Chapter 3
System Catalog

Section 1.0
Option Catalog Numbers

The following is a list of system options requiring site planning work for the LightSpeed 5.X system. *Contact your local GE Healthcare Sales representative for a complete list of all system options or visit us at http://www.gehealthcare.com. Refer to the instruction manuals supplied with specific options for respective details.*

CATALOG NUMBER	OPTION DESCRIPTION
International Dolly Set B7850LD	For International customers, if dollies are required.
Rear Cable Cover B7850RC	Optional rear cable cover for LightSpeed 5.0 upgrades
GE Color Printer E7014LA, E8203JA	GE Color Printer includes Cassette
Remote Color Monitors B7710WM B7530RC	LightSpeed 5.0 LCD monitor. Remote 20 inch diagonal color monitor.
SmartScore Option B7850KC	EKG Monitor and Recording Device
Performance Network Kits K9000L	6 Node, 10/100 Mbit Auto Sensing
Switched Network Kit B7500PM	ConnectPro Option provides a direct interface to HIS/RIS
Bar Code Reader B7540RB	LightSpeed 5.0 Bar Code Reader
Seismic Kit R4390JC	Earthquake zones
Long Cable Set B7816PH	
Modem B7700MK	Global Modem Kit

Table 3-1 System Options Catalog (Part) Numbers

Section 2.0
Base Scanner System

2.1 Application

The CT scanner system includes hardware and software to support patient data acquisition and image analysis for whole-body computed tomography.

2.2 Configuration

The base scanner system is configured as shown. All scan and analysis functions are controlled from the operator console (not shown).

Figure 3-1 Base Scanner System

Chapter 4
Room Planning

Section 1.0
Required Systems Clearances

NOTICE If your system was installed before December 2006, Limited Access is not available for your system. You must use the Preferred Room Size to reinstall it.

Consult your local GE Sales and Service Representative about your specific needs.

Some possible room size dimensions are shown in the table below. These room size dimensions are table dependent.

System Configuration	Preferred Scan Room Size	Typical Scan Room Size	Recommended Ctrl Room Size	Clearance Requirements
GT 1700 mm Table	4420 mm x 7772 mm (14 ft. 6 in. x 25 ft. 6 in.)	4115 mm x 6096 mm (13 ft. 6 in. x 20 ft.)	2743 mm x 4420 mm (9 ft. x 14 ft. 6 in.)	see Figure 4-1
GT 2000 mm Table	4420 mm x 7772 mm (14 ft. 6 in. x 25 ft. 6 in.)	4115 mm x 6706 mm (13 ft. 6 in. x 22 ft.)	2743 mm x 4420 mm (9 ft. x 14 ft. 6 in.)	see Figure 4-1

Limited Access Rooms (width only):

Minimum Width: 3556 mm (356 mm cover to wall)
11 ft. 8 in. (14 in. cover to wall)

Average Width: 3912 mm (711 mm cover to wall)
12 ft. 10 in. (28 in. cover to wall)

Additional component dimensions are available in Figure 4-7 through Figure 4-11 of this document. Consult your local General Electric Project Manager of Installation (GE PMI) for your appropriate room specifications.

For equipment clearance requirements, refer to Section 2.0. Remember, sufficient Regulatory and Service clearances must be maintained around equipment for full operation, service and safety.

Cable length is an important consideration in room layout. The LightSpeed 5.X system is shipped with standard short length cables, with a set of longer cables (B7816PH) available as an option. See the electrical page of the GE print for your specific requirements.

Note also, that where possible, the cables should enter the gantry from the rear, utilizing the rear cable cover assembly. Alternate cable entry is possible at the center of the gantry (refer to the Installation template).

- Excess cable length cannot be stored behind the operator console or PDU.
- A long cable must not be cut or shortened.
- Excess cable may be stored in conduits, a cable storage box if present, or the floor duct, provided sufficient space is available. Observe the fill rate for each option. If there are questions regarding local electrical or building codes, consult the project electrical contractor or electrician.
- All NEC 70-E Electrical Regulations regarding conduit or duct fill must be observed.

Section 2.0
Regulatory and Service Clearances

2.1 Regulatory Clearances

MINIMUM CLEARANCES UNDER U.S. FEDERAL REGULATIONS AND NATIONAL STANDARDS:

29 CFR 1910 (OSHA),

NFPA 70E (STANDARD FOR ELECTRICAL SAFETY IN THE WORKPLACE)

NFPA 101 (LIFE SAFETY CODE):

Figure 4-1 is a map of clearance requirements for U.S. regulatory compliance. See clearance tables on the following pages for detailed dimensional clearances. Please note: all systems installed in the United States must comply with all Federal and local regulations. For installations outside the United States, country-specific or other local regulatory clearance requirements must be met.

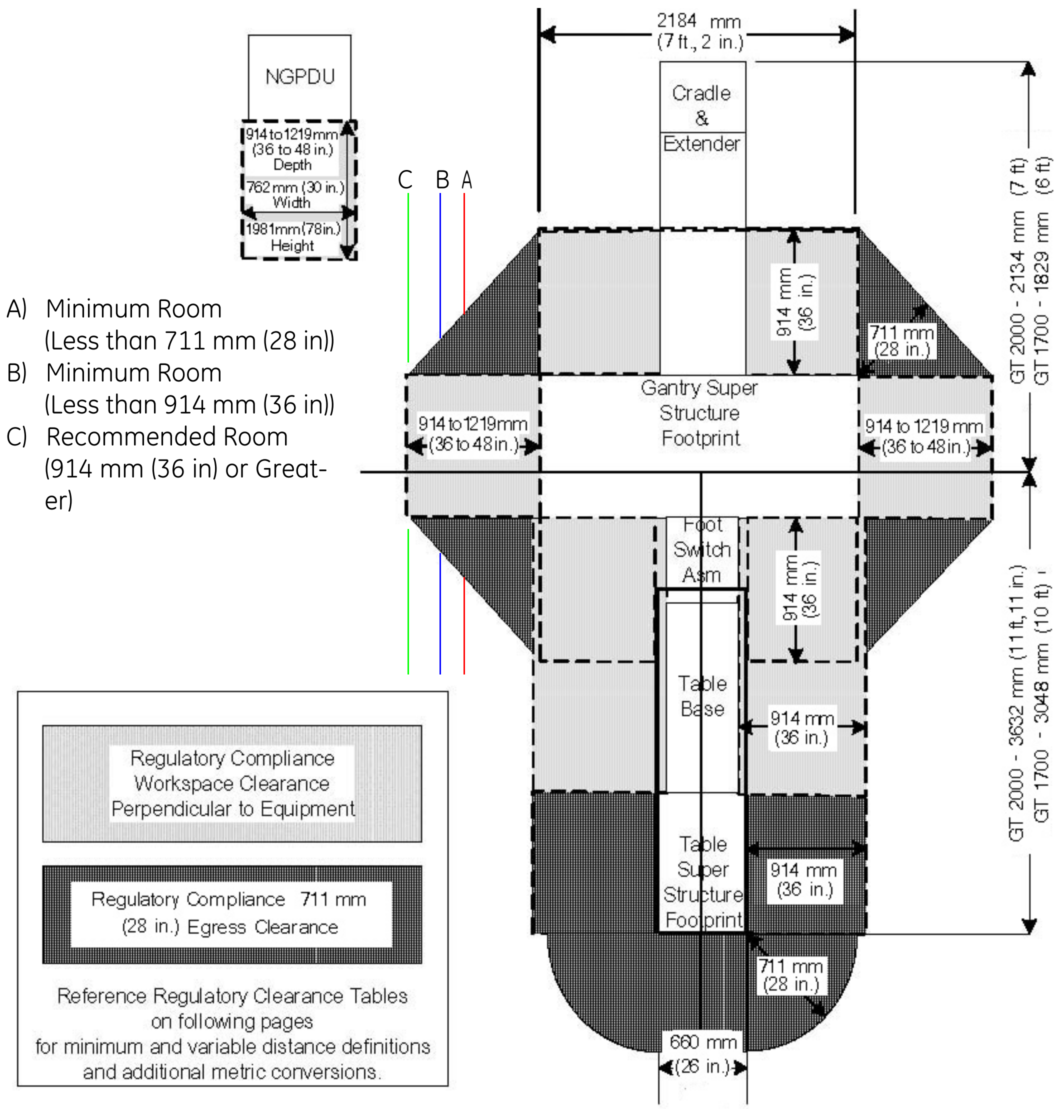


Figure 4-1 Regulatory Clearance Requirements for LightSpeed Pro16 System Configurations

Note: See Section 4.0 for Service Clearances

2.1.1 Regulatory Minimum Working Clearance by Major Subsystem

- Requirements apply to equipment operating at 600 V or less, where examination, adjustment, servicing, or maintenance is likely to be performed while live parts are exposed.
- Direction of Service Access is defined as perpendicular to the surface of the equipment being serviced.
- Required regulatory clearance distances must be maintained and may not be used for storage. This includes normal system operation as well as service inspection or maintenance.

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access: front of console	914 mm (36 in.)	There are no exposed live part hazards with the cover in place. If the console is placed under a counter, the front edge of the console must be even with the vertical edge of the console workspace. Note: This component is typically serviced from the front with access to the rear.
Service access width: Front of console	762 mm (30 in.)	This is the width of the workspace in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head clearance	1981.2 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981.2 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 4-1 Console Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of NGPDU)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.

Table 4-2 NGPDU Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 4-2 NGPDU Subsystem

- For the gantry and table, distances are measured from the enclosure, not the finish covers.

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (All Sides)	914 mm (36 in.)	If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) on both sides of workspace with the operator between is required. If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.

Table 4-3 Gantry Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Table Head or Foot)	914 mm (36 in.)	There are no exposed live parts hazards with the cover in place. This component is typically serviced form all four sides. this is the width of the workspace on each side of the equipment. A minimum of 914.4 mm (36 in.), or the width of the equipment, whichever is greater, is required.
Direction of Service Access (Table Sides)	914 mm (36 in.)*	*This distance can be reduced to 711 mm (28 in.) provided a written and signed approval is obtained by the local team from the local AHJ (Authority Having Jurisdiction). The signed document must be on file with GE.

Table 4-4 Table Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Table Foot)	711 mm (28 in.)	For the front gantry cover removal, a minimum of 457 mm (18 in.) is allowed only if an unobstructed egress space of 711 mm (28 in.) is maintained around the equipment for room exit. This also means no trip hazards exist along the path of egress.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.

Table 4-4 Table Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of UPS)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Right side and length of UPS)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 4-5 UPS Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of A1 Disconnect)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Right side and length of A1 Disconnect)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 4-6 A1 Disconnect Subsystem

2.1.2 Terms and Definitions

EGRESS

The path of exit from within any room. U.S. regulatory requires a minimum of 711 mm (28 in.) of continuous and unobstructed space including trip hazards along the path of exit.

WORK SPACE

This is the dimensional box required for safe inspection or service of energized equipment. It consists of depth, width, and height. The depth dimension is measured perpendicular to the direction of access. U.S regulation is minimum of 914 mm (36 in.). Additional conditions can increase the minimum requirement. FCT defines this as the envelope of the component superstructure. For the NGPDU it is with the front panel removed. For the gantry and table, it is with the patient or external covers removed.

SERVICE ACCESS WIDTH

This is the width of the working space in front of the equipment, a minimum of 762 mm (30 in.), or the width of the equipment, whichever is greater.

HEAD CLEARANCE

This is the height dimension of “Work Space”. The height of the workspace measured from floor at the front edge of equipment to ceiling or overhead obstruction(s), 1981 mm (78 in.) or height of equipment, which ever is greater.

GROUNDED WALL

Any wall that can be electrically conductive to earth ground. Masonry, concrete, or tile, are considered conductive. Additional commonly found aspects of a wall should also be considered as grounded. This is not an all-inclusive list:

- Medical Gas ports
- Metal door and window frames
- Water sources and metallic sink structures
- Metallic wall-mounted cabinets
- A1 disconnect panel
- Equipment Emergency Off panels
- Industrial equipment, such as air conditioners and vents
- Expansion joints

The following are not considered as grounded elements of a common wall:

- Standard wall outlet
- Light switches
- Telephones
- Communication wall jacks

MINIMUM

The lowest limit permitted by law or other authority.

DIMENSIONS AND CLEARANCES

Consisting of, or representing, the lowest possible amount of degree for freedom permissible for equipment siting. This relationship must meet all safety, service, and regulatory requirements to be acceptable.

PRE-INSTALLATION ESCALATION

Process to consult with CT Engineering, the Design Center or EHS regarding pre-installation issues related to your siting concerns.

Section 3.0 Additional Regulatory Clearance Information

3.1 Minimum Room Size (Limited Access)

The CT Gantry Left Side Limited Access Initiative provides the capability to reduce the minimum room size for CT Systems while still meeting all installation requirements and specifications. This adds left side flexibility, allowing the CT system to be sited in rooms with widths 559 mm (22 in.) smaller than the current minimum room width. Left-side access and egress may be restricted. Refer to your site's installation print for your room's detail.

Wall duct and conduit on walls within the regulatory clearance areas shall be 1067 mm (42 in.), measured from the covers to the obstruction. Servicing of the CT System can be safely performed within the regulatory envelopes, however sufficient space must be maintained to remove system covers, and replace large system components. To achieve this clearance for the gantry, clear space must be available to maneuver the gantry covers mounted on the service dollies. Surface floor raceway cannot be used in the egress route areas. OSHA ramps are available. The FE lifting the rear or front cover to avoid floor obstructions is not an EHS-approved service procedure.

One Service Engineer shall be able to accomplish all service component replace tasks listed without the need for special tools or equipment, such as a tube change, detector change, and HV tank.

3.1.1 Regulatory Caution

Site prints are required for all system installations including relocation and moves. CT room layout, as shown on your site print, shall meet all regulatory requirements as described in the installation manual. Additional room components, such as cabinets, reduce room size. Equipment not shown on the site print may void the caution statement, making the room non-compliant. Actual site measurements before installation will be taken to determine room size and compliance.

3.1.2 Egress Clearance

Egress requires a clear, unobstructed route out of the room, either around the back of the gantry or around the back of the table. If your egress route is not around the back of the table, maintain 457 mm (18 in.) of clearance between the back of the table, with a continuous width of 3200 mm (126 in.), 1600 mm (63 in.) on each side of the table center line, on each side to any obstruction so that the front cover can be removed. Refer to the Pre-Installation manual for more details on service clearances.

Exceptions

Rooms smaller than 3556 mm x 5867 mm (11 ft. 8 in. x 19 ft. 3 in.) require construction to meet the minimum requirements. The design center or your GE PMI may have additional recommendations for your room size.

3.1.3 Operational Caution

In a minimum room layout 356 mm - 686 mm (14 in. - 27 in.), the customer should consider workflow, customer access for patient care, and critical-care operations space requirements. Additionally, there may be limited equipment access on the gantry left side when loading patients or when positioning patient equipment in the room between the gantry and the wall. Detailed customer installation tasks are detailed in the product Pre-Installation manual, Chapters 1-4.

3.1.4 System Specifications (Pro16)

System Configuration	Recommended Room Size	Typical Room Size	Minimum Room Size
Pro16 with HP1600	4420 x 7772 mm[1] (14 ft. 6 in. x 25 ft. 6 in.)	4115 mm x 5867 mm[1] (13 ft. 6 in. x 19 ft. 3 in.)	3556 mm x 5867 mm[2] (11 ft. 8 in. x 19 ft. 3 in.)

Table 4-7 System Specifications (Pro16)

[1] Same Regulatory requirements apply

[2] Same Regulatory requirements apply, with the addition of no energized left side service.

3.1.4.1 Preferred Room Size

The "preferred room size" configuration offers the most flexibility for future upgrades. It has sufficient workspace and space to add mill work, while meeting all regulatory requirements. This room is compatible with most two-step future installations.For Upgrades to CT-PET refer to Section 3.1.5.

3.1.4.2 Typical Room Size

The "typical room size" configuration allows for some future upgrades. It has sufficient workspace, but limited space to add mill work and meet all regulatory requirements. This room may be compatible with some two-step future installations.

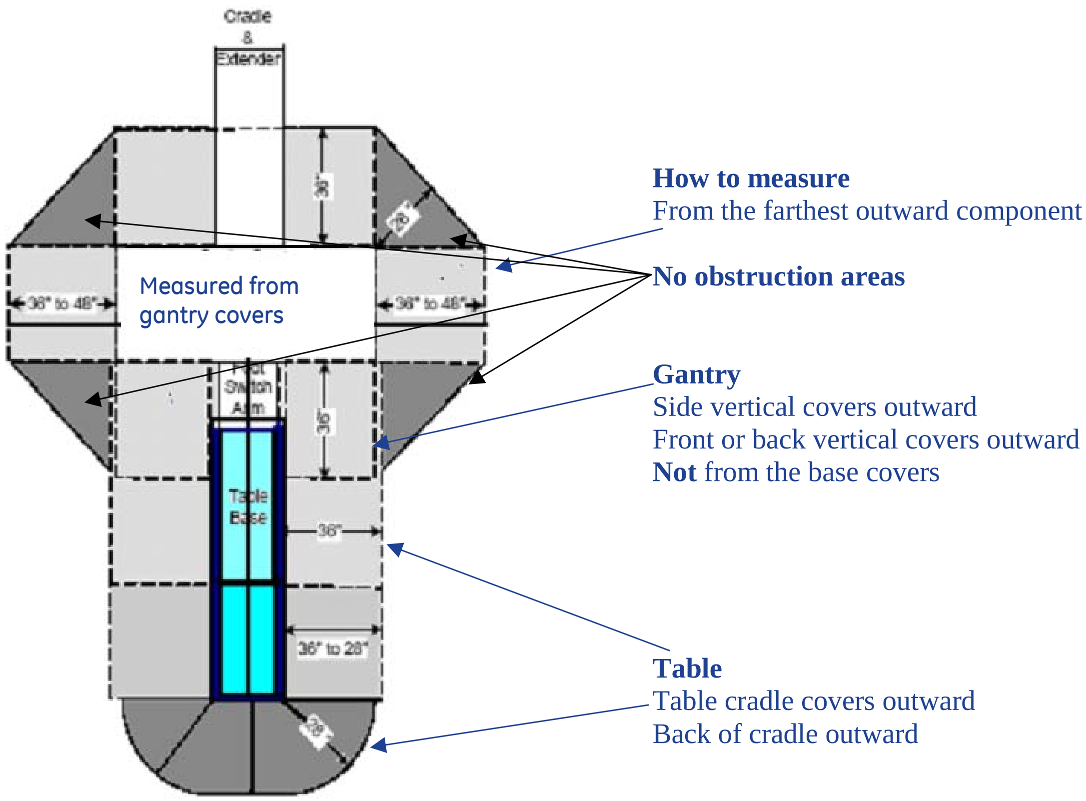
3.1.4.3 Minimum Room Size

The "minimum room size" configuration allows for no future upgrades. It has limited workspace and no in-room mill work, but meets all regulatory requirements. This room is not compatible with two-step future installations.

3.1.5 Room Size for PET Upgrades

To complete a future upgrade to PET, you need to meet the requirements, including room size, to support the CT PET configuration. To upgrade to Discovery VCT, you need a room size 7772 mm x 4420 mm (25 ft. 6 in. x 14 ft. 6 in.), which, depending on your layout, allows space to add mill work while meeting all regulatory requirements. This room would be compatible with most PET two-step future installations.

3.2 How to Measure

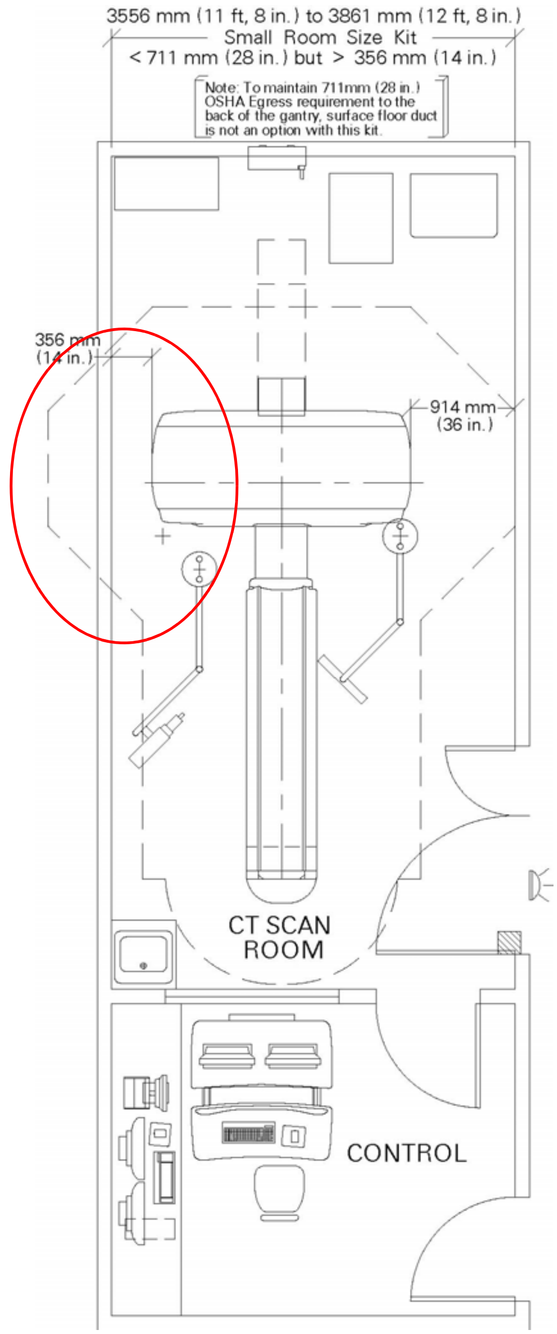


3.3 Minimum Room Size & Requirement Layouts

All gantry covers shall be removed by one person without lifting. Allow sufficient space to remove the front and back covers so that they can be "parked" during equipment service. In the limited access (Room A) configuration, carefully review user operational compatibility and placement of emergency equipment within the room.

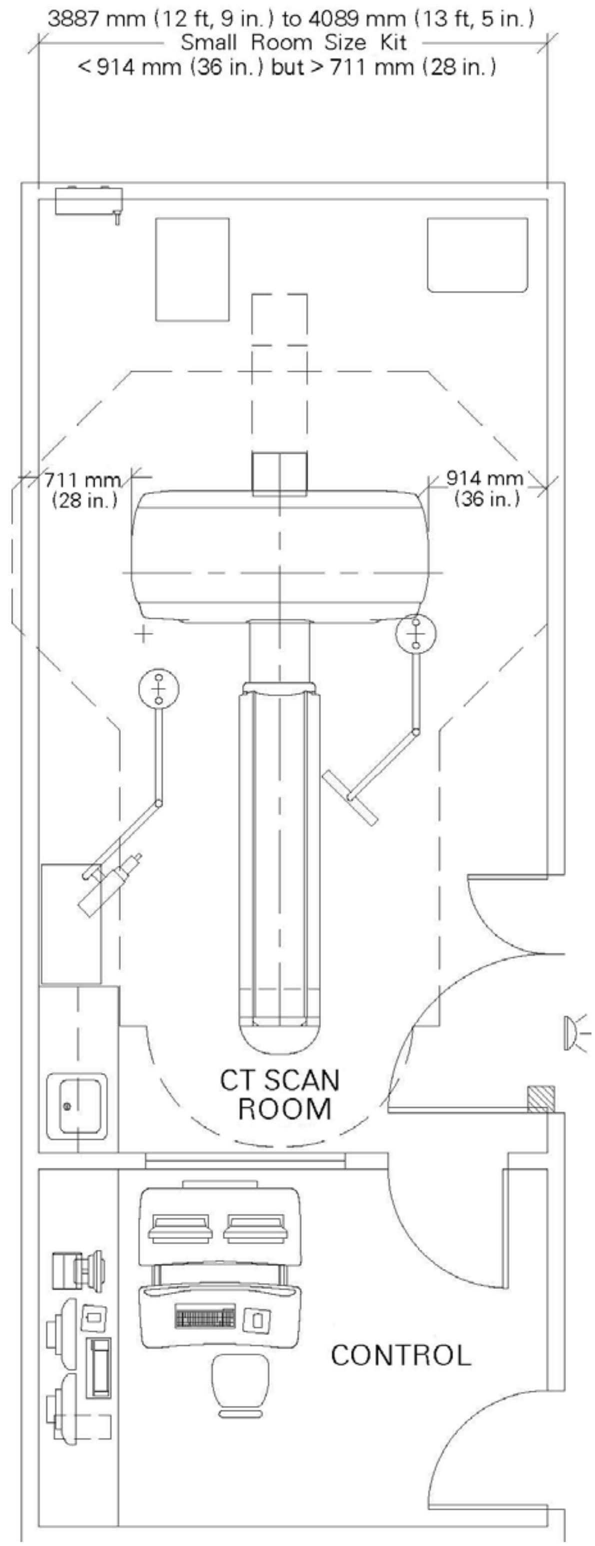
Room A - Less than 711 mm (28 in.), but greater than 256 mm (14 in.), measured from the covers to the left sidewall.

In this configuration service, egress and workspace are compromised around the gantry's left side.



Room B - Less than 914 mm (36 in.) but greater than 711 mm (28 in.) measured from the covers to the left sidewall.

In this configuration service, egress, and workspace are acceptable around the gantry.



3.4 Preferred and Typical Room Size & Requirement Layouts

Figure 4-2 and Figure 4-3 show preferred and typical room layouts, with and without in-room cabinets. You need to be aware of locations for med gas, surface duct-work, or other items that make a grounded wall.

Note: Your room layout may meet the preferred or typical room requirements but look different than Figure 4-2 or Figure 4-3. Equipment shown in these details may not represent the product you are siting. Contact your sales person to have a detail room layout completed for your site.

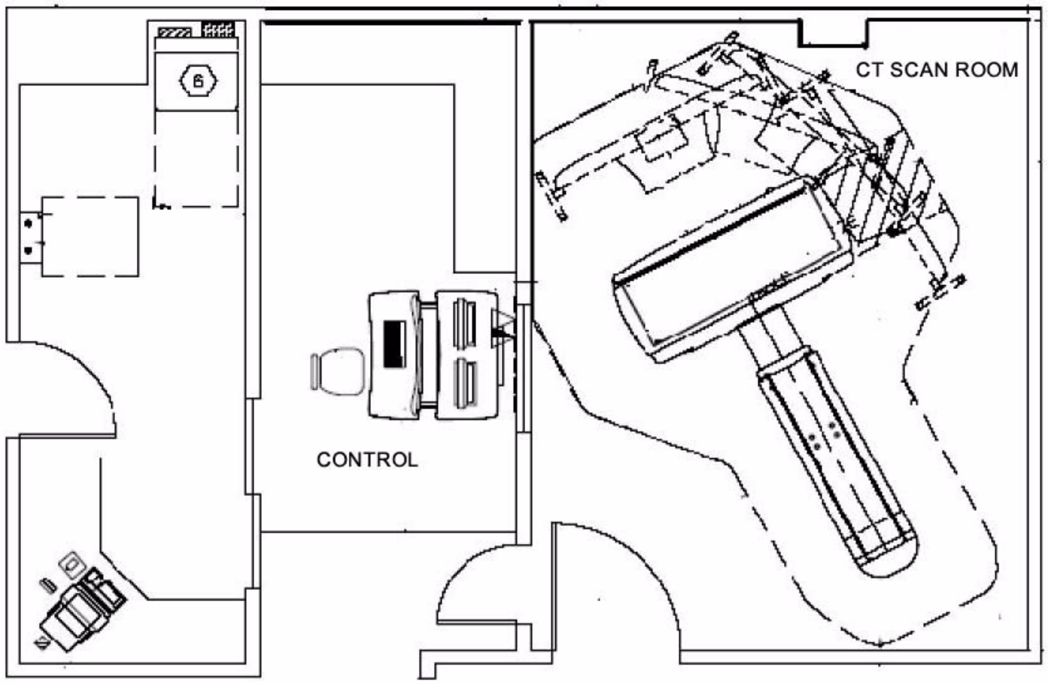


Figure 4-2 Room Layout with cabinets

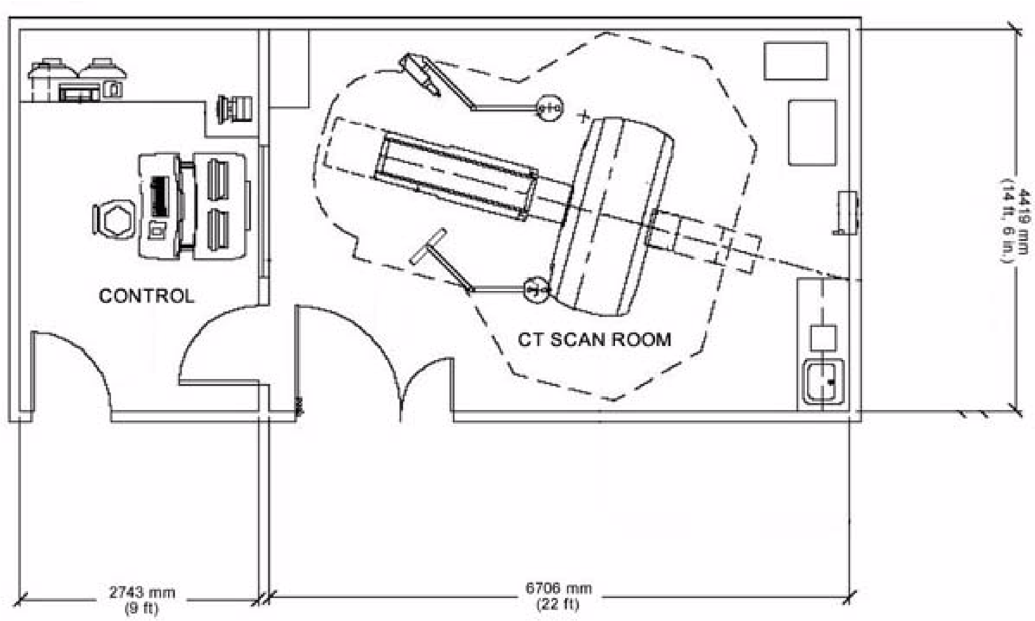


Figure 4-3 Room Layout without cabinets

Section 4.0
Service Clearances

Servicing the CT System can be safely performed within the regulatory envelopes defined in Section 2.1, however sufficient space must be maintained to remove the covers from the system. To achieve this clearance for the gantry, clear space must be available to maneuver the gantry covers mounted on the service dollies. One Service Engineer can accomplish this. If such space is not available, then two Service Engineers are required for gantry cover handling. This must be clearly communicated with the Customer.

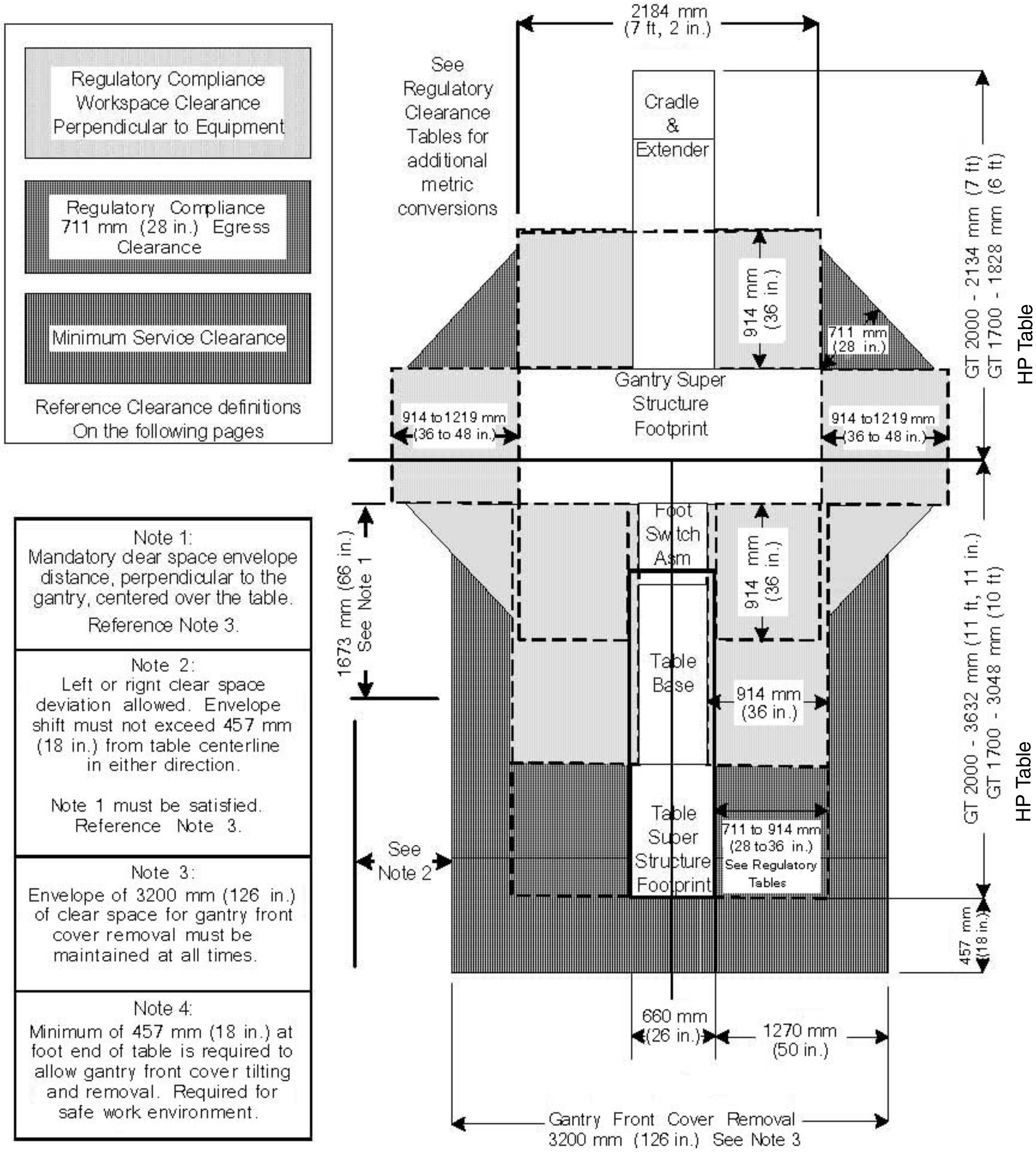


Figure 4-4 Minimum Service Clearance

4.1 Service Clearances for Single Service Engineer

- Gantry front cover removal requires the use of the Tilting Cover Dollies. These dollies allow the Service Engineer to separate the cover from the gantry, tilt the cover 90 degrees, roll the cover to the foot end of the table, and then tilt the cover an additional 90 degrees, such that the front cover is now upside down relative to the normal system-mounted condition. Figure 4-4 illustrates the minimum clear space required to achieve this operation of 3200 mm (126 in.). The gantry front cover must be removed to a position that satisfies the minimum regulatory clearances.
- The gantry rear cover with service dollies installed, requires a width of 2388 mm (94 in.) and a depth of 584 mm (23 in.) of clearance for removal as shown in Section 6.0, Figure 4-6. Sufficient space must be calculated to move the cover either straight back or to one side of the table to satisfy the minimum service clearance shown in Figure 4-4. This means the rear cover cannot violate the workspace on the rear, or on either side of the gantry.
- If gantry service requires both the front and rear covers be removed, then these covers must be positioned within the room in such a manner as to not violate the egress clearances on any side of the gantry. This may necessitate removing the covers from within the suite. This should be discussed with the customer and provisions made to accommodate this potential event.
- A single Service Engineer can safely perform servicing of the table. Sufficient clear space must be available to maintain egress clearances when the table covers or cradle are removed.
- In your room layout design, service shall have clear and unobstructed access to the gantry tube change area for all major component replacements. These components must be able to reach the service area without lifting or rigging by one service engineer. Major components include:
  - CT X-ray tube in crate
  - High voltage tank(s) in crate
  - Slipring in crate
  - Detector assembly

  Be aware of cabinet placement, and how surface floor ducts are used in room configurations.

4.2 Power Distribution Unit (NGPDU) Service Clearance

Positioning of this component must be considered for regulatory compliance as defined in Section 2.1, Regulated Minimum Working Clearance by Major Subsystem.

See Regulatory Tables.

4.3 Console Service Clearance

The operator console does not present an exposed live parts hazard. However, a minimum working space depth of 1219.2 mm (48 in.) and full width of the operator console shall be maintained at all times for service activity. Additionally, sufficient space needs to be provided for repositioning of the operator console, and side clearance for rear service access. See Figure 4-5 for a typical control room layout.

Section 5.0
Recommended Layouts

5.1 Storage Cabinet

A storage cabinet is provided by GE Healthcare to store all supplied service equipment (see Table 4-8 for equipment list). This storage cabinet (457 mm D x 914 mm W x 1067 mm H) (18 in. D x 36 in. W x 42 in. H; ~90 lbs) should be located in the scan room suite area, for easy service access.

ITEM	SIZE	WEIGHT (TOTAL)	
QA Phantom (water filled)	20 cm x 15 cm (7.9 in. x 5.9 in.)	6 kg	12 lb
35CM Phantom	35 cm x 7 cm (13.8 in. x 2.8 in.)	8 kg	18 lb
48CM Phantom	48 x 7 cm (18.9" x 2.8)	11 kg	25 lb
Phantom Holder	25 cm x 25 cm (9.8 in. x 9.8 in.)	4 kg	8 lb
FE Box	30 cm x 38 cm x 30 cm (11.8 in. x 15 in. x 11.8 in.)	7 kg	15 lb
Rear Cover Dollies	158 cm x 82 cm (62.2 in. x 32.3 in.)	11 kg	25 lb
Front Cover Dollies	85 cm x 20 cm and 85 cm x 15 cm (33.5 in. x 7.9 in. and 33.5 in. x 5.9 in.)	16 kg	35 lb
Install Support Kit (box)	30 cm x 30 cm x 38 cm (11.8 in. x 11.8 in. x 15 in.)	9 kg	20 lb
Tube Hoist Kit	77 cm x 8 cm and 38 cm x 15 cm (30.3 in. x 3.1 in. and 15 in. x 5.9 in.)	91 kg	20 lb
Balance Weight Kit		33 kg	73 lb

Table 4-8 Equipment to be stored in storage cabinet

5.2 Advantage Workstation (AW)

Refer to Pre-Installation Manual 2111833 and Installation/Service Manual 2111831. For document access, please refer to GE Healthcare's Common Documentation Library web site.

5.3 Control Room Considerations

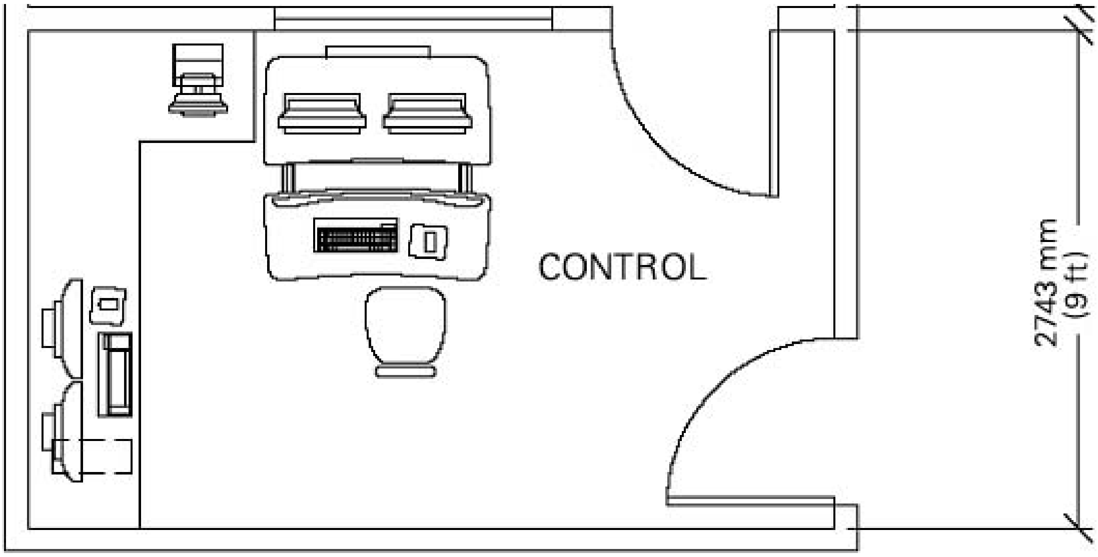


Figure 4-5 Typical Control Room Layout

- The control room must provide a suitable operating environment for the operator console electronic, and operator working comfort.
- The operator console cannot be dismantled of have components removed or rearranged in configurations other than as shipped.
- A suitable work area, which is within reach of the operator console, should be provided for placement of the injector control. Injector controls differ in dimensions depending on the brand selected.
- A PACS, workstation, image printer, or filming device are often placed in the operator console control room area, and sometimes may be directly linked to the operator console.
- Additional components although linked via network or ethernet cable, are not powered from the CT operator console.
- Additional room power and network connection should be considered when reviewing the operator console work space.

Section 6.0
Component Dimensions

DESCRIPTION	WIDTH		DEPTH		HEIGHT	
	MM	INCH	MM	INCH	MM	INCH
System						
Gantry	2200	86.6	1000	39.4	1895	74.6
Table	617	24.3	2387	94		
Power Distribution Unit (NGPDU)	711	28	559	22	1067	42
Operator's Console/computer	1219 1238	48 48.75	991 1020*	39 40.15*	85 680*	33.5 26.78*
Remote Color Monitor (LCD)	413	16.25	203	8	406	16
Color printer	584	23	457	18	178	7
UPS	762	30	406	16	711	28
Advantage Windows						
Workstation	419	16.5	409	16.1	79	3.1
Monitor	495	19.5	483	19	471	18.6
A1	508	20	203	8	787	31
* Minimum Dimension						

Table 4-9 Dimensions of Components

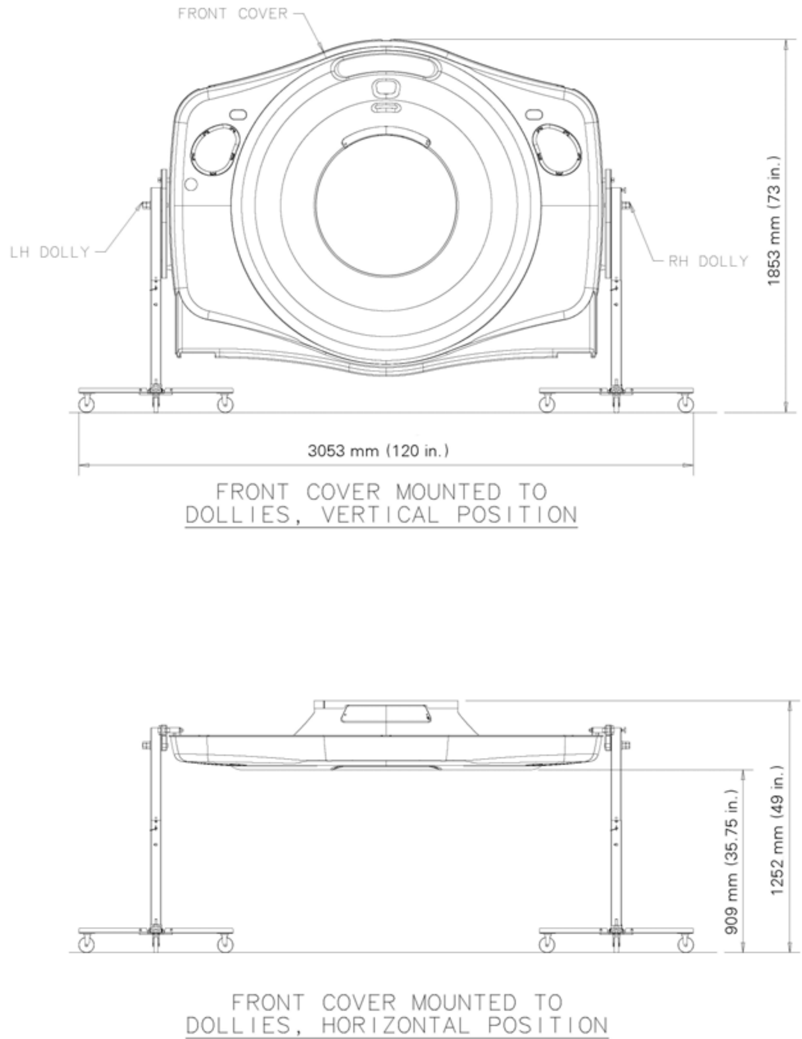
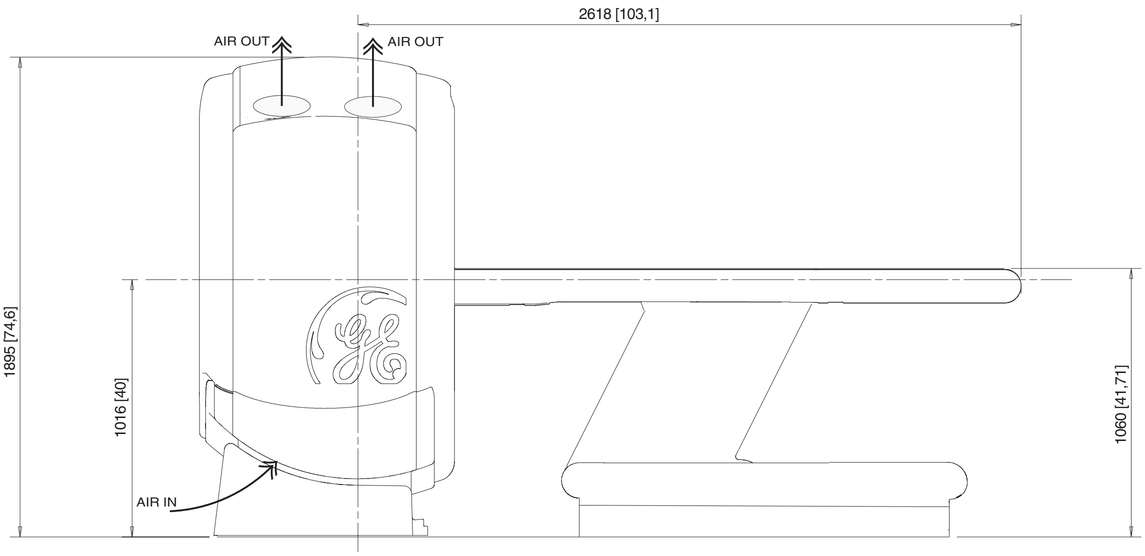


Figure 4-6 Gantry Front Cover with Service Dolly Dimensions

6.1 Table and Gantry



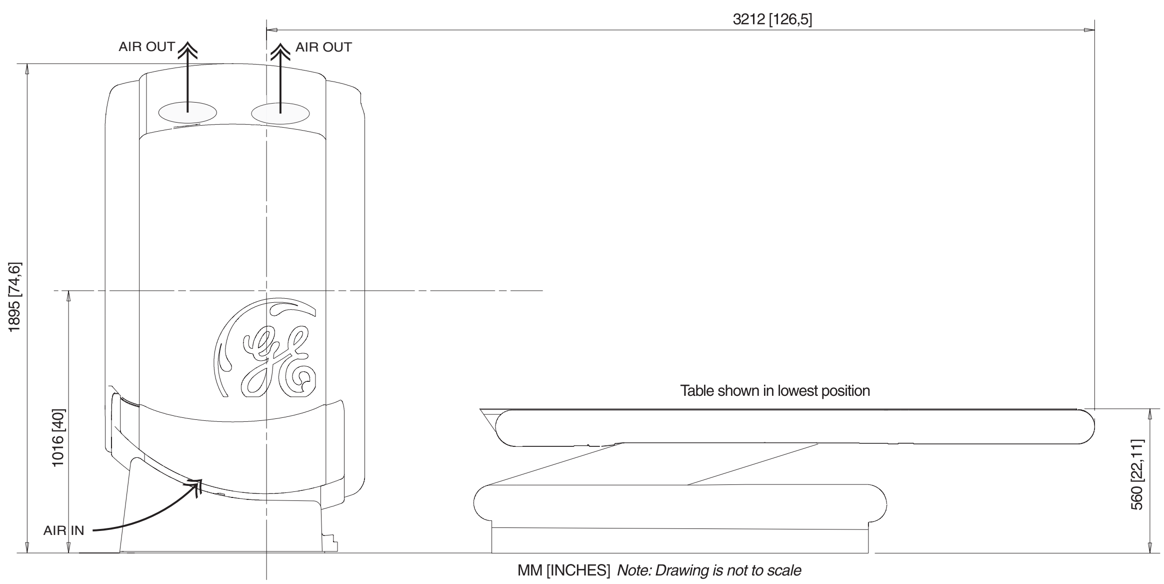


Figure 4-7 Table and Gantry (Side View)

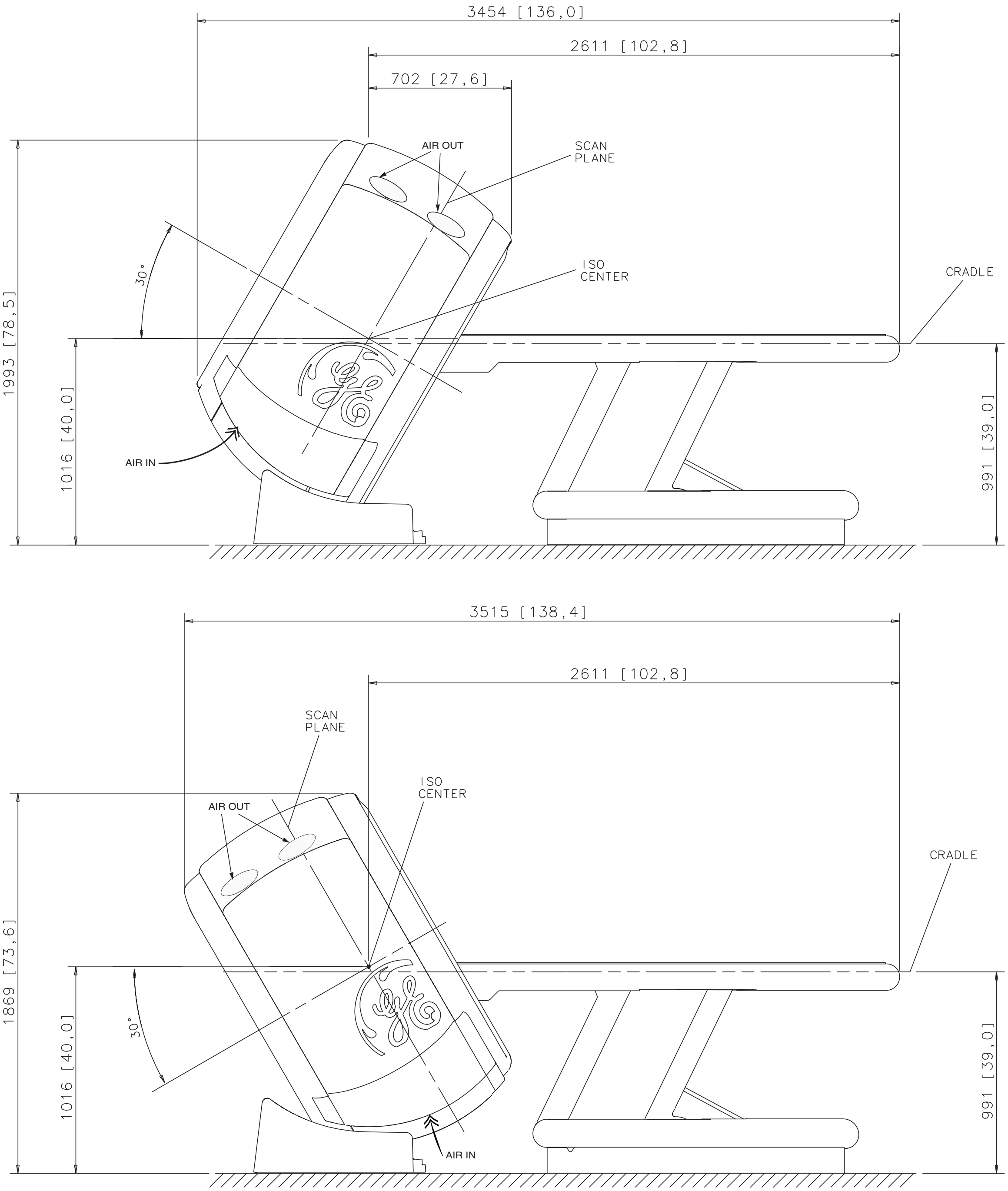


Figure 4-8 Gantry shown tilted +30º (top) and -30º (bottom)

6.2 Power Distribution Unit

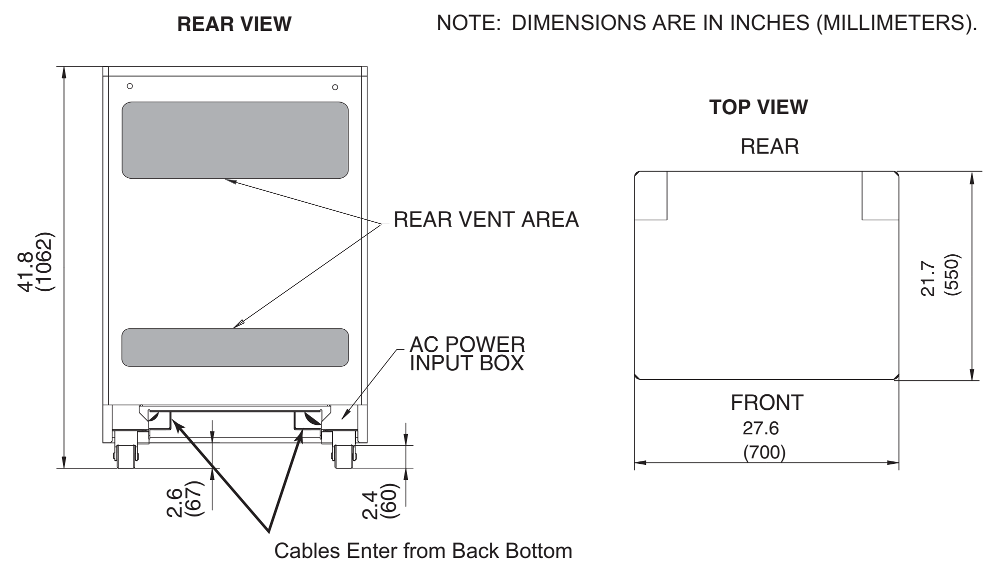


Figure 4-9 Power Distribution Unit (NGPDU-4)

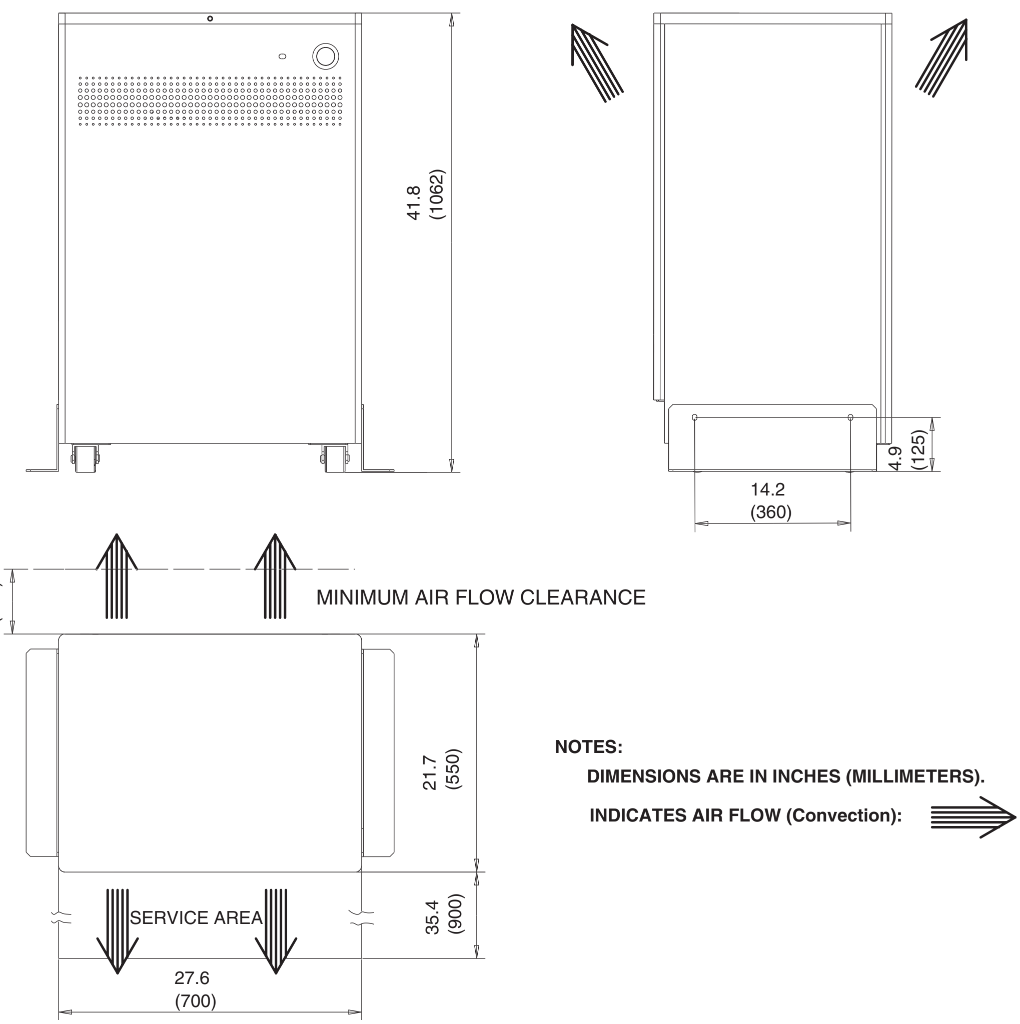
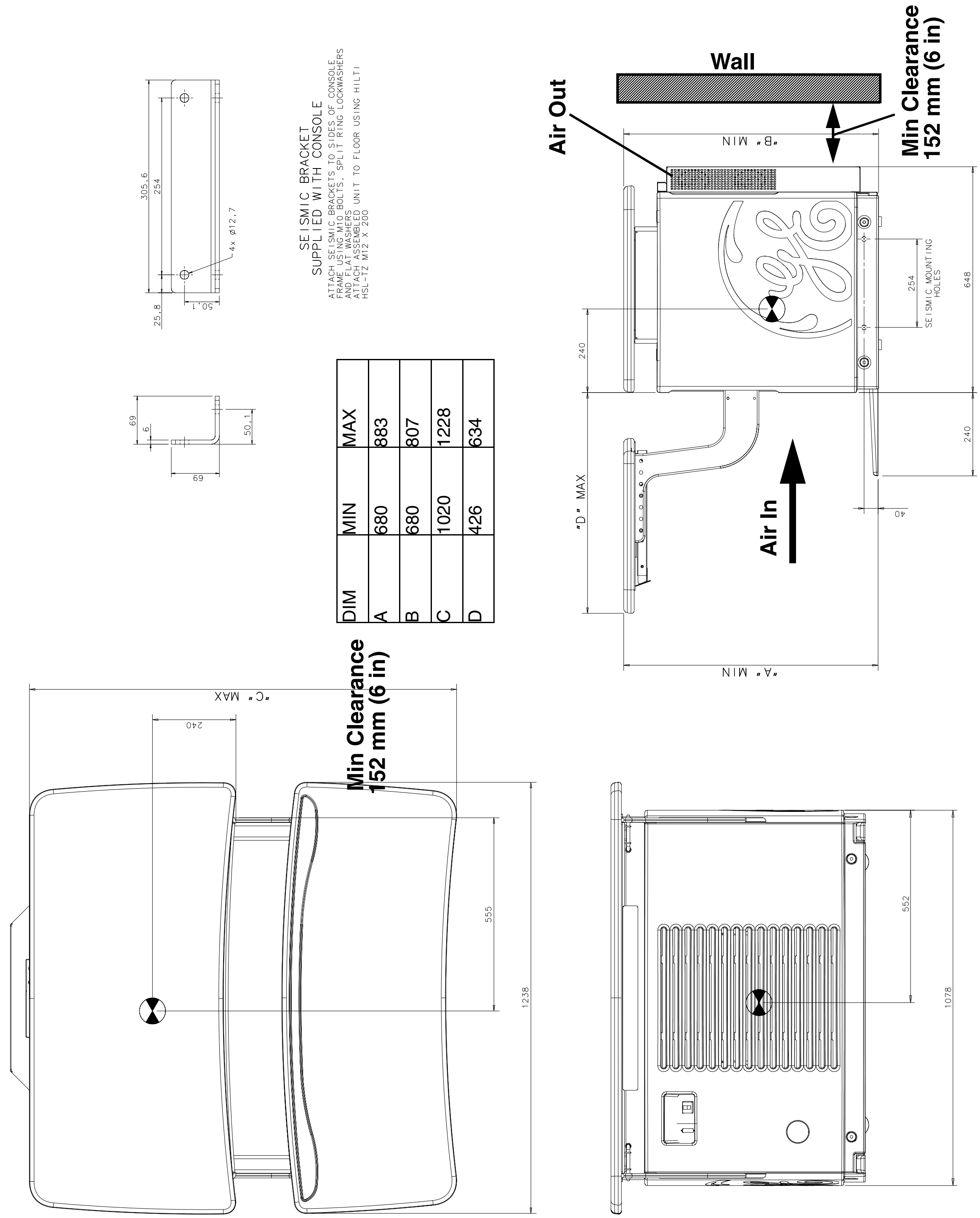


Figure 4-10 Power Distribution Unit (NGPDU-4)

6.3 Operator Console



DIM	MIN	MAX
A	680	883
B	680	807
C	1020	1228
D	426	634

Figure 4-11 Operator’s Console

Section 7.0
Minimum Dimensions and Clearances

System Operation	MM	Inches
Ceiling Pedestal mount (optional) (lowest point to floor injector or monitor)	2439 mm	96 in.
Finished ceiling to floor (recommended)	2743 mm	108 in.
Finished ceiling to floor (minimum)	2439 mm	96 in.
Table max extension head end w/extender from Center Line	2030 mm	80 in.
Table extension head end w/extender to obstruction	152 mm	6 in.
Table in lowest pos. w/cradle @home pos. to Center Line	3209 mm	126.5 in.
Back of Console to wall	152 mm	6 in.
Back of PDU to wall	152 mm	6 in.

Table 4-10 Minimum Dimensions and Clearances

7.1 Injector Control

A suitable work area, which is within reach of the operator console, should be provided for placement of the injector control. See Figure 4-12.

Wall mounted, ceiling mounted, and pedestal units need cables to be routed from the gantry area to the operator console area. The supplied cable is 15 m (50 ft.) in length. Injectors require AC power that is not supplied by the CT system

Note: Injector cables should not be routed with the system cables.

Mounts are available in different configurations and lengths. Refer to injector documentation for detailed installation instructions See Figure 4-12 for injector placement information.

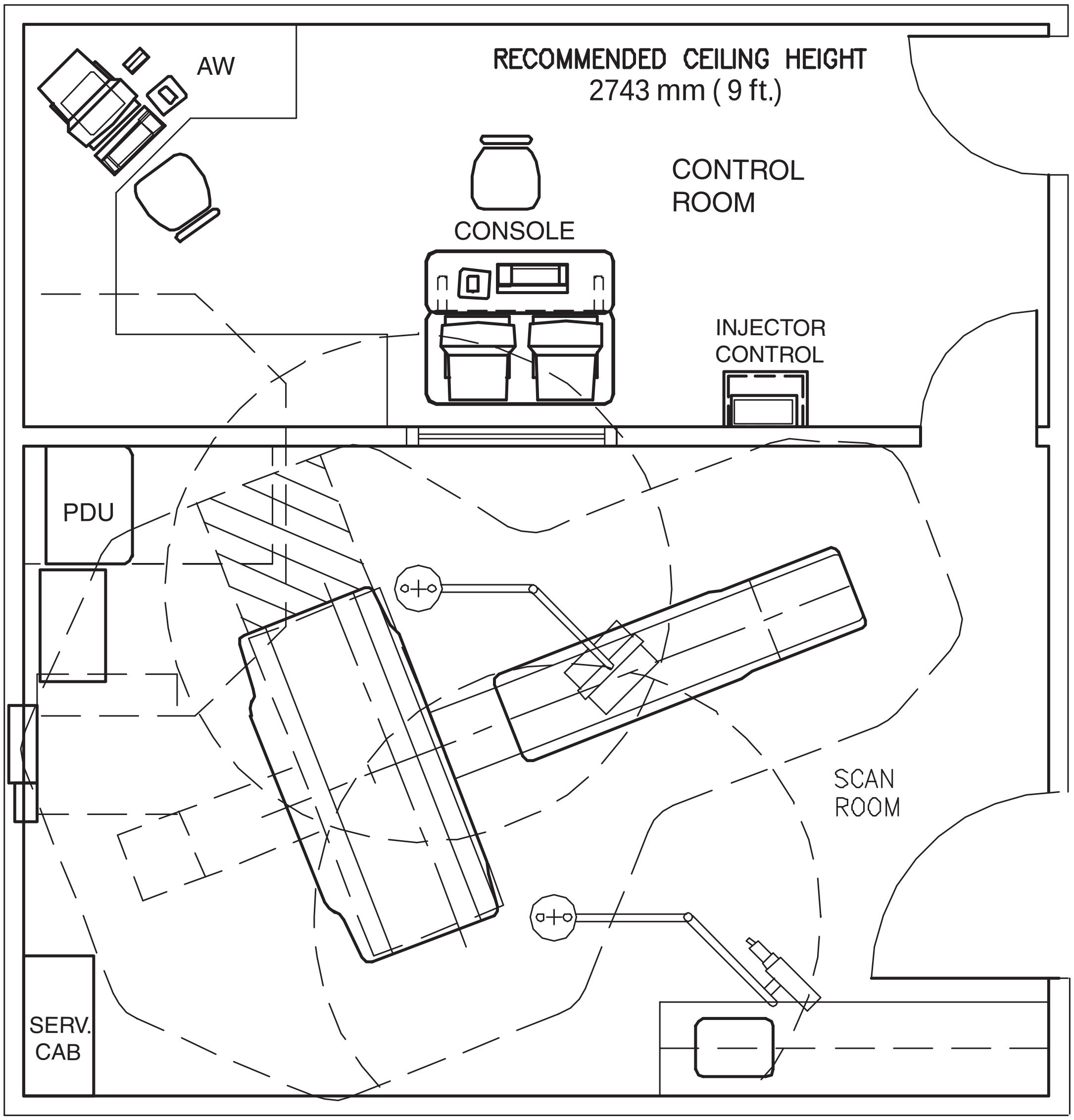


Figure 4-12 Room Layout, with Injector Option

Section 8.0
Structural Requirements

8.1 Table and Gantry Mounting Requirements

WARNING **POTENTIAL FOR PATIENT INJURY.**

IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.

PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

NOTICE **It is the purchaser's responsibility to provide an approved support structure and mounting method for all floor types other than those listed. General Electric is not responsible for any failure of the support structure or method of anchoring, including seismic requirements.**

GE is not responsible for methods other than those listed.

Table and gantry mounting dimensions are shown in Figure 6-2, Figure 6-3, and Figure 6-5. Refer to Chapter 6 for additional details of floor loadings, component weights, and gantry and table installation and anchoring.

Anchor the gantry and table to the floor by a means that maintains their relative alignment and meets applicable building and other local codes, including seismic structural mounting requirements.

The floor structure must be capable of withstanding the occupied weight of the table and gantry, and the individual contact area loading of these components.

Support areas of the patient table and gantry must rest on solid concrete, not resilient tile or carpeting, which will slowly yield over a period of time and disturb the alignment of the table to the gantry.

Factors that could cause misalignment between Gantry and Table due to floor sag should be considered. The cradle can potentially carry a 205 kg (450 lb) patient. Center of gravity changes as cradle cantilevers.

Take into consideration all other moving weights such as gurneys or personal equipment. Refer to Chapter 6 for gantry and table mounting details.

No part of the floor surface within the table and gantry, nor the two interface areas between the table and gantry, should be higher than the support area for the table and gantry.

8.2 Minimum Floor Thickness

For any installation on a floor with a rating less than a Class “A,” consult a structural engineering specialist to determine any necessary enhancements.

Support areas of the patient table and gantry must rest on at least102 mm (4 in.) of solid concrete, not resilient tile or carpeting, which will slowly yield over a period of time and disturb alignment of table to the gantry.

Factors that could cause misalignment between gantry and table due to floor sag should be considered. The cradle can carry a 205 kg (450 lb)patient. The center of gravity changes as the cradle cantilevers.

Take into consideration all other moving weights such as gurneys or personal equipment. Refer to Chapter 5 for gantry and table mounting details.

8.3 Floor Anchors

A qualified person must verify that the site and method of anchoring are adequate to support loads and maintain table-to-gantry alignment.

Location of supporting beams and columns may dictate position of table-to-gantry assembly. Use of flush floor duct or conduit in the floor may significantly affect floor strength. The method and placement of anchoring through bolts must not reduce the structural strength of floor. New floor anchors must be located at lease102 mm (4 in.)from existing floor penetrations or concrete edges.

Provided floor anchors are designed for use ONLY on concrete floors that meet the 102 mm (4 in.) concrete floor requirements. At the customer's expense, all other anchoring methods (on floor types other than the 102 mm (4 in.) concrete minimum) must be determined by their structural contractor to meet the stated GE minimum load requirements. The customer's contractor is responsible for the installation of all anchors other than those shipped with the system.

8.4 Non-Concrete Floors

If installing the Lightspeed system on a type of floor that fails to meet the 102 mm (4 in.)concrete floor requirements, the customer, at their expense, shall provide acceptable anchoring and mounting methods that meet all the structural requirements listed in this Site Planning Guide.

8.5 Floor Strength

Concrete floors must have a minimum strength of f'c = 2000 psi (1.4 X 10^7 Pa) at 28 days for (curing time) mounting floor anchors. It is the responsibility of each customer to have appropriate tests performed to determine and measure concrete strength.

Note: If installing the GE LS scanner on a floor type thinner than a 102 mm (4 in.) concrete floor, the purchaser, at their expense, shall provide acceptable anchoring and mounting methods that meet all structural specifications provided in sections 8.1 through 8.7 of this Pre-Installation Manual.

8.6 Floor Levelness

The CT Room floor levelness requirement is important for accurate patient positioning. Floor levelness in the Scan Room must not be greater than 3 mm (0.125 in.) between depression and high spots over any 3048 mm (120 in.) distance within the area of the gantry/table template (see the envelope shown in Figure 2-1, on page 37).

Note: **The floor must meet levelness specification to properly align the table/gantry. Minimum gantry height at this specification is 15 mm (1/2 in.) to prevent cable crushing.**

Table level may not be achievable if the overall floor levelness is greater than the specification. The overall floor level must be 0 in. to use under gantry cable entrance. The minimum gantry height is 20 mm (3/4 in.) with this option to prevent cable crushing.

The use of floor shims is not suitable to achieve floor levelness. It is recommended that the concrete be leveled to meet this requirement.

8.7 Floor Vibration

The CT equipment may be sensitive to vibration in the frequency range of 0.5 to 20 Hz depending on the amplitude of the vibration. It is the customer's responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specific limits, However, it is ultimately the customer/architect/engineer responsibility to design the site solution.

8.7.1 Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed 10^{-3} m/s^2 rms maximum single frequency above ambient baseline from 0.5 to 80 Hz (measured in any 1 hour during a normal operating period).

8.7.2 Transient Vibration

The behavioral characteristics must be such that any measurable transient disturbance must also be minimized to less than 0.01 m/s^2 peak-to-peak.

8.7.3 Equipment Location

To minimize the interference, the CT equipment should be placed on a solid floor, located as far as possible from the following vibration sources:

- Parking lots
- Roadways
- Subways
- Trains
- Hallways
- Elevators
- Heliports
- Hospital power plants containing pumps, motors, air handling equipment and air conditioning units

8.8 Walls: Scan Window

The recommended patient viewing window dimensions are 1219 mm wide x 1067 mm high (48 in. x 42 in.). The location of the window is dependent on the position of the operator workspace position. Consult Section 10.0 of this chapter and a **qualified radiological health physicist** for radiation protection requirements of the window glass (lead content and thickness).

Note: The operator at the operator workspace must be able to view the patient during a scan.

Section 9.0
Network Connections

Broad-band is the standard network connection and is required for all LightSpeed 5.X systems. (A dial-up modem is optional.) A 1000 baseT high-speed network is desired, with 100 baseT network service acceptable. Broad-band connections should use one of the following Category 5 patch cables:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

The CT system is connected to the network through the console.

- A patch cable (not to exceed 3 m (10 ft.)) should be provided by the customer to connect the operator console to a wall box. (See Notes on Figure 9-3)
- Some customer-site units may require cable duct-work or conduit to route connecting network cables to the workstation, camera, and operator console.
- The run from the hospital switch to the CT wall outlet must not exceed 88 m (290 ft.). Bandwidth performance is degraded when the length reaches 91 m (300 ft.) or greater.
- For the optional modem: **Two phone lines should be provided by the facility**. One line is for use with a modem and must be an analog line. The second line is a voice-only line.

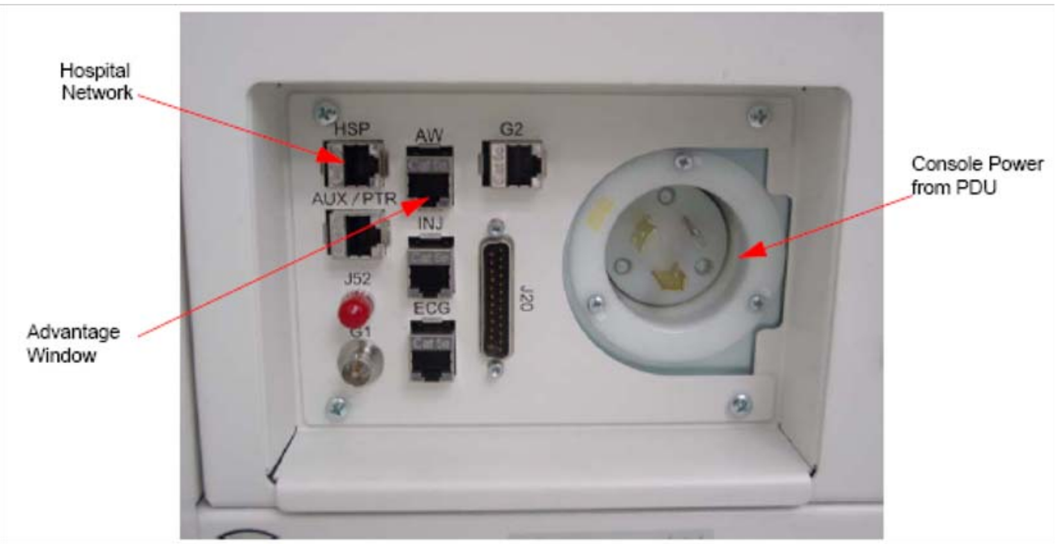


Figure 4-13 Console Rear Bulkhead

4 – Room Planning

9.1 US Broad Band Process Overview

The United States network connectivity requirement for this product is broad-band. The US process relies on the GE PMI to select a Customer Champion and identify an IT contact for the site. Together, those individuals then complete a site assessment to gauge what tasks are needed to fulfill the connection.

Anyone can contact the GE Connectivity team at 800.321.7937, Option #3, with questions.

9.2 Customer Broad-Band Responsibilities

Provide the GE PMI with an accurate site address, telephone number, contact name, and e-mail address for the:

- Customer Champion
  - Coordinate VPN activities between Radiology/Cardiology and the Information Technology (IT) departments.
  - Act as a focal point in assuring site broad-band infrastructure meets GE Healthcare requirements for connection as determined by a mutual assessment with the GE Healthcare Connectivity team.
- IT Contact
  - Complete an equipment assessment with the GE Healthcare Connectivity team to determine site readiness for broad-band.
- Contact your GE PMI, for the name of the zone broad-band specialist to:
  - Work with the Customer Champion to complete any identified infrastructure changes.
  - Provide an IP addresses for new CT equipment.
  - Provide a VPN compatible appliance that will support the IPSec tunneling protocol and 3DES data encryption.
  - Utilize an Internet Service Provider that supports static routing.

GE Medical Systems

Remote Service Broadband - Customer Site Assessment

Site Name: ______________________ **FE Name:** ______________________
City, State: ______________________ **FE Phone:** ______________________
Date: ______________________ **FE Email** ______________________

	Yes	No
1. Does your site currently have a persistent (24x7) Internet connection?	☐	☐
2. Is the GEMS Diagnostic Imaging equipment on the Local Area Network and will it be accessible to the Internet?	☐	☐
3. Does your site have a VPN device today?	☐	☐
4. Is the VPN device one of the models below? *If **Yes** , please select the model from the options below.*	☐	☐

- ☐ a) Cisco Pix Firewalls
- ☐ b) Cisco Routers
- ☐ c) Cisco 3000 Series (Altiga acquisition)
- ☐ d) Checkpoint Firewalls Software Version 4.1 and higher
- ☐ e) Nortel Contivity Software Version 3.2 or higher
- ☐ f) Redcreek
- ☐ g) Symantec (Raptor) firewalls
- ☐ h) Firebox
- ☐ i) Linux S/WAN
- ☐ j) Sidewinder
- ☐ k) Netscreen
- ☐ l) None
- ☐ m) Other ______________________

If **No , the GEMS Connectivity Support Team can help determine device compatibility.*

	Yes	No
5. Does your VPN device support "triple DES" Encryption?	☐	☐
6. Has approval been given to install this VPN connection?	☐	☐

Site Approver's Name ______________________

7. Provide your VPN Installer information, this is the person who will be contacted to schedule the VPN install.

Customer Installer Name: ______________________
Installer Telephone Number: ______________________
Installer e-mail address: ______________________

Notes: ______________________

Field Engineer needs to provide compatible system informaiton:

System ID	IP Address	Gateway Address

Additional System and IP Address Spaces Available on Page 2

If you have questions or need assessment support contact your Zone Champ or:

Joe Gracz - HQ Support 1-262-524-5261
Joseph.Gracz@med.ge.com

Once you have completed both pages of this form, please send it to:
a) Judy Heyer judy.heyer@med.ge.com
b) Judy Heyer Fax# **414-918-4707**

Use the send button on page 2

GE Medical Systems

System ID List & Compatibility Matrix

MR	Software Version	CT	Software Version
Include all Signa(LX, Ovation, OpenSpeed and CV/I) MR's System Ids and IP Addresses in assessment.		HiSpeed CT/e	Minimum 6.0
		HiSpeed CT/e dual	Minimum 2.06
		HiSpeedNX/I	Minimum 5.5
Prior to completing an InSite checkout systems must be upraded to Version 9.0 or 9.1 unless shown in the compatible list below.		LightSpeed QX/I	Minimum 1.3
		LightSpeed Plus	Minimum 2.1
Signa TwinSpeed	Release 9.0	LightSpeed Ultra	Minimum 3
Signa	Release 9.1	HiSpeed X/I	
Signa	Release 10.x	CT/I	Minimum 6.2
Signa CV/I	CNV4	Lxi, Dxi, Fxi, Zxi	Minimum 6.01
Signa 3T			
Signa Profile	Minimum 7.66		
Signa Contour	Minimum 7.66		
Magnet Monitor	Minimum 2.3 software		

Network Products	Software Version	Nuclear/Pet	Software Version
AW 4.0	Version 4.0 or Above	Advance	Minimum 5.1
AW 4.0P		eNTEGRA	Minimum 2.03

X-Ray	Software Version
Cardiac	
INNOVA	All
Mammography	
Seno2000D	All
Digital Radiology	
Revolution XQ/I	Version 10.12.5 or above
Revolution XR/d	Version 18.0

Multi Vendor

MR	PSI	CT	PSI
Philips ACS 1.5T	MZP 400	Picker MX-TWIN	CZM 400
Philips NT 2000	MZP 401	Picker MX 8000	CZM 401
Philips NT 3000	MZP 402	Picker PQ	CZM 500
Philips NT ACS	MZP 403	Picker PQ 2000	CZM 501
Picker Outlook 0.23T	MZM 301	Picker PQ 5000	CZM 502
Picker Edge 1.5T	MZM 800	Picker PQ 6000	CZM 503
Picker Vista 1.0T	MZM 801	Picker PQS	CZM 504
Picker Eclipse 1.5T	MZM 900	Siemens AR STAR	CZS 100
Picker Polaris 1.0T	MZM 901	Siemens AR.C	CZS 101
Siemens Harmony 1.0T	MZS 100	Siemens AR.HP	CZS 102
Siemens Impact 1.0T	MZS 200	Siemens AR.SP	CZS 103
Siemens Impact Expert 1.0T	MZS 201	Siemens AR.T	CZS 104
Siemens SP 1.0T	MZS 400	Siemens PLUS-4 EXP	CZS 400
Siemens SP 1.5T	MZS 401	Siemens PLUS-4 POW	CZS 401
Siemens SP4000 1.5T	MZS 403	Siemens PLUS-4 VOL	CZS 402
Siemens Symhony 1.5T	MZS 500		
Siemens Vision 1.5T	MZS 600		

Field Engineer needs to provide compatible system informaiton:

System ID	IP Address	Gateway Address

Once you have completed this form, please send it to:

Judy Heyer **email** **judy.heyer@med.ge.com**
Judy Heyer **Fax #** **414-918-4707**

Send Now

Section 10.0 Radiation Protection

NOTICE **Scanner-room shielding requirements should be reviewed by a qualified radiological health physicist taking into consideration:**

- **Scatter radiation levels within the scanning room (see Figure 4-14)**
- **Equipment placement**
- **Weekly projected work-loads (# patient/day technique (kvp*ma))**
- **Materials used for construction of walls, floors, ceiling, doors, and windows**
- **Access to surrounding scan room areas**
- **Equipment in surrounding scan room areas (e.g., film developer, film storage)**

Figure 4-14 depicts measurable radiation levels within the scanning room while scanning a 32 cm CTDI phantom (body) and a 20 cm water phantom (head) with the technique shown.

Note: Actual measurements can vary. All measurements have an accuracy of ± 20% because of measurement equipment, technique, and system-to-system variation.

Use the correction factors shown in Table 4-11 to adjust exposure levels to the usual scan technique at your site.

CHANGED PARAMETER	MULTIPLICATION FACTOR
mAs	new mAs/710
80 kV	0.21
120 kV	0.71
140 kV	1.0
4 x 4 mm images	0.82
16 x 0.625 LD 8 x 1.25 LD Fluro 5 mm	0.59
4 x 1.25 LD 5 mm (1i) Fluro 3 mm	0.40
1 x 1 mm images	0.20
4 x 0.625 LD 1 x 1.25	0.10

Table 4-11 Shielding Requirements Scaling

NOTICE **The units of measure used for radiation levels have been changed in this publication, from mR (millirads) to µGy (micrograys). The conversion factor is:**

1 mR = 8.69 µGy

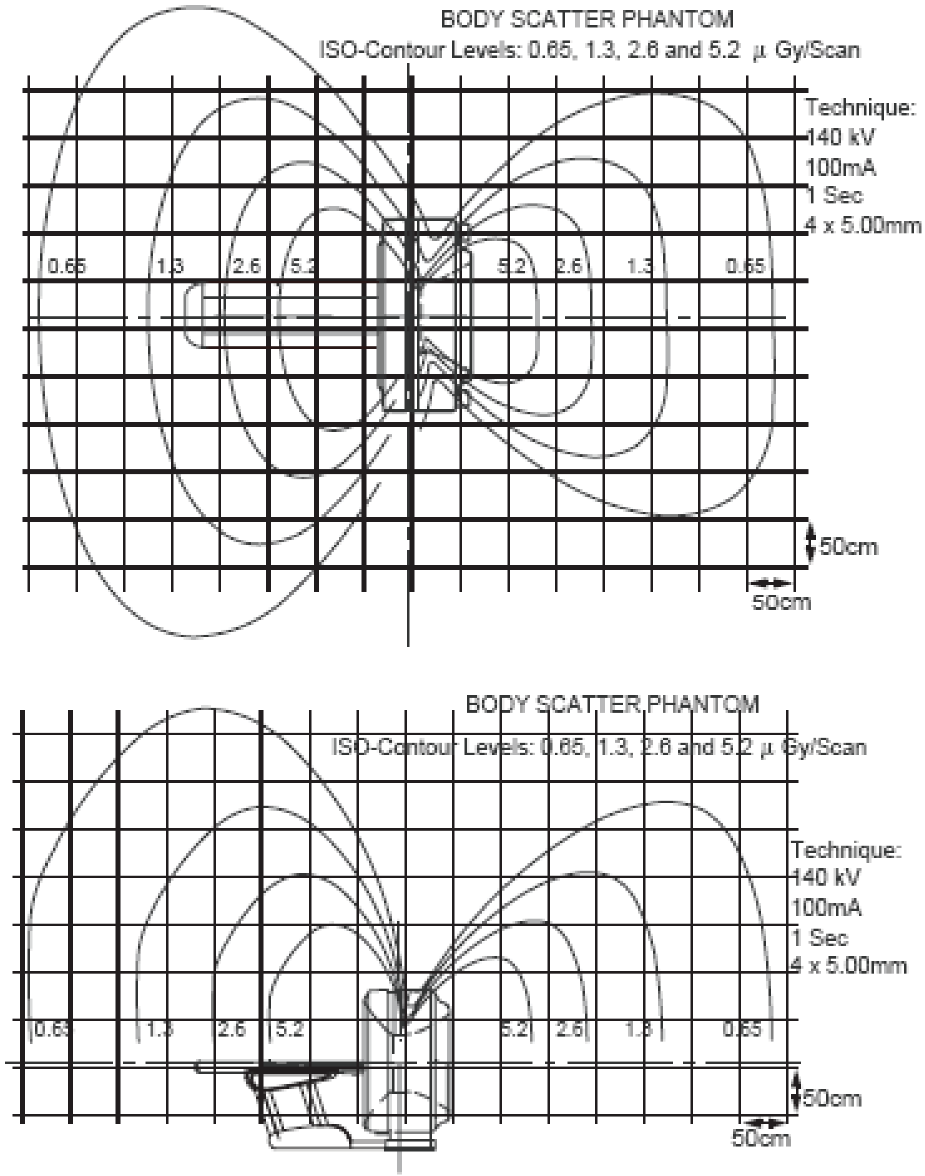
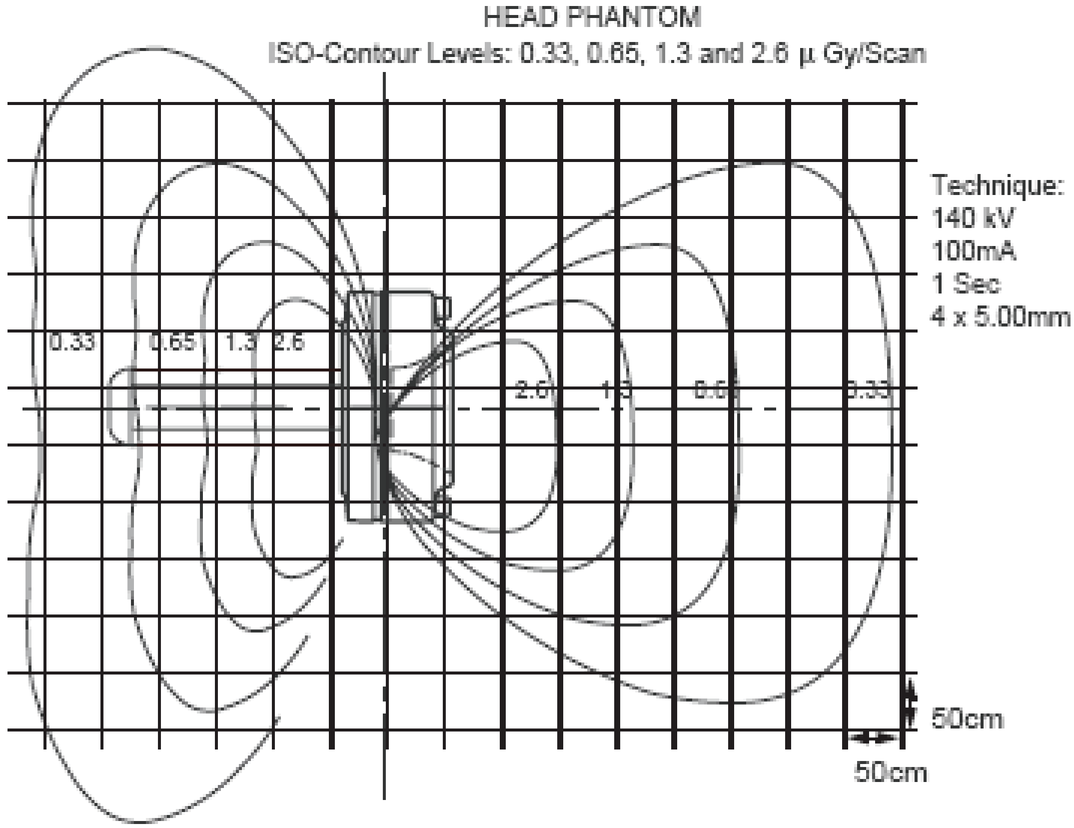
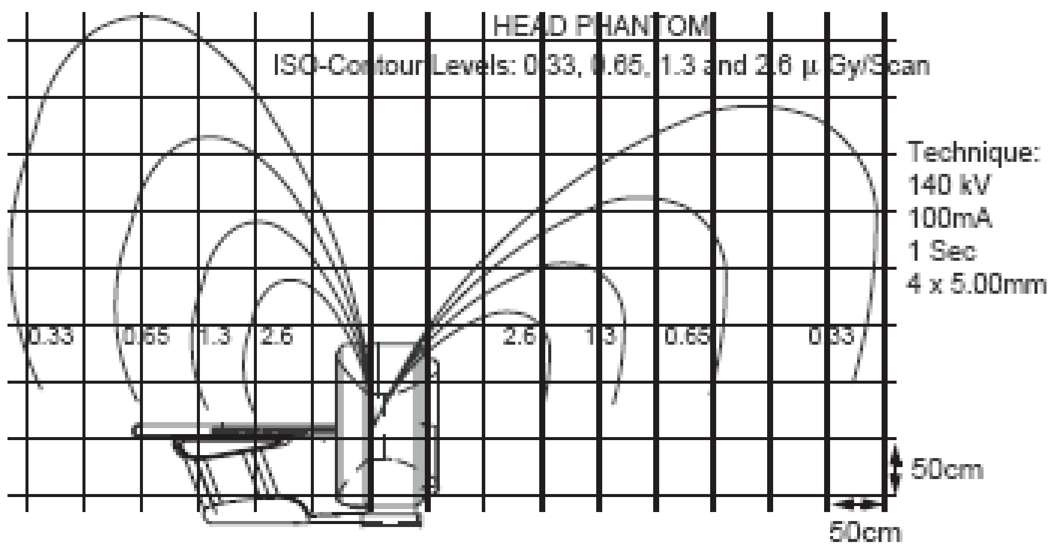


Figure 4-14 Typical Scatter Survey (Body Filter)



F



Typical Scatter Survey (Head Filter)

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Chapter 5 Environmental Conditions

Ratings and duty cycles of CT subsystems apply if site environment meets the standards of this section. Maintain environmental conditions listed below at all times – including, for example, overnight, weekends and holidays. Shut down the CT system if air conditioning is not working. When system is shut down for major repair, air conditioning may be shut down also.

Section 1.0 Temperature and Humidity Specifications

NOTICE Potential Equipment Failure **Do not operate (i.e., "Power ON") gantry or console subsystems with ambient room temperatures exceeding 75° F. Scan room or control room temperatures in excess of 75° F can result in the failure of gantry or console components.**

- Ambient Temperature: (Fahrenheit and Celsius)
  - Scan Room: Maintain a temperature of 22° C (72° F ±3° F), for patient comfort. When scan room is unoccupied, table and gantry temperature limitations are 21° - 26° C (70° - 75° F).
  - Control Room (including Console/Computer): Maintain 22° C (73° ± 3° F).
  - Equipment Room: If a separate equipment room is used to house the PDU, the allowable temperature range is 15° - 26° C (60° - 75° F).
- Store media (cartridges) in long-term storage in same temperature range as host computer.
- Store media in the host computer environment for one-half hour before use.
- Maintain relative humidity of 30%-60% (non-condensing) during operation (all areas).
- The maximum temperature rate of change is 3° C/hr. (5° F/hr.).
- The maximum relative humidity rate of change is 5% RH/hr.

Environmental specifications apply to the table, gantry, Power Distribution Unit, and operator console.

NOTICE **System operation and image quality may be affected, if environmental specifications are exceeded.**

1.1 Temperature (Scan Room & Control Room)

Maximum allowable ambient room temperature:	26° C (79° F)
Recommended ambient room temperature:	22° C (72° F)
Minimum allowable ambient room temperature:	18° C (64° F)
Maximum allowable ambient room temperature rate of change:	3° C per hour (5° F per hour)

Note: Any cooling equipment cycle control range must be taken into account, such that the maximum and minimum ambient room temperatures shown above are not exceeded during room thermal cycling. For example, if the HVAC is capable of ± 2° C control, then the limits would 20° C - 24° C to maintain absolute limits.

1.2 Humidity (Scan Room & Control Room)

Maximum allowable non-condensing relative humidity:	60%
Minimum allowable non-condensing relative humidity:	30%
Maximum allowable relative humidity rate of change:	5% per hour

1.3 Other Guidelines

- To help determine the hospital room environmental conditions, a temperature and humidity recorder may be temporarily installed close to where the system will be installed. Note readings before and after installation to verify the true temperature and humidity in your environment.
- Consider heating, ventilating, air conditioning (HVAC) needs, and redundancy. An air conditioner with two compressor units, rather than one, may prevent system downtime. A back-up (redundant) air conditioner permits CT system operation during an extended repair of the primary air conditioner.

Section 2.0
Cooling Requirement

Use Table 5-1 to assist in cooling requirements planning. Over half the cooling used by your scanner is required for gantry operation. Locate a wall air-conditioning vent at floor level beside and behind gantry to meet both gantry cooling needs and provide patient comfort. Do not locate any cooling vents directly above the gantry. Air returns above the gantry are recommended.

SYSTEM COMPONENT	BTU/HR	WATT
1. Gantry recommended (See NOTE 1)	43,000	12,600
2. Table	700	200
3. Power distribution unit	3400	1000
Recommended Scan Room Subtotal (see notes):	**47,100**	**13,800**
4. Operator's Console/computer with one IG	7361	2165
Additional IG, each	1360	400
LCD Monitor, each (x2)	170 (340)	50 (100)
SCSI Tower	425	125
Injector	425	125
DASM	425	125
Recommended Control Room Subtotal (includes OC w/1 IG, 2 monitors & SCSI Tower)**:**	**8126**	**2390**
System Total (Recommended) (See NOTE 1)	**55,226**	**16,190**
Option: UPS	2500	732
Option: Remote Color Monitor (LCD)	170	50
Option: Advantage Windows	256	75
Option: UPS	2900	850
Option: TV camera	34	10
Option: TV monitor	300	88
ROOM TOTAL (SEE NOTE 2)		

NOTE 1: With 75 scan rotations per patient:
Recommended BTU/hr. provides for up to six patients per hour. It is also needed during calibration of the system.

NOTE 2: Cooling requirements do not include cooling for room lighting, personnel or non-CT equipment.

Table 5-1 Cooling Requirements (Worksheet)

Please refer to Chapter 4, Section 6.0, for component air flow requirements.

5 – Environmental Conditions

Recommended placements of HVAC vents (intake and output) and Thermostat are shown for the Scan Room (Figure 5-1) and the Control Room (Figure 5-2).

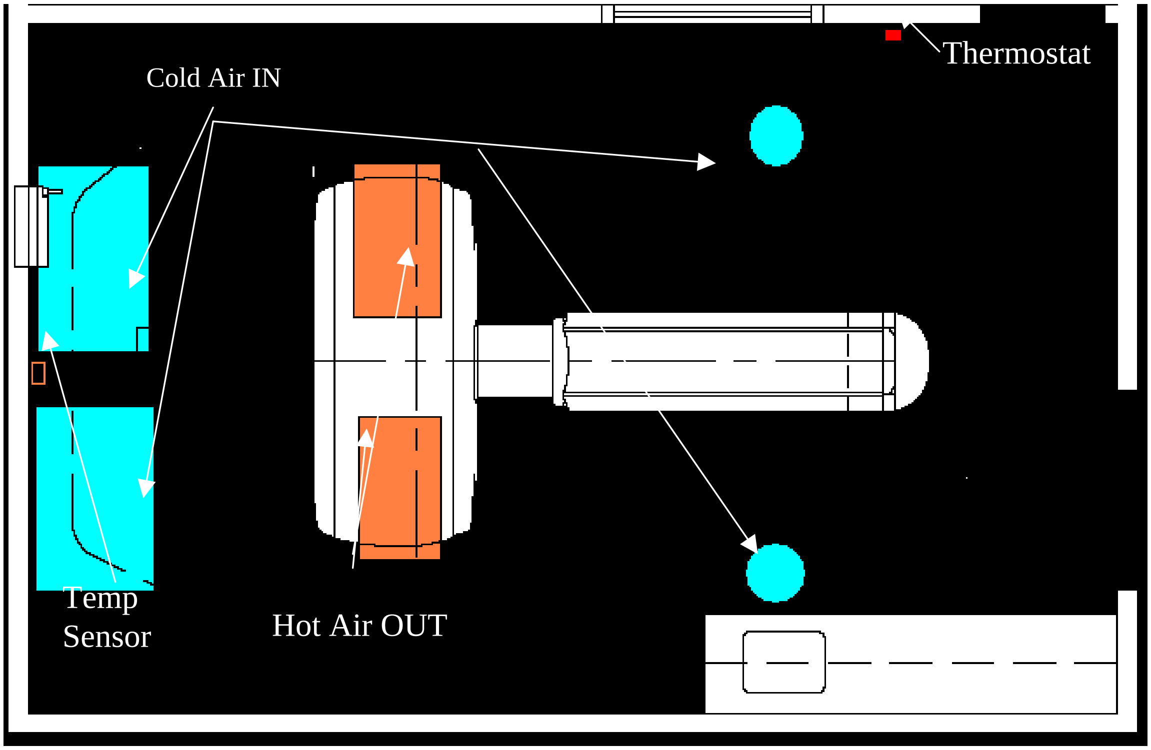

Figure 5-1 HVAC Air Vent Placement - Scan Room

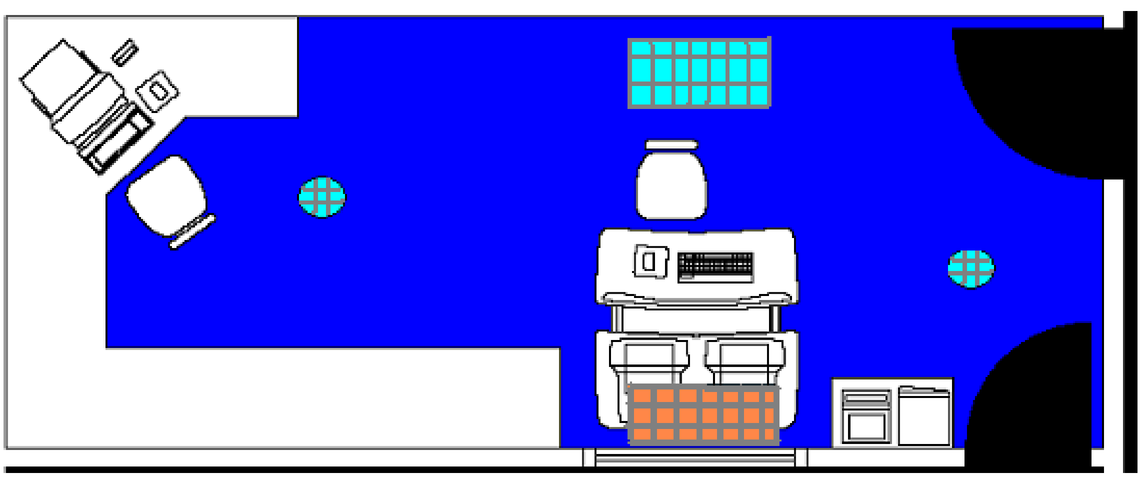

Figure 5-2 HVAC Air Vent Placement - Control Room

Section 3.0 Altitude

System operating altitude is from mean sea level to 3050 m (10,000 ft.).

Section 4.0
Electro-Magnetic Interference (EMI)

Note: If power sub-stations exist under or above the scan room, or near the control room, consider EMI testing to determine if your proposed room meets the published acceptable EMI room limits. This also includes high voltage lines under the scan or control room floor.

4.1 Gantry

Locate the gantry in ambient static magnetic fields of less than 10^{-4} tesla (1,000 milligauss) to guarantee specified imaging performance. Ambient AC magnetic fields must be below 10^{-6} tesla (10 milligauss) peak.

4.2 Color Monitor

Locate color monitors in ambient static magnetic fields of less than 5 x 10^{-5} tesla (100 milligauss) to guarantee color purity and display geometry. See Figure 5-3.

4.3 Console / Computer Equipment

Locate computer equipment in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss) to guarantee data integrity. See Figure 5-3.

4.4 Magnetic Media

Locate magnetic media in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss).

4.5 PDU

The PDU produces an electromagnetic field that radiates outward from its cabinet in all directions. Do not place sensitive electronics (e.g., operator console or computer equipment - the UPS is not classified as sensitive electronics) within one-and-a-half meters (2 m) (5 ft.) of the Power Distribution Unit., in any direction (including above or below). See Figure 5-3.

4.6 EMI Reduction

If fields of excessive EMI are known or suspected to be present, consult GE Healthcare Sales & Service for recommendations. Consider the following if you attempt to reduce EMI:

- External field strength decreases rapidly with distance from source of the magnetic field.
- External leakage magnetic field of a three-phase transformer is much less than that of a bank of three single-phase transformers of equivalent power rating.
- Large electric motors are a source of substantial EMI.
- High-powered radio signals are a source of EMI.

Maintain good screening of cables and cabinets.

4.7 UPS

The Uninterruptable Power Supply (UPS) provides a consistent power supply to various electrical components of the system. Also, it continues to provide electrical power to components during a site-wide power outage so components can be safely shut down. The UPS should be kept at least one meter (1m) (3.28 ft.) away from sensitive electronics (the PDU does not include sensitive electronics).

For UPS interconnect information, please refer to Chapter 9, Section 5.0 on page 113.

4.8 Equipment EMI “Envelopes”

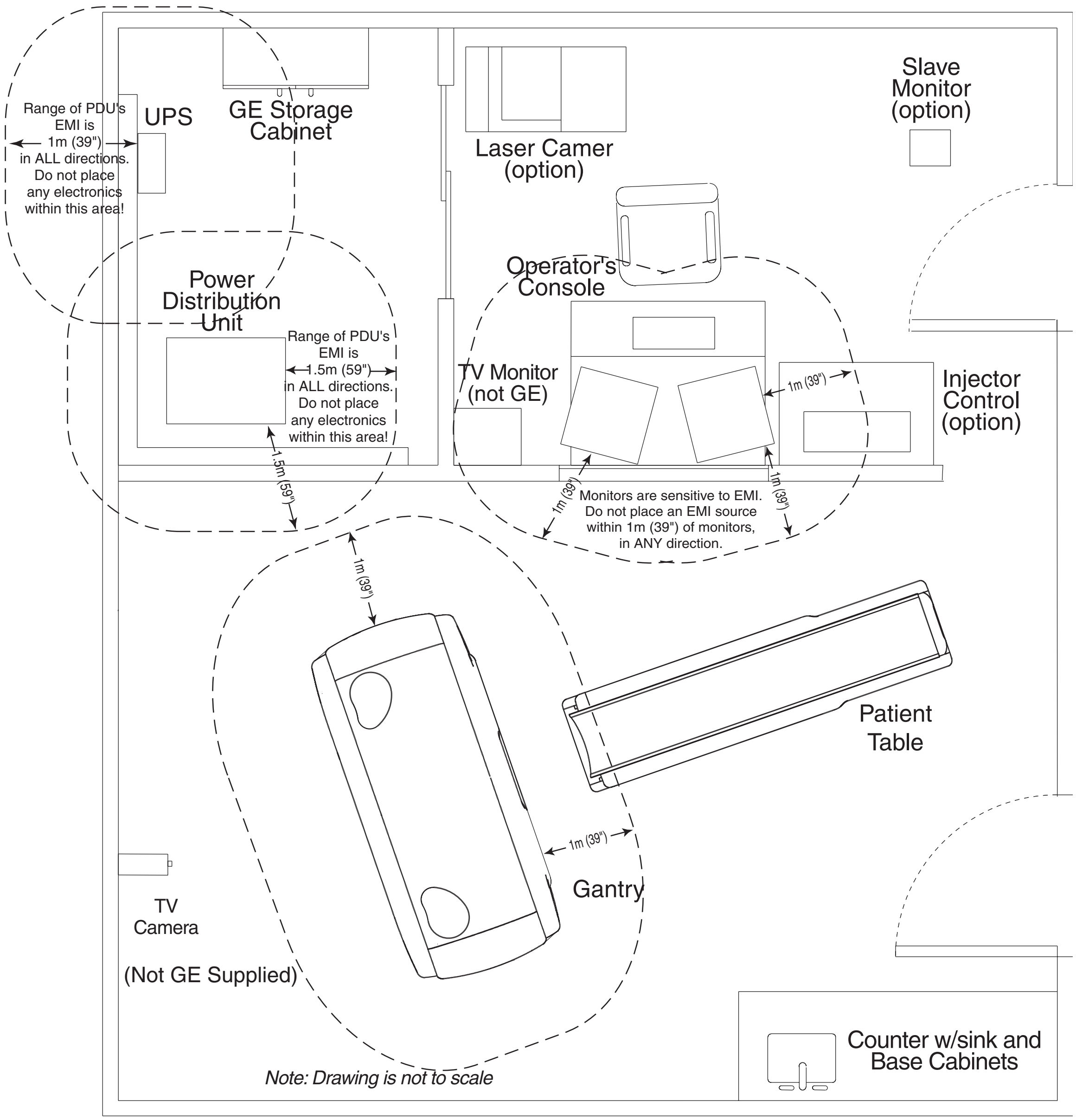

Figure 5-3 Sample Room Layout, showing approximate EMI requirements

4.9 EMC Edition 2 Compliance

4.9.1 General Scope

This equipment complies with IEC60601-1-2 Edition 2 EMC standard for medical electrical equipment.

The LightSpeed Pro16 System is suitable to be used in the electromagnetic environment, as per the limits and recommendations described in your technical reference manual.

Adhering to the distance separation recommended in tables found in the technical reference manual, between 150KHz and 2.5GHz, reduces disturbances recorded at the image level but may not eliminate all disturbances. However, when installed and operated as specified herein, the system maintains its essential performance by continuing to acquire, display, and store diagnostic quality images safely.

(*)For example, a 1W mobile phone (800MHz to 2.5GHz carrier frequency) shall be put 23 m (75.46 ft.) apart from the LightSpeed Pro16 (in order to avoid image interference risks).

4.9.2 USE LIMITATION: External Components

The use of accessories, transducers, and cables other than those specified may result in degraded ELECTROMAGNETIC COMPATIBILITY of the EQUIPMENT and/or SYSTEM

4.9.3 INSTALLATION REQUIREMENTS & ENVIRONMENT CONTROL

In order to minimize interference risks, the following requirements shall apply.

CABLE SHIELDING & GROUNDING

All interconnect cables to peripheral devices must be shielded and properly grounded. Use of cables not properly shielded and grounded may result in the equipment causing radio frequency interference.

THIS PRODUCT COMPLIES THE RADIATED EMISSION AS PER CISPR11 GROUP1 CLASS A STANDARD LIMITS.

The LightSpeed Pro16 is predominantly intended for use, in non-domestic environments, and not directly connected to the Public Mains Network.

If the LightSpeed Pro16 is used in a domestic environment (e.g. doctors' offices), in order to avoid interferences, it is recommended to use a separated AC power distribution panel & line, with a X-ray shielded room.

SUBSYSTEM & ACCESSORIES POWER SUPPLY DISTRIBUTION

All components, accessories subsystems, systems which are electrically connected to the LightSpeed Pro16, must have all AC power supplied by the same power distribution panel & line.

STACKED COMPONENTS & EQUIPMENT

The LightSpeed Pro16 should not be used adjacent to, or stacked with, other equipment. If adjacent or stacked use is necessary, the LightSpeed Pro16 should be observed in order to verify normal operation in the configuration in which it will be used.

LOW FREQUENCY MAGNETIC FIELD

In case of a digital LightSpeed Pro16, the gantry (digital detector) shall be apart 1m (3.28 ft.) from the generator cabinet, and 1m (3.28 ft.) apart from the analog (CRT) monitors. These distance specifications minimize the low frequency magnetic field interference risk.

STATIC MAGNETIC FIELD LIMITS

In order to avoid interference on the LightSpeed Pro16 system, static field limits from the surrounding environment are specified.

The static field is specified as less than <1 Gauss in examination room and in the control area.

Static field is specified as less than <3 Gauss in the technical room.

ELECTROSTATIC DISCHARGE ENVIRONMENT & RECOMMENDATIONS

In order to reduce electrostatic discharge interference, install a charge dissipative floor material to avoid electrostatic charge buildup. The relative humidity shall be at least 30 percent.

The dissipative material shall be connected to the system ground reference, if applicable.

Chapter 6
Floor Loading and Weights

Section 1.0
Floor Loading

The LightSpeed 5.X system has a total floor load of approximately 3021 kg (6660 lbs). About 2500 kg (5485 lbs), including patient (204 kg (450 lbs)), is concentrated in the table-gantry assembly.

Table 6-1 lists CT components with weight, size, floor loading and normal mounting requirements.

ITEM	NET WEIGHT LB (KG)	OVERALL W X D INCH (MM)	WEIGHT/AREA LB/SQ. FT. (KG/M²)	LOAD PATTERN IN. (MM)	NORMAL METHOD OF MOUNTING IN. (MM) (GE SUPPLIED)[1]
Gantry (no covers)	~4050 (~1850)	86.6 X 39.4 (2200 X 1000)	300 (1460)	Rectangular base plate 24 X 81 (610 X 2057) with four round pads, each 2.5 (63.5) in contact with floor. Individual pad loadings are: 910 lb, 960 lb, 1040 lb, and 1090 lb (see Figure 6-3).	Hilti Kwik-Bolt II 1/2in (13 mm) diameter by 8in (203 mm) long per P/N 2106573 at four leveling pads into concrete floor.
Dollies (each)	250 (114)				
Top Cover (each)	24.6 (11.2)				
Side Cover (each)	25 (11.3)				
Front Cover	95 (43)				
Rear Cover	100 (45)				
Patient Table	1185 (538) Includes 450 (204) Patient	24.3 X 94 (617 X 2387)	175 (852)	Rectangular base 17 X 57.5 (432 X 1460) with five round pads, each 2.5 (64) in contact with floor. Worst-case cantilever pull on any bolt is 1235 lb (561 kg).	Hilti Kwik-Bolt II 1/2in (13mm) diameter by 8in (203 mm) long per P/N 2106573 at three leveling pads into concrete floor.
Dolly	215 (98)				
Power Distribution Unit	~700 (~336)	28 X 22 (711 X 559)	161 (788)	Four Casters support area of 28 X 22 (711 X 559).	Casters are for positioning and service. Set on floor. May be anchored to floor using angle brackets[2] in seismic zones.
Console w/HP & w/o monitors	450 (204)	48 X 39 (1219 x 991)	140 (681)	Four Casters or Leveling Feet support area of 46 X 19 (1168 X 483).	Casters are for positioning. Set on floor. Console may be anchored to floor using angle brackets[2]
Monitor - CRT (ea)	80 (36)				
Monitor - LCD (ea)	22 (10)				

Notes:

1.) Use the GE Supplied mounting hardware ONLY IF APPROVED by qualified personnel.
[See statements in 8.1 - Table and Gantry Mounting Requirements, on page 66.]

2.) Angle brackets are included on the shipping skid, and are also available in the OPTIONAL Seismic Kit (R4390JC).

Table 6-1 LightSpeed Pro16 System Floor Loading

Section 2.0 Mounting Data, Including Seismic

WARNING **POTENTIAL FOR PATIENT INJURY.**

IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.

PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

The following pages show center-of-gravity information for system components:

- Gantry: Figure 6-2
- Table: Figure 6-4
- Power Distribution Unit: Figure 6-5
- Operator's Console/Computer: Figure 6-6

Floor mounting hole locations for components that don't have templates are also in this section.

An optional Seismic Kit (R4390JC) is available, for facilities in areas that require one.

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The customer is responsible for seismic mounting. Refer to all applicable codes in your area.

GE-provided floor anchors (Figure 6-1) are designed to be used ONLY on concrete floors that meet the concrete floor requirement. Supplied floor anchors must be installed by a trained contractor, and shall be set to a minimum depth of 76 mm (3 in.) at each anchor point. Any anchors having more than 25 mm (1 in.) of thread showing above the nut, when torqued to 75 N-m (55 ft.-lbs), shall have a second anchor installed in the closest adjacent hole. the second anchor shall be installed to the standard depth and torque specifications.

MOUNTING REQUIREMENTS	GANTRY	TABLE
Minimum Floor Thickness:	102 mm (4 in.)	102 mm (4 in.)
Recommended Drilling Depth:	95 mm (3.75 in.)	95 mm (3.75 in.)
Average Anchor Embedment:	89 mm (3.5 in.)	89 mm (3.5 in.)
Minimum Anchor Embedment:	76 mm (3 in.)	76 mm (3 in.)
Available Alternate Anchor Locations:	Yes	Yes
Shipped Anchor Size:	203 mm (8 in.)	203 mm (8 in.)
Alternate Anchoring Methods:	Yes (see notes, below)	Yes (see notes, below)
FLoor Levelness Requirement:	8 mm (0.3125 in.) over 3 m (10 ft.)	8 mm (0.3125 in.) over 3 m (10 ft.)

Table 6-2 Gantry and Table Mounting Requirements

Note: If the Installers cannot set at least three anchors for the table, the installer must inform the customer that the minimum anchoring cannot be met, and a structural engineering contractor is required to determine the anchoring method and certify that their anchoring meets the stated GE minimum load requirement and torque specifications.

Note: All other anchoring methods on floor types other than the concrete minimum must be determined at the customer's expense by a structural engineering contractor, and anchoring method must be certified to meet the stated GE minimum load requirement and torque specification.

Note: If installing the GE LS scanner on a floor type other than a 102 mm (4 in.) concrete floor, all structural specifications in this document must be reviewed and met.

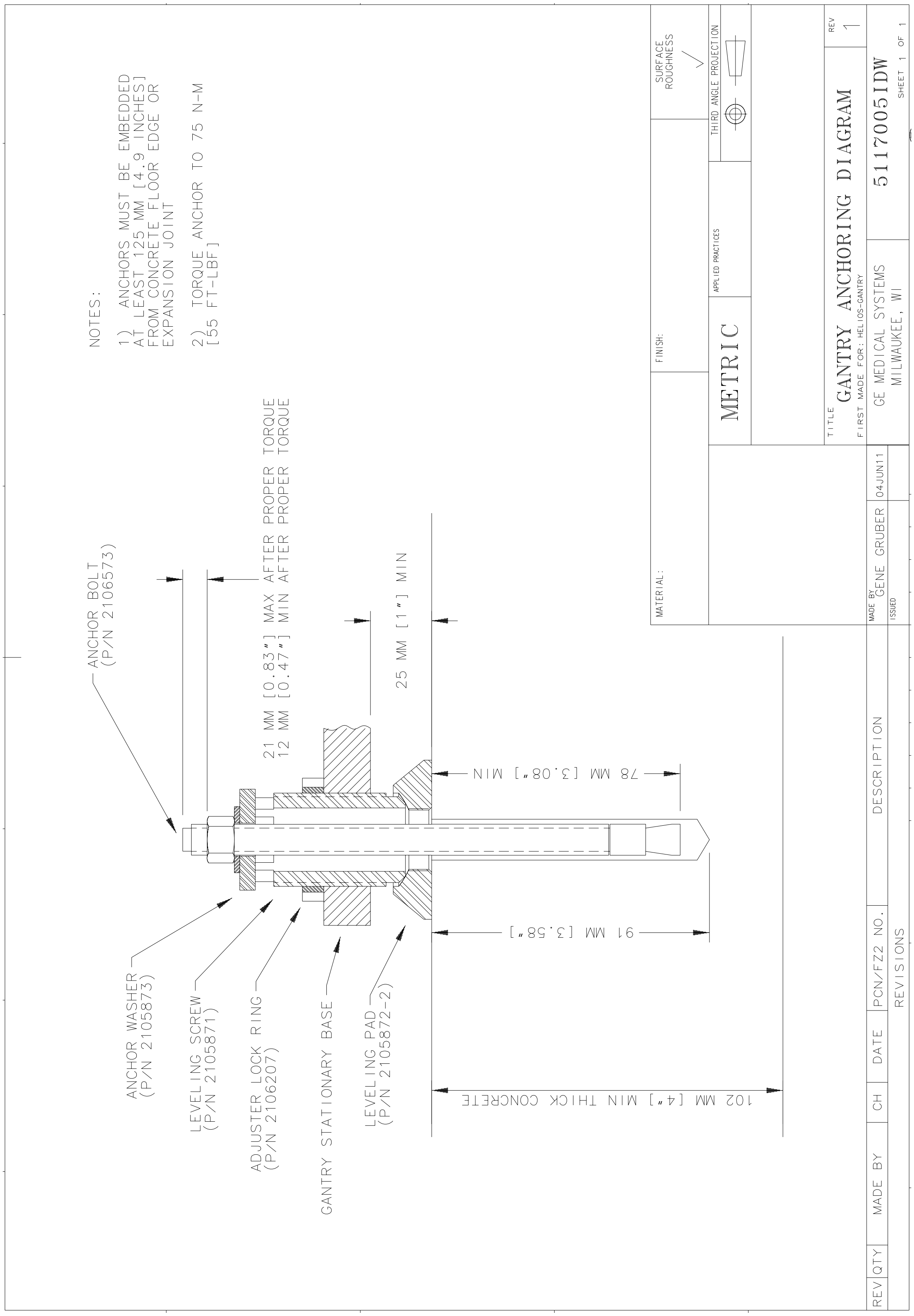


Figure 6-1 Typical Floor Anchor, Gantry and Table

6 – Floor Loading

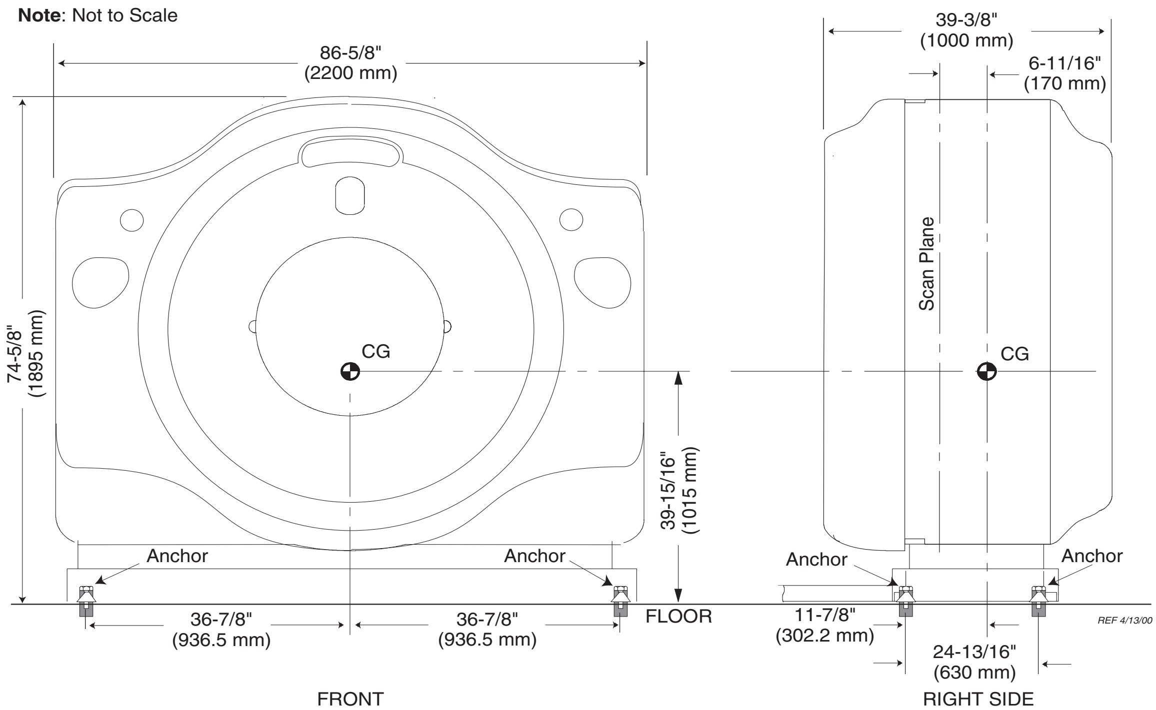
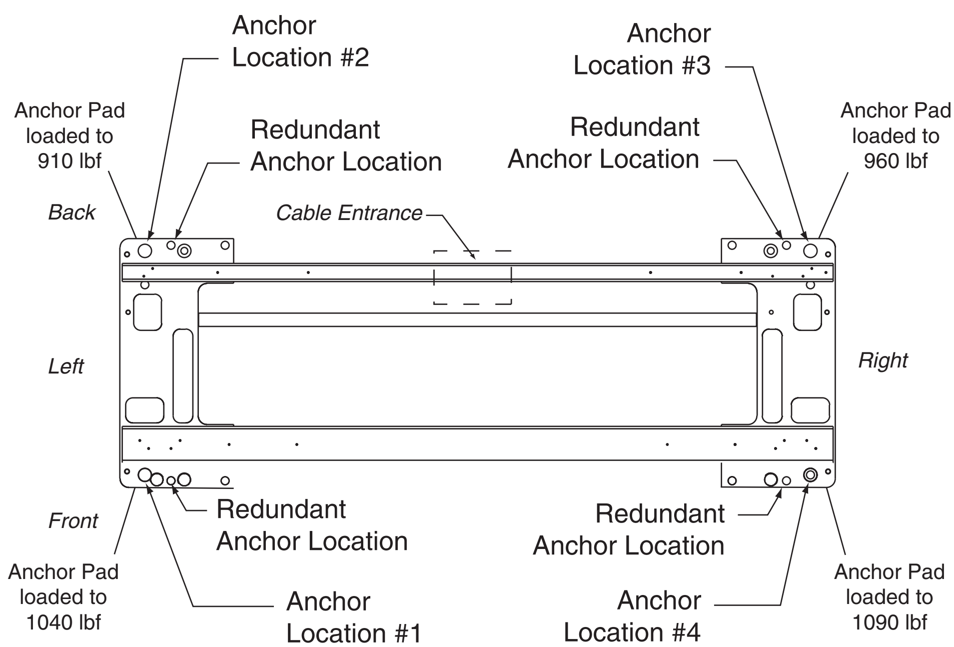


Figure 6-2 Gantry (CT2)



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm). Void between adjuster and anchor must be filled according to local building codes for siesmic application.

Figure 6-3 Gantry Anchor Locations

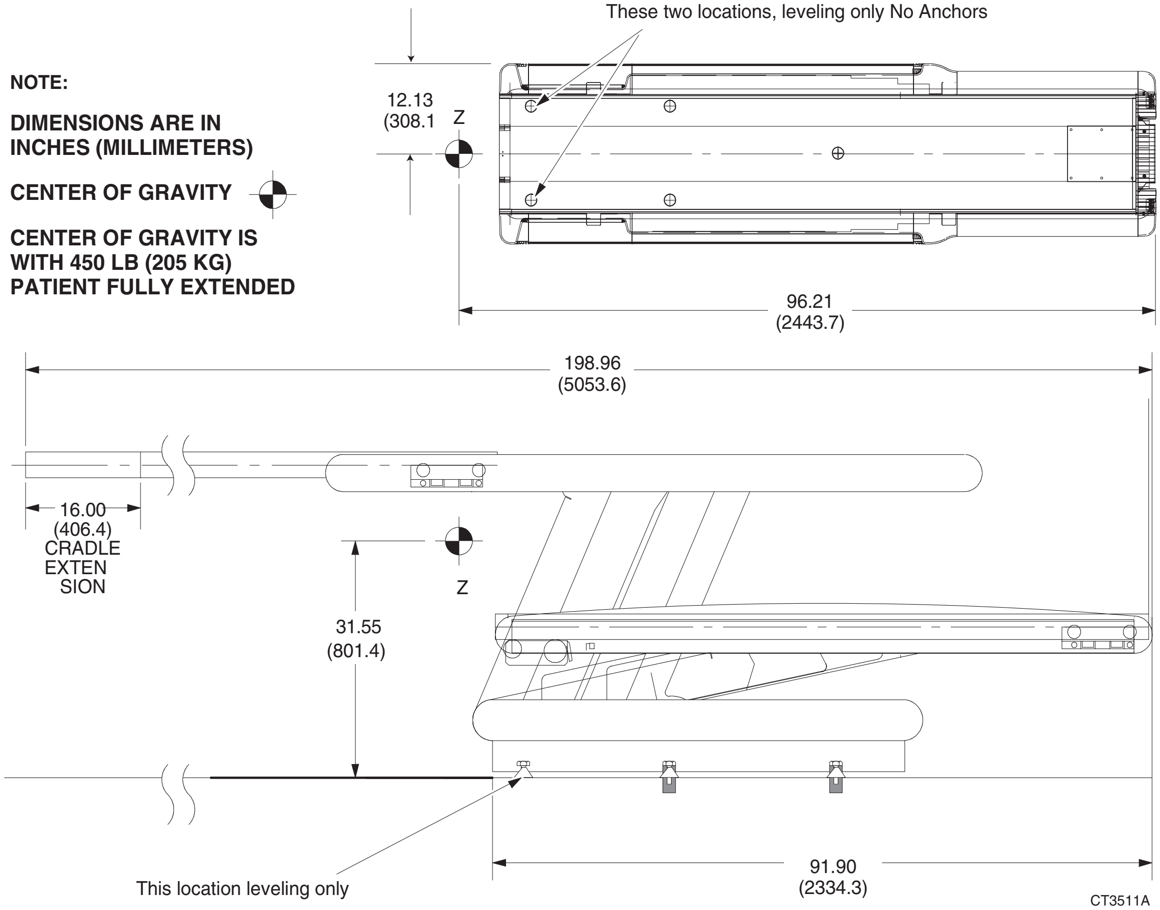


Figure 6-4 Patient Table (CT1)

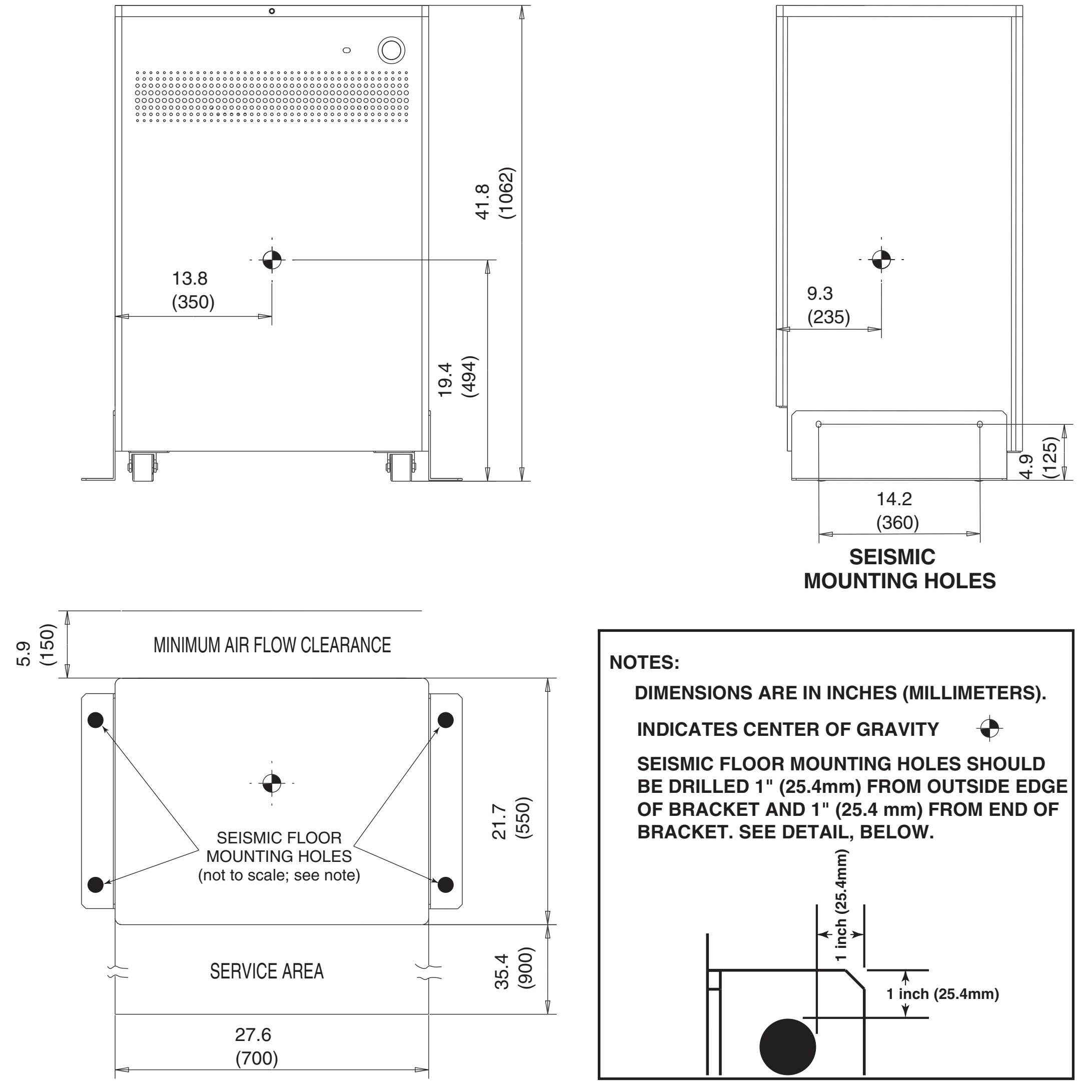


Figure 6-5 Power Distribution Unit (NGPDU)

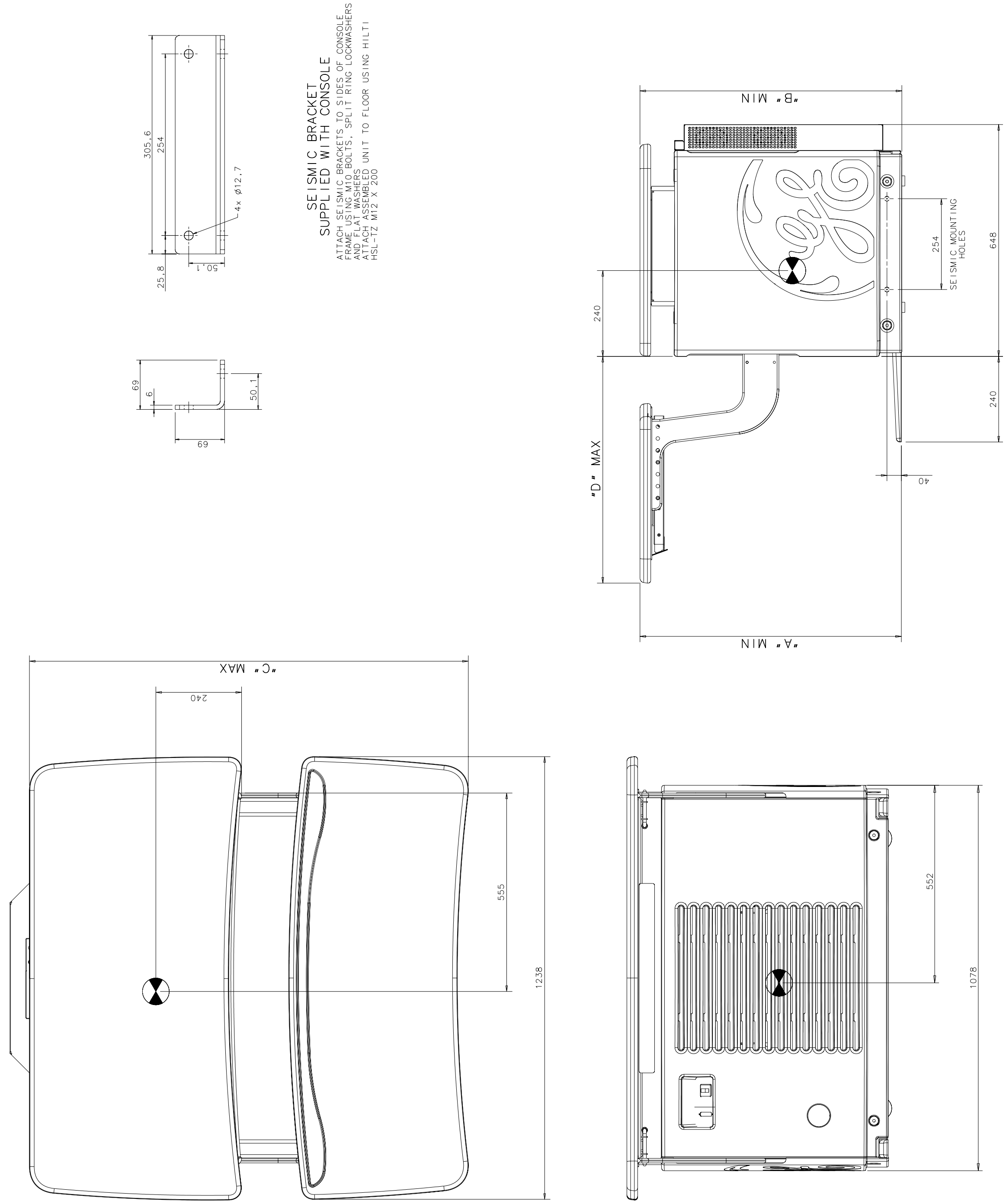


Figure 6-6 Operator Console

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Chapter 7
Delivery Data

WARNING SOME ASSEMBLIES ARE TOP-HEAVY. BE CAREFUL NOT TO TIP.

Section 1.0
Van Delivery

The CT system is packed for van shipment with minimum tear-down of components. It consists of approximately 20 shipping containers, which include dollies, skids and boxes without skids.

Section 2.0
Delivery/Shipping Requirements & Considerations

The LightSpeed 5.X System is not designed to tolerate excessive mishandling, including dropping, shock, vibration, tipping, or hoisting.

The gantry, console, table and PDU must NEVER be dropped. A drop from a height greater than 13 mm (0.5 in.) may induce structural damage to the frame or other major components. Damage resulting from a drop (e.g., bent frame, misalignment) may not be obvious until after system installation is complete.

To avoid dropping the gantry, dock-to-dock shipment is recommended. Other methods are acceptable, provided that the system is not dropped or otherwise mishandled. For example, the system may be moved via flat-bed wrecker or by rolling it across SMOOTH sidewalks or other paved surfaces.

When moving the gantry off of a flat-bed wrecker, attach the straps to the lowest point possible on the dolly. Lower the gantry at the slowest reasonable rate. See Figure 7-1.

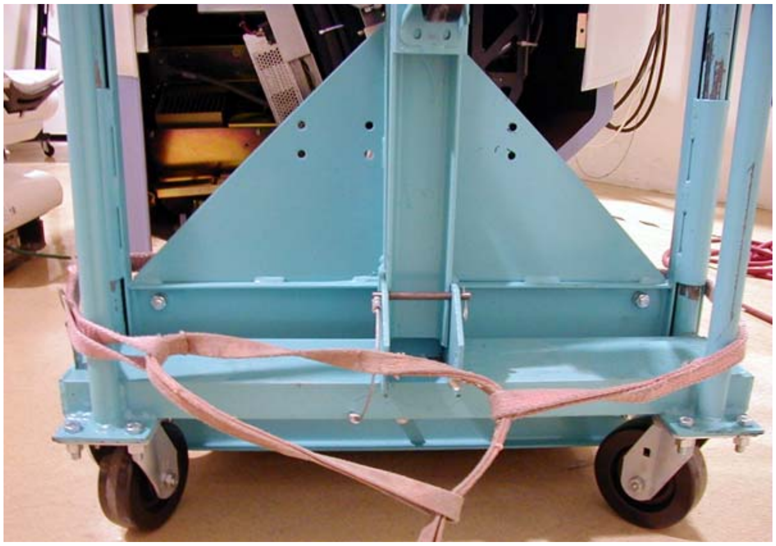

Figure 7-1 Gantry Strap Location

The LightSpeed 5.X System including the gantry, console, table and PDU, is not designed to tolerate the excessive shock or vibration that may occur during unloading. For example, rolling the console across a “washboard” style ramp may vibrate components to the extent of loosened or broken connections, etc. Damage resulting from shock or vibration (e.g., monitor, CD-ROM, hard-drive, or octane failure) may not be evident until after system installation is complete.

All system components must remain upright at all times, and must not be tipped. The gantry should not be hoisted; it should be moved by rolling it on its dollies only. Movement through hallways, doorways, elevators, etc., must be done without tipping or lifting the gantry.

Protection for flooring along the move path (from dock to scan room) is advised.

Lifting a gantry requires engineering approval for each occurrence. Your GE PMI should contact CT Engineering for all special lifting requirements.

- **Do Not lift the gantry with a forklift.**
- **Unauthorized gantry lifting will cause gantry bearing damage.**

Section 3.0 Site Environmental Considerations

3.1 Dust/Dirt Contamination

LightSpeed 5.X systems (consisting of: console, PDU, table, and gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site. Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.

The act of installing a GE CT scanner in a construction (i.e., unfinished) site will likely result in the following adverse effects:

- **Increased installation time**
- **Decreased installation quality**
- **Increased scanner downtime, due to increased service calls**

3.2 Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of LightSpeed 5.X CT scanner components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer’s requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

Section 4.0 Storage Requirements

NOTICE Failure to adhere to Storage Requirements will likely result in equipment damage.

4.1 Short-term Storage (Less than Six Months)

If storing the CT system before installation for less than six months, store it in a temperature- and humidity-controlled warehouse. Protect it from weather, dirt, and dust.

Meeting the following requirements prevents rust and corrosion from forming on bearing surfaces due to condensation:

- Storage temperature should not exceed 4º to 27º C (40º to 80º F).
- Maintain relative humidity (non-condensing) between 20% and 60%.
- Maximum rate of relative humidity change measures 5%/hr.
- Maximum rate of temperature change measures 3º C/hr. (5º F/hr.)
- Storage longer than 6 months is not recommended

NOTICE Between delivery qualifies as short-term storage. Van storage must meet the same specifications listed above.

LONG TERM STORAGE (6 MONTHS OR MORE)

If the CT system is to be stored before installation, store in a temperature and humidity controlled warehouse. Protect from weather, dirt, and dust.

Meeting these requirements prevents rust and corrosion from forming on bearing surfaces due to condensation.

- Storage temperature should not exceed 10º to 32º C (50º to 90º F).
- Maintain relative humidity (non-condensing) between 20% and 70%.

If you are storing a system longer than six (6) months, but less than 12 months, contact CT Engineering for installation start-up instructions. Storage longer than 12 months is not recommended.

Section 5.0 Extreme Temperature Transportation and Deliveries

NOTICE Failure to adhere to extreme temperature requirements during delivery and storage can result in equipment damage.

Avoid extreme temperatures during system transportation and delivery.

Extreme temperatures consist of temperatures below -18º C (0º F), or above 49º C (120 º F), without humidity control.

When transporting the CT system, prevent extended exposure of the system to temperatures or humidity outside of the following specifications:

- Temperature: -40º to +70º C (-40º to +158º F)
- Humidity: 5% to 95%

NOTICE Component freezing occurs when exposing the CT system to temperatures below -18º C (0º F) for a period longer than two (2) days. Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature prior to installation.

Section 6.0
Gantry Considerations

The gantry is shipped with most covers installed. The assembly is mounted between two dollies. See Figure 7-2. Two side rails are bolted to the dollies to stabilize dollies and protect gantry. Use dolly elevating casters to lift the gantry off its base and roll it into position.

Figure 7-2 Gantry with Shipping Dollies and Side Rails

Door Openings. Clear door openings for moving equipment into building must be 1067mm X 2083 mm (42 X 82 inches) minimum, 2439 mm (8 ft.) corridor width is helpful.

Elevator requirements. To move the gantry from the receiving location to the scanning room, consider elevator capacity and size. By removing side rails and one dolly after the gantry is placed in the elevator, the gantry width/length and elevator depth requirements are reduced. Contact a representative of elevator manufacturer if the gantry weight exceeds elevator capacity.

CONFIGURATION	LENGTH	WIDTH	HEIGHT
Dollies On, Side Rails On	2900 mm (114 in.)	1290 mm (51 in.)	2000 mm (79 in.)
Dollies On, Side Rails Removed	2900 mm (114 in.)	1000 mm (39.4 in.)	2000 mm (79 in.)
Dollies Off, Covers Off	2060 mm (81 in.)	860 mm (34 in.)	1850 mm (73 in.)

Table 7-1 Size of Gantry & Dollies, with and without Side Rails

Minimum hallway and door size for Gantry with covers and dollies attached, but side rails removed, is 1016 mm (40 in.).

For alternative lifting arrangements and instructions, contact GE Installation Support Services.

Dollies: Typically, dollies are used on the gantry, table, and operator console for domestic shipments. Once the gantry, table and operator console are installed at the site, return the dollies to GE using the shipping document located in Box #1.

Dollies can be purchased for international shipments (B7850LD) to use at the customer site. After the system is removed from the crates, dollies shipped with international shipments **only** are not to be shipped back to GE in Milwaukee, WI, USA, but to be sent to the local GE office or warehouse.

Zero clearance dollies are available from UMI. For information on ordering zero clearance dollies, please go to: http://www.umi-dollyshop.com

Section 7.0 Table Considerations

The Table is shipped without side covers installed. Covers are shipped in four separate boxes. The Table is mounted between two dollies.

Table shipping dimensions are 2565 mm (101in.) long, 81 mm (32 in.) wide, and 119 mm (47 in.) high.

Section 8.0 Console Considerations

The operator console is shipped without the keyboard table installed. The keyboard table is shipped with the operator console.

- The operator console is shipped on a skid equipped with ramps for unloading.
- Do not remove the operator console from the shipping skid until it is in the CT equipment room.
- Console shipping dimensions (on the shipping skid) are 1016 mm (40 in.) deep, 1346 mm (53 in.) wide, and 1041 mm (41 in.) high.

Note: he dimensions of the operator console alone (as shipped) are 888 mm (34.95 in.) deep, 1238 mm (48.74 in.) wide, and 750 mm (29.5 in.) high.

NOTICE **Do not lift the Operator Console by the monitor table top to remove from the shipping skid.**

Figure 7-3 Console ready to be unloaded from shipping skid.

4 – Room Planning

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Chapter 8
Power Requirements

Section 1.0
Introduction

Note: Power requirements for LightSpeed 5.X differ from those for LightSpeed 4.X and previous systems.

The Power Distribution Unit (PDU) supplied with the LightSpeed 5.X system transforms and distributes power to all system components. The PDU is the only power entry point required to operate the system.

To minimize voltage regulation effects, power wiring between the facility main distribution panel and the PDU should be kept as short as possible.

When routing the power wiring, all three-phase wires and ground must be run in the same conduit or raceway duct. Power wires should be routed separately from the system control and signal cables using a separate conduit or trough in the raceway duct.

A metallic conduit, floor duct, or surface raceway may be used for running cables, depending upon local codes and practices. However, cable passageways should be large enough to install any cable with all other cables already installed. The use of non-metallic conduit is not recommended.

Section 2.0
System Input Power

2.1 Power Source Configuration

The LightSpeed 5.X CT Scanner is designed to operate on a three-phase, four-wire wye power source. A solidly grounded wye source is preferred. The neutral wire does not need to be run to the system, i.e., four-wire connection. If a neutral wire *is* run, then it should be terminated in the A1 box.

A dedicated feeder from the nearest Main Distribution Panel (MDP) should supply power to the scanner. In accordance with the National Electric Code (U.S.), and similar applicable national and local codes, a protective disconnect device must be provided in the power line supplying the PDU. It must be located within 10 m (32 ft.) of the PDU, visible to PDU service personnel, and must have lockout /tagout provisions. This disconnect device is identified as “A1” in the interconnection schematic diagrams.

2.2 Rating

The LightSpeed 5.X system operates on three-phase power meeting the following specifications.

Voltage	380 to 480 VAC
Capacity	150 kVA
Frequency	50 or 60 Hz +/- 3 Hz

- Maximum power demand = 150 kVA @ 0.85 PF at a selected technique of 140 kV, 715 mA.
- Average (continuous) power demand at maximum duty cycle = 25 kVA.

The A1 disconnect device referenced above must provide overcurrent protection for the system and facilitate multi-point remote “Emergency Off” control of system power. A disconnect utilizing undervoltage release control is preferred over shunt trip devices. The rating of the A1 disconnect device depends on the nominal line voltage at the site. Refer to Section 3.0: “Recommended Power Distribution System”, for minimum rating requirements and suggested disconnect devices.

2.3 Regulation

Total load regulation as measured at the PDU input terminals must not exceed 6%. The capacity of the facility transformer and size & length of feeder wires directly affect the load regulation presented to the system. Refer to Section 3.0: “Recommended Power Distribution System”, for recommended single-unit installation specifics.

2.4 Phase Imbalance

The difference between the highest line-to-line voltage and lowest line-to-line voltage must not exceed 2% of the lowest line-to-line voltage.

2.5 Sags, Surges & Transients

Sags and surges of the power line must not exceed the absolute range limits shown in Table 8-1. Maximum transient voltages should be limited to 1500V peak.

2.6 Grounding

Metal conduit, raceway or the armor of armored cable used to power the system should be bonded to the PDU cabinet. However, in addition to such mechanical grounding, a dedicated 1/0 (55 mm^2) or larger insulated copper ground wire must be run with the phase wires from the main distribution panel to the PDU.

Note: The shield or armor of armored cable is not sufficient for this purpose.

The ground wire should be bonded to intermediate distribution panels through which it passes in accordance with local codes. The resistance between the PDU ground and the facility earth ground must not exceed 0.5 ohm. In addition, the total resistance between the PDU ground and earth must not exceed 2 ohms.

Section 3.0
Recommended Power Distribution System

A dedicated feeder run from the facility main isolation transformer is recommended to power the LightSpeed 5.X CT scanner. If the scanner must be powered from an existing distribution transformer and secondary feeder, such as the equipment distribution panel of an X-ray department, installation with other X-Ray equipment that use rapid film changers should be avoided. These changers use a large number of high powered, closely-spaced exposures, which may coincide with the CT scan and produce image artifacts.

If a dedicated distribution transformer is provided for the scanner, the minimum recommended transformer size is 225 kVA, rated 2.4% regulation at unity power factor. For this configuration, the minimum recommended feeder size and overcurrent protection device based on line voltage is shown in Table 8-3 Minimum Feeder Wire Size.

In all cases, qualified personnel must verify that the transformer and feeder, at point of take-off, plus the run to the LightSpeed 5.0 CT scanner meet all the requirements stated in this document.

SYSTEM CHARACTERISTICS:

- Maximum power demand = 150kVA @ 0.85 PF: at a Selected Technique of 140 kV, 715 mA.
- Continuous (average) power demand at maximum duty cycle = 25kVA.
- Maximum allowable total source regulation is 6%.
- Minimum recommended transformer size: 225 kVA, with 2.4% rated regulation at unity power factor. Resultant maximum allowable feeder regulation is 3.4%.

The nominal line voltage must fall within one of the ranges listed below

Nominal Line Voltage	380	400	420	440	460	480
Hi-Line Limit, +10%	418	440	462	484	506	528
Lo-Line Limit, -10%	342	360	378	396	414	432
Continuous Line Current	38	36	34	33	31	30
Momentary Line Current	228	217	206	197	188	180
Maximum Line Current	253	241	229	219	209	200
Minimum Recommended Circuit Protection Rating	150	150	150	125	125	125

Table 8-1 Nominal Line Voltage

OPTION	CAT NUM
225 kVA Transformer	E4500AW
Isolation Transformer	E4500BC

Table 8-2 Purchasable Options

FEEDER LENGTH (MDA TO A1)	MINIMUM FEEDER WIRE SIZE, AWG OR MCM (SQ. MM)/ VAC					
	380 VAC	400 VAC	420 VAC	440VAC	460VAC	480VAC
15 m (50 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
30 m (100 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
46 m (150 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
61 m (200 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
76 m (250 ft)	2/0 (70)	2/0 (70)	1/0 (55)	1/0 (55)	1 (45)	1 (45)
91 m (300 ft)	3/0 (85)	3/0 (85)	2/0 (70)	2/0 (70)	1/0 (55)	1/0 (55)
107 m (350 ft)	4/0 (100)	3/0 (85)	3/0 (85)	2/0 (70)	2/0 (70)	1/0 (55)
122 m (400 ft)	250 (125)	4/0 (100)	3/0 (85)	3/0 (85)	3/0 (85)	2/0 (70)

Table 8-3 Minimum Feeder Wire Size

SUB-FEEDER LENGTH (A1 TO PM)	MINIMUM SUB-FEEDER WIRE, AWG OR MCM (SQ. MM)					
	380 VAC	400 VAC	420 VAC	440VAC	460VAC	480VAC
9.7536 m (32 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)

Table 8-4 Minimum Sub-Feeder Wire Size

1.) Table 8-1, Table 8-3, and Table 8-4 (above) are based on the use of copper wire, rated 75 C and run in steel conduit. Ampacity is determined in accordance with the National Electrical Code (NFPA 70), Table 310-16 (2002)
2.) The minimum feeder size is determined by the ampacity of the circuit protection device listed above, except where a larger size is necessary to meet total source regulation limits.
3.) A 1/0 (55 sq. mm) ground wire is recommended in all cases.

Section 4.0
Ground System

The LightSpeed 5.X CT Scanner has been designed to use an equal potential grounding system. The required ground system is shown in Figure 7-1. There are three primary grounding points:

- A system power ground point located in the PDU.
- A reference ground point located between gantry and table base.
- A patient ground point located at the front of the Table base.

All exposed metal surfaces in the patient vicinity are grounded to the reference ground point.

For additional information, refer to Electrical Safety Equipment, Direction 46-014505.

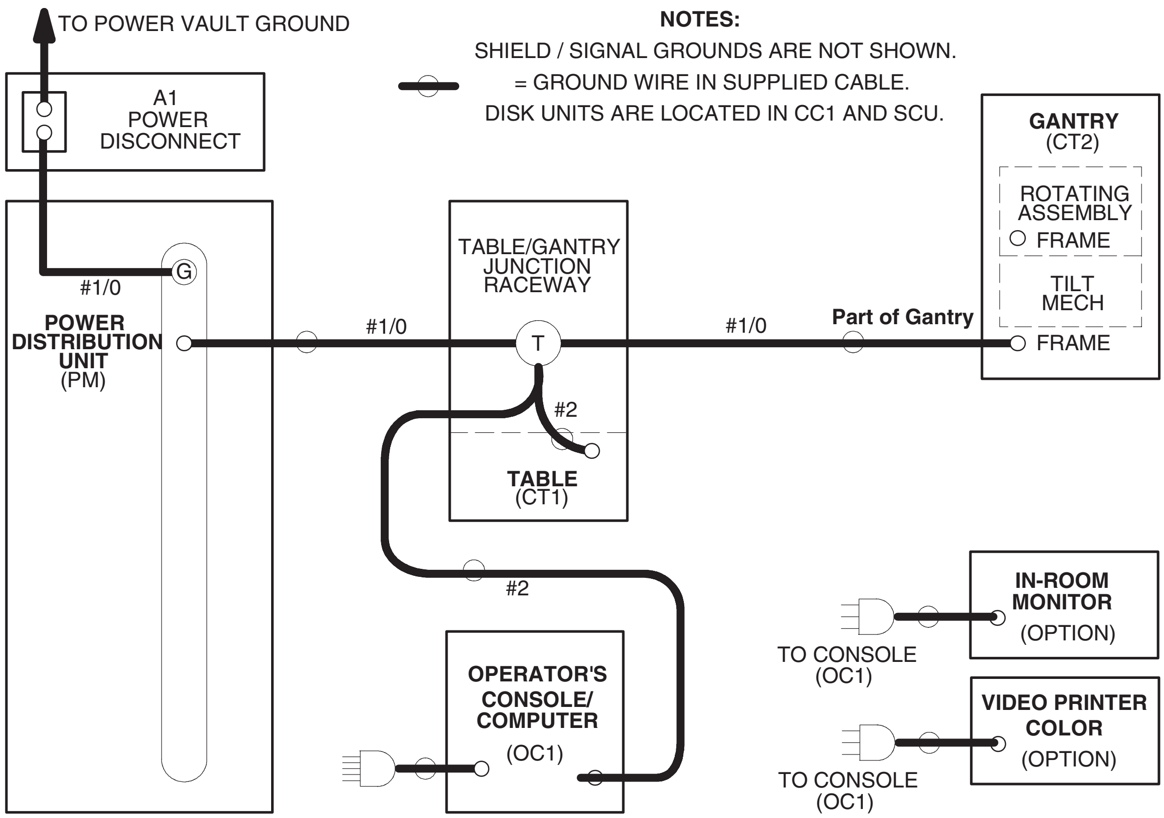

Figure 8-1 System Ground Map

8– Power Req's

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Chapter 9
Interconnection Data

Section 1.0
Introduction

Figure 9-3 shows interconnection runs for a 50/60 Hz system.

Table 9-1 shows component designators for supplied equipment and options and wall power outlets.

Table 9-4 lists customer-installed wiring and supplied cables. Actual length of each run is less than the length of supplied cables to allow for routing inside equipment. Cable diameters and sizes of connectors are provided to aid in sizing conduit and access plates.

Table 9-2 and Table 9-3 list details for connection to LightSpeed Series equipment using standard (short) length and non-standard (long) length cables, respectively. Details are listed for the following types of runs as appropriate:

- Flush-floor duct
- Computer floor
- Through-wall bushing
- Junction box
- Surface floor duct
- Through-floor duct
- Wall duct
- Conduit

The need for additional junction boxes is minimized by using either a cable raceway system or a raised computer floor. LightSpeed series systems use prefabricated cables with large plugs. Therefore, conduit or pipe is not recommended for cable runs.

Section 2.0
Component Designators

DESIGNATOR	APPLIES TO	SOURCE
A1	Primary power disconnect	**Contractor supplied**
CT1	Patient Table	System
CT2	Gantry	System
ITL	InSite telephone lines	**Contractor supplied**
OC1	Operator's Console/computer	System
PDU	Power distribution unit	System
SEO	System emergency off	**Contractor supplied**
SM	Slave monitor	Option
WL	“X-ray on” warning light	**Contractor supplied**
DS	Door Interlock Switch	**Contractor supplied**
BBNC	Broad-Band Network Connection	**Contractor supplied**

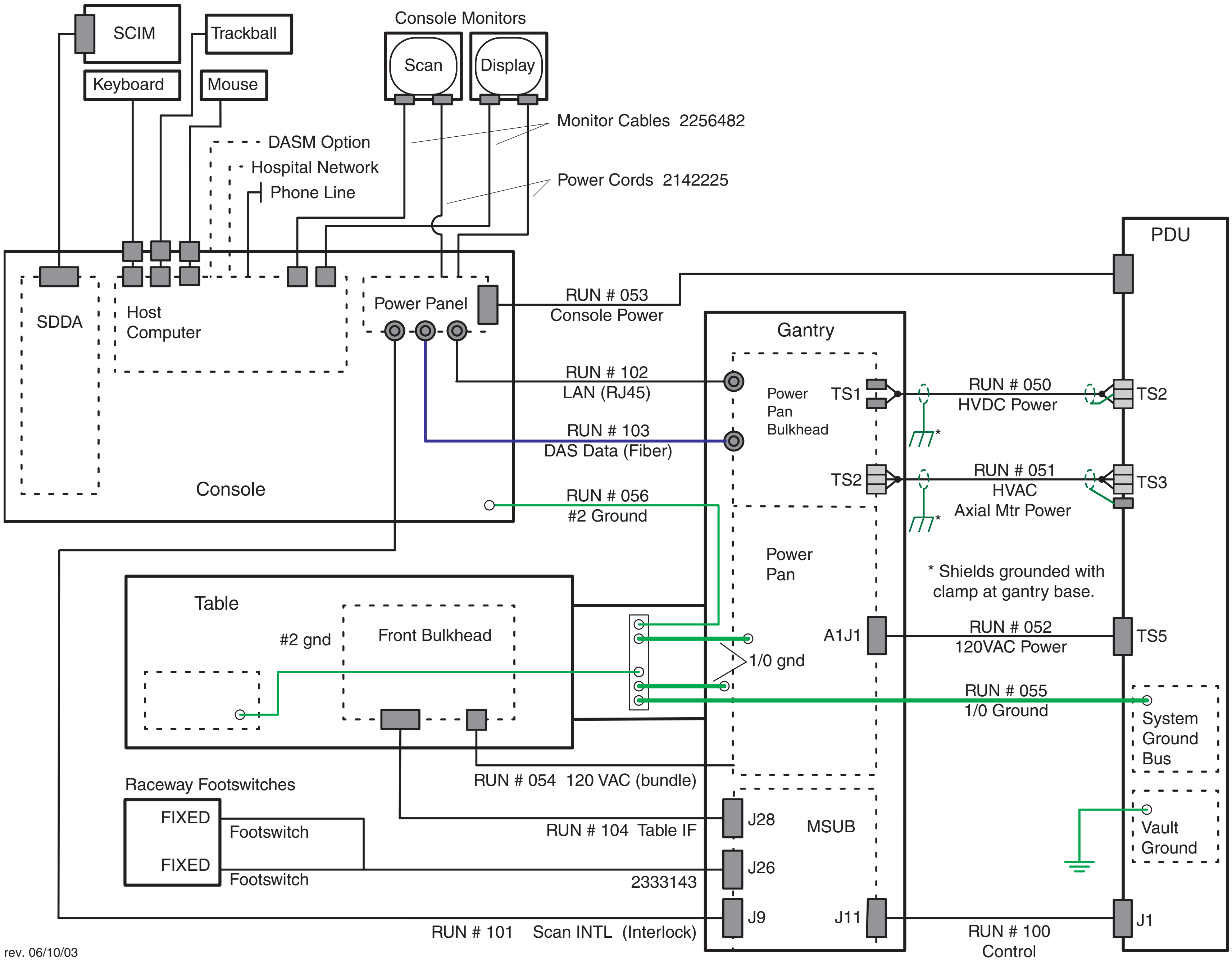
Table 9-1 Component Designators

Section 3.0
Interconnect Runs, Wiring and Cables

3.1 GE Healthcare Supplied (Standard Length)

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	28 (20)	9 (6)	2343529-2	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 (.751)	3	(2) 4 (1) 8	22 (.87) Dia
051	28 (20)	9 (6)	2343530-2	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15 (.604)	4	14	11 (.44) Dia
052	28 (20)	9 (6)	2343528-2	LVAC, PDU to Gantry	2587	FT4	600	208Y/120	90	14 (.542)	5	8	56 (2.22) Dia
053	65 (60)	20 (18)	2343531-2	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12 (.483)	3	10	56 (2.22) Dia
054			n/a	LVAC, Gantry to Table	1015		600	120VAC			3	14	
055	43 (35)	13 (11)	2371450-2	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	16 (.608)	1	1/0	16 (.62) Dia
056	71 (60)	22 (18)	2371450-4	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	12 (.467)	1	2	12 (.48) Dia
100	32.5 (20)	10 (6)	5120646-2	Signal, Gantry MSUB to PDU		FT-4	300	<30VDC	80	11 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
101	71 (60)	22 (18)	5120645-2	Signal, Gantry MSUB to OC		FT-4	300	<30VDC	80	11 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	68 (60)	21 (18)	2352714-3	Signal (LAN), Gantry to OC			1900	<30VDC		6 (.234)	8	24	15 (.59) Dia
103	68 (60)	21 (18)	2117848-7	Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (.39) Dia
104			n/a	Signal, Gantry to Table		FT-4	300		80		25	22	

Table 9-2 GE Healthcare Supplied Cables (Standard Run) - UL Information



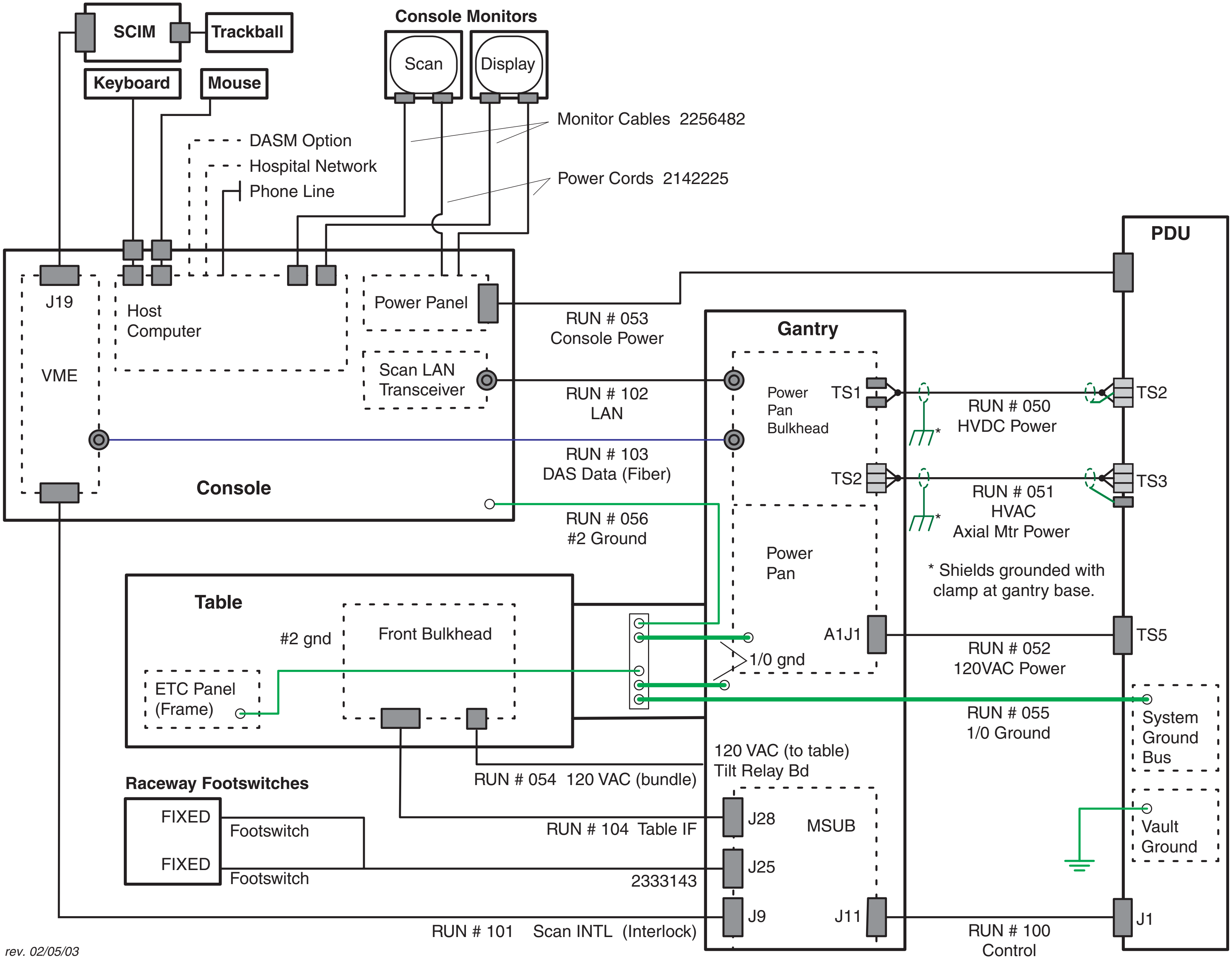
rev. 06/10/03

Figure 9-1 System Interconnect Diagram

3.2 GE Healthcare Supplied (Optional, Long Run)

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	63 (55)	19 (17)	2343529	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 (.751)	3	(2) 4 (1) 8	22 (.87) Dia
051	62.5 (55)	19 (17)	2343530	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15 (.604)	4	14	11 (.44) Dia
052	60 (55)	19 (17)	2343528	LVAC, PDU to Gantry	2587	FT4	600	208Y/120	90	14 (.542)	5	8	56 (2.22) Dia
053	80 (75)	25 (23)	2343531	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12 (.483)	3	10	56 (2.22) Dia
055	63 (55)	19 (17)	2371450	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	16 (.608)	1	1/0	16 (.62) Dia
056	83 (75)	26 (23)	2371450-3	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	12 (.467)	1	2	12 (.48) Dia
100	63 (55)	19 (17)	5120646	Signal, Gantry MSUB to PDU		FT-4	300	<30VDC	80	11 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
101	83 (75)	26 (23)	5120645	Signal, Gantry MSUB to OC		FT-4	300	<30VDC	80	11 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	80 (75)	24 (23)	2373436-1	Signal (LAN), Gantry to OC			1900	<30VDC		6 (.234)	8	24	15 (.59) Dia
103	80 (75)	24 (23)	2117848-2	Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (.39) Dia

Table 9-3 GE Healthcare Supplied Cables (Optional, Long Run) - UL Information



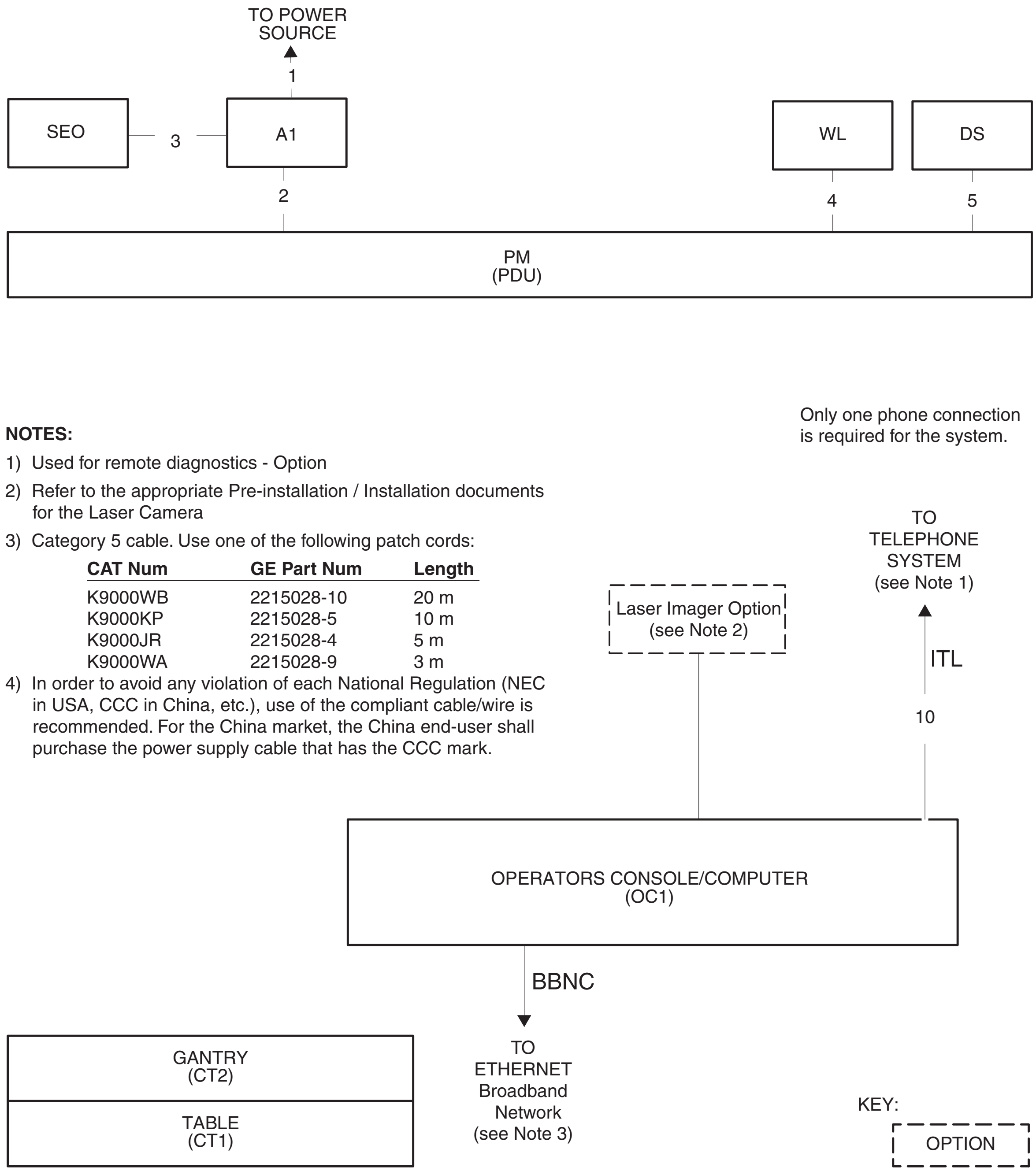
rev. 02/05/03

Figure 9-2 System Interconnect Diagram

3.3 Contractor (Customer) Supplied

CUSTOMER INSTALLED WIRING		DESCRIPTION	CABLES SUPPLIED			PLUG PULLING DIMENSIONS		WIRE & CABLE PIGTAILS FT. (M.)	
QTY	**SIZE AWG (MM^2)**		**PART NO**	**LENGTH FT. (M.)**	**DIA. IN (MM)**	**FROM**	**TO**	**FROM**	**TO**
RUN NO. 1 FROM PRIMARY POWER SOURCE TO FACILITY DISCONNECT (POWER SOURCE - A1) Maximum Run Length *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
RUN NO. 2 FROM FACILITY DISCONNECT TO POWER MODULE (A1 - PM) MAXIMUM RUN LENGTH *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
RUN N0. 3 FROM FACILITY DISCONNECT TO SYSTEM EMERGENCY OFF (A1 - SEO)									
2	14 (2)	POWER						6 (2)	6 (2)
1	14 (2)	GROUND						6 (2)	6 (2)
RUN NO. 4 POWER MODULE TO WARNING LIGHT CONTROL (PM - WL)									
2	14 (2)	WARNING LIGHT 24 VOLT CONTROL A3J2-1,2,3,4							
RUN NO. 5 POWER MODULE TO SCAN ROOM DOOR INTERLOCK (PM - DOOR SWITCH)									
2	14 (2)	SCAN ROOM DOOR INTER LOCK A3J6-1,2							
*	REFER TO Table 8-3 on page 102 FOR AWG (MM2) WIRE SIZES.								

Table 9-4 Runs 1, 2, 3, 4 and 5 Connections



NOTES:

1) Used for remote diagnostics - Option
2) Refer to the appropriate Pre-installation / Installation documents for the Laser Camera
3) Category 5 cable. Use one of the following patch cords:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

4) In order to avoid any violation of each National Regulation (NEC in USA, CCC in China, etc.), use of the compliant cable/wire is recommended. For the China market, the China end-user shall purchase the power supply cable that has the CCC mark.

Figure 9-3 Interconnection Runs

Section 4.0
Contractor Supplied Components

REFERENCE	ASSOCIATED EQUIPMENT	MATERIAL/LABOR SUPPLIED BY CUSTOMER CONTRACTOR	USA VENDOR / CAT NO. GE CATALOG
A1 380 - 480V 50/60 Hz	Circuit Breaker with Magnetic Contactor	3 Pole, 380V - 480V, Combination breaker with magnetic contactor. Includes control transformer, optional UPS interface, On/Off controls and auto-restart feature	Recommend*: • E4502AE (125A) • E4502AF (150A) Optional remote operator control available from GE Supply, Cat No. GESCTR0CS1
BBNC (required)	Broad-Band Network Connection	Broad-Band network connection wall jack, located within 1m (39inches) of console location, for internal hospital networking and InSite Broad-Band connectivity. Cabling to conform to facility's IT standards.	
ITL (optional)	In-suite Telephone Lines	Supply 2 voice-grade telephone lines. One line must be a direct number from outside the facility – do not route this line through a telephone switchboard. Telephone line operating charges are paid by customer.	
	System Components	Reference the system installation drawings supplied by Installation Support Services within your geographic area.	
*Refer to Table 9-6 on page 113.			

Table 9-5 Contractor-Supplied Components

Section 5.0
UPS Interconnect

Figure 9-4 Typical UPS

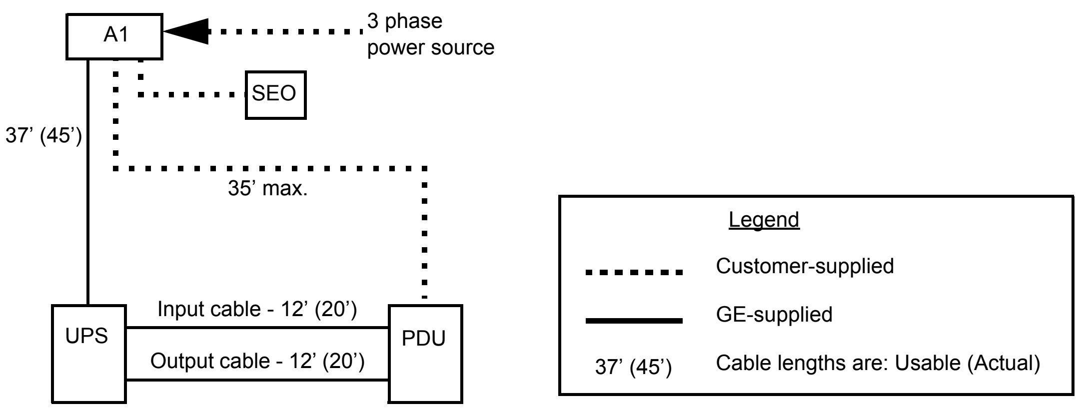


Figure 9-5 Typical UPS Interconnect

PDU Type & Model #	Max. Mom. kVA Rating	**Required Main Disconnect (A1) Cat #**		Optional Partial UPS Kit Cat #
		Europe & Asia (380-400V or 420V)	North America (440V or 460-480V)	
NGPDU 2326492	150kVA	**E4502AF** (150A) (incl. Auto Restart & Integrated UPS Control) (Additional A1 Disconnects are available for Europe through the European Sales Team.)	**E4502AE** (125A) (incl. Auto Restart & Integrated UPS Control)	**B7999PP** (incl. 12.5 kVA UPS & 40 A hardware kit) NOTE: REQUIRES one of the A1 Panels shown at left.

Table 9-6 LightSpeed 5.X Partial UPS Back-up Options

Section 6.0 Typical Customer Supplied Wiring

6.1 Primary Power Disconnect

Installing this system requires a Lockout/Tagout-compatible disconnect. If a UPS is required, a GE Disconnect is strongly recommended for safe operation. The GE disconnect and UPS are designed to work together.

Figure 9-6 Primary Power Disconnect (A1)

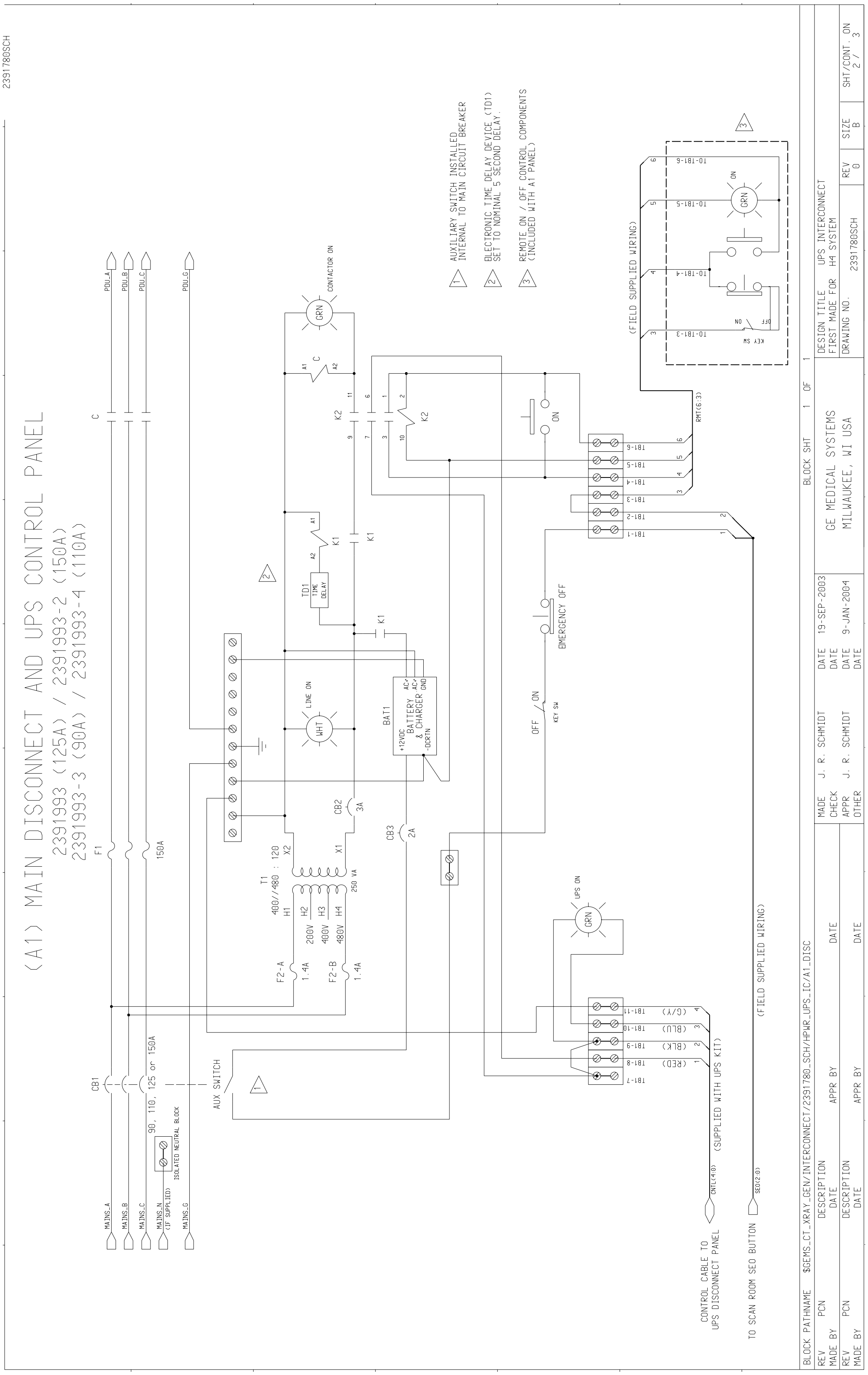


Figure 9-7 Primary Power Disconnect (A1) – Fusible Disconnect and Magnetic Contactor

6.2 Scan Room Warning Light & Door Interlock

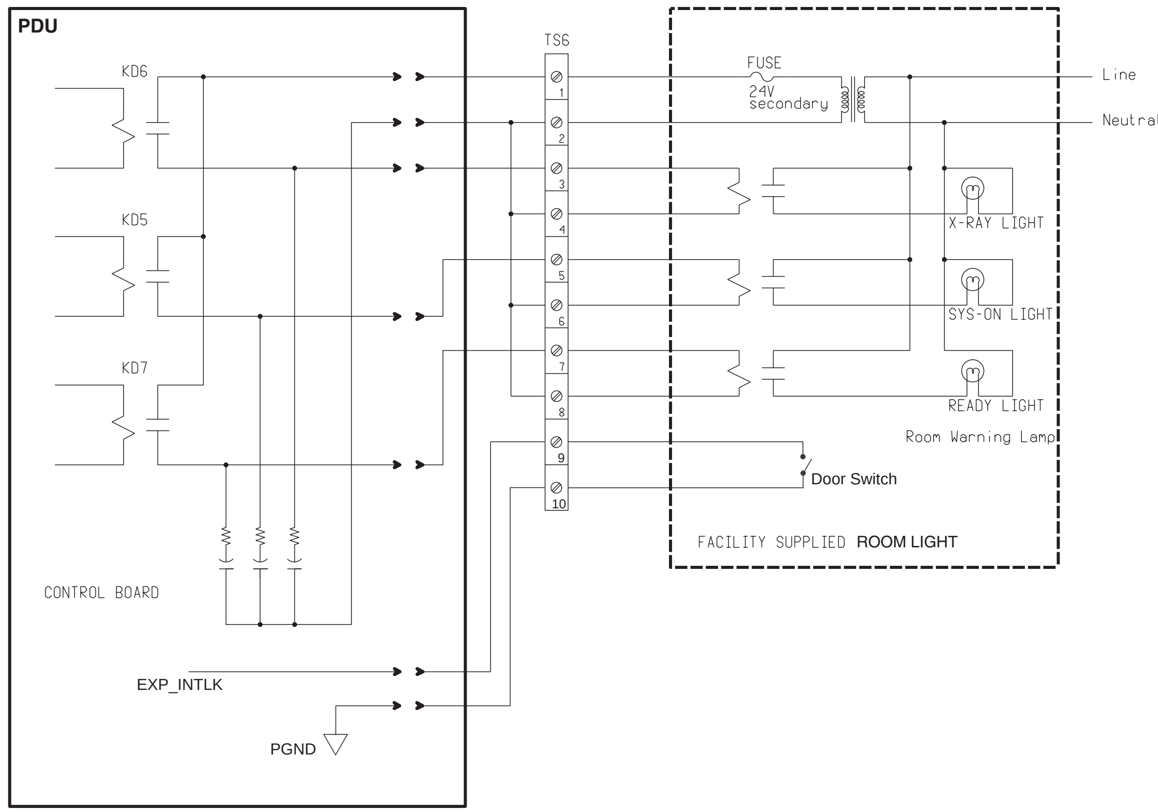
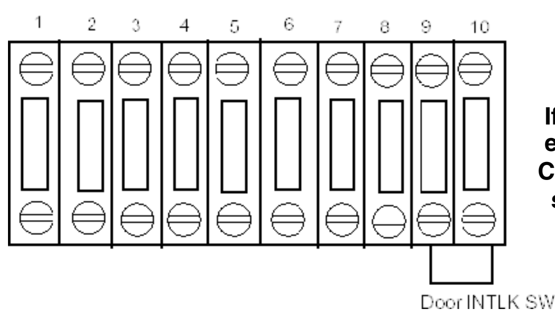
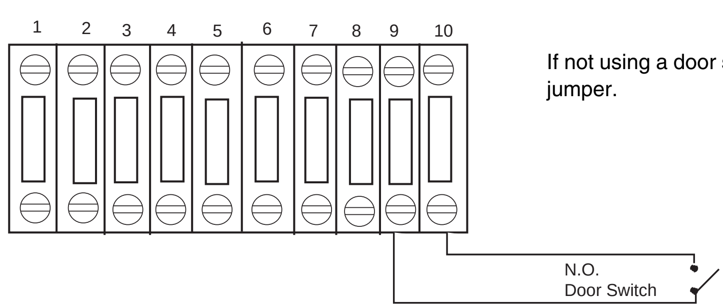


Figure 9-8 TS6 X-Ray Warning Light Connections



If jumper is not in place, exposures will not be made. Check this jumper if you get scan interlock errors.

Figure 9-9 TS6 Room Door Interlock Connections - Without a Door Interlock



If not using a door switch, add a jumper.

Figure 9-10 TS6 Room Door Interlock Connections - With a Door Interlock

GE Medical Systems, a General Electric Company, going to market as GE Healthcare.
3000 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com